Cardinal episode scripts

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Cardinal — Episode 1: The Ledger (audio script)

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Source: reviewed script, Oct 18, 2026. Narrator = Kore. Documentary and official quotes = Charon. Personal letters and Dana = Puck. [HARD] = music gated out. [SILENCE] = 2 s of silence before a reveal. Pronunciation notes and the fact-check log are in the review file, not narrated. Money is Minneapolis Fed CPI, said as "roughly."

01. The Tall Stick

[CUE: singing-bowl chapter chime] [MUSIC: hebrides from 30s, ducked] [AMBIENCE: wind.mp3]

Narrator (Kore): November, 1769. Sixty-four men, by the standard count of the party that left San Diego on July 14, and a string of mules, come down onto a wide valley at the south end of a bay no European had ever seen from land. Spanish soldiers, two Franciscan friars, an engineer, Christian Indians from the Baja missions, and a commander named Gaspar de Portolá. They were sent to find Monterey. They walked past it without recognizing it. On November fourth their scouts climbed a ridge and saw a body of water nobody expected. Narrator (Kore): That ridge is called Sweeney Ridge now. It sits between San Bruno and Pacifica, inside the Golden Gate National Recreation Area, with a discovery-site marker on the trail. The day they stood there was a Saturday, and it was a saint's day. The expedition diary records that November fourth was the day of San Carlos Borromeo, the saint in whose honor they had come north to found a presidio and a mission at Monterey, and also the name-day of the king, Carlos the Third. They said Mass on the beach. They ate mussels off the rocks. Then they turned up into the hills, and from the summit they saw what one of them called an immense arm of the sea. Narrator (Kore): A day later, Portolá's own diary records the view flatly. Before them Quote (Charon, flat documentary): extended a great arm of the

sea, sixteen to twenty leagues Narrator (Kore): in extent. Narrator (Kore): On November sixth they make camp beside a creek that runs down from the hills to the bay. They will stay five nights while the scouts go out again. On November eleventh a council of officers votes to turn around. Narrator (Kore): The story goes that they camped beneath one enormous redwood, a tree so tall and alone on the plain that the Spanish called it el palo alto. The tall stick. Narrator (Kore): [SILENCE] [HARD] That tree is still standing, on San Francisquito Creek, at the border of Palo Alto and Menlo Park, about a hundred and ten feet tall, commuter trains a few yards from its roots. It sprouted around the year 940. It named a city, and it sits in the middle of the Stanford seal, between two Latin words, semper virens: ever green. Narrator (Kore): Here is the first thing this series is going to teach you, and it is a small one. The story goes means legend. The record shows means a document. And the record on this tree is more interesting than the story. Narrator (Kore): The friar who kept the expedition's most detailed diary, Juan Crespí, did write about redwoods. A month earlier, near today's Watsonville, he described very high trees of a red color, not known to them, and wrote that because none of the expedition recognized them, they were named redwood from their color. That is the first written mention of the coast redwood. But the first time this particular tree shows up in writing is five years later, in 1774, when another friar, Francisco Palóu, put up a wooden cross near a big tree on the creek and marked it as a possible mission site. Two years after that, in 1776, Father Pedro Font measured it with a surveying instrument: fifty varas high, about a hundred and thirty-seven feet. And some local historians think Portolá's actual campsite tree was a different one, farther downstream, since cut down. Narrator (Kore): Even Malcolm Harris, who grew up in Palo Alto, has them camped five days under that tree. The diaries support the five nights by the creek. They do not name this tree. A famous tree, a famous name, and a gap where the legend lives. This whole hour lives in that gap. Narrator (Kore): No European ship had sailed into the water they saw from the ridge. The first one would not come for nearly six years, and it would carry the saint's name from the day of the sighting. We will meet that ship. Narrator (Kore): From 1769, when that column walked onto this plain, to 1891, when Stanford University opened its doors on the same creek, is one hundred and twenty-two years. In those years, the ground under the campus belonged, in turn, to the Ohlone, to the King of Spain, to the Franciscan missions, to the Republic of Mexico, to Mexican rancheros, and finally to the United States. Each handoff was made with paper. A baptismal book. A land grant. A hand-drawn map. A treaty. Narrator (Kore): This is Cardinal. I'm your host. Episode one: The Ledger.

02. The Broken Link

[CUE: singing-bowl chapter chime] [MUSIC: brahms4_2 from 0s, ducked] [AMBIENCE: wind.mp3]

Narrator (Kore): The Broken Link. Narrator (Kore): Before there was a tall stick, there were people who did not need a name for it, because they knew exactly where they were. Narrator (Kore): Along the lower San Francisquito Creek, where the campus now sits, lived a community the Spanish records call the Puichon. They were one of roughly forty small, independent tribal groups that outsiders would later lump together as

Ohlone, or, in older books, Costanoan, from the Spanish word for coast people. Each group held a small territory, usually a few hundred people in a cluster of villages, connected to neighbors by marriage, trade, and language. Narrator (Kore): At Jasper Ridge, the biological preserve that Stanford owns in the hills just west of campus, the university's own chronology records more than five thousand years of human occupation. Narrator (Kore): Let me give you a number to hold that against. Five thousand years of people on this ground. Spanish, Mexican, and American rule together, from 1769 to today, is two hundred fifty-seven years. That is about five percent of the human story on this land. If the whole five thousand years were a single day, Europeans would have walked onto that plain at about a quarter to eleven at night. Narrator (Kore): The settlers got this completely wrong. Ohlone communities did not plant fields of wheat. Later American newcomers looked at that and called them lazy, or worse, diggers, for the sticks they used to pry up edible bulbs. Malcolm Harris points out what the settlers missed. The Ohlone moved, he writes, Quote (Charon, flat documentary): in response to abundance rather than scarcity. Narrator (Kore): The ripening of acorns, the spawning of fish, the arrival of birds on the bay marshes: short bursts of intense work, followed by long stretches of what Harris calls Quote (Charon, flat documentary): social luxury. Narrator (Kore): That is not the absence of an economy. It is an economy built on knowledge instead of storage, on knowing which oak drops first and which creek runs in which month. California before 1769 held one of the densest concentrations of human linguistic diversity anywhere on earth. Historians' usual estimate is more than five hundred distinct tribal groups, speaking about a hundred languages. The total population of what the Spanish called Alta California was just over three hundred thousand. About forty thousand of them lived in the Bay Area alone. Narrator (Kore): Harris went to a public school in Palo Alto called Ohlone Elementary, named for what the children were told was the tribe that used to live on the peninsula. Quote (Charon, flat documentary): At Ohlone, we learned about the people who lived in Palo Alto before we did, and we practiced grinding acorns the way they did. Narrator (Kore): The next year, in fifth grade, the school ran a gold-rush simulation. The teachers painted little nuggets gold and hid them in the dirt of the school farm. On the first day, the kids lined up at the gate, someone yelled go, and they all ran in to grab as much as they could before anyone else got it. Quote (Charon, flat documentary): The kids who played Indian the year before now played settler. Narrator (Kore): And then the line that stops the sweetness. Quote (Charon, flat documentary): That year, there were no Ohlone. We practiced forgetting about them, too. The gold helped. Narrator (Kore): Harris opens his history of Palo Alto with this sentence. Quote (Charon, flat documentary): To speak of the Ohlone is to speak of the broken link between the Bay Area's indigenous people and the Bay Area itself. Narrator (Kore): A broken link. Some places change rulers and keep their people: the language, the place names, the memory of where the water is. Other places rupture. The people who knew the land are separated from it, and the new arrivals start the clock over. Narrator (Kore): Harris is careful about who broke the link and how. Disease, he writes, rather than extermination campaigns, killed about half of the roughly three hundred thousand Native people of Alta California under Spain and Mexico. We will get to how the missions made that worse. And then his turn. Quote (Charon, flat documentary): But when the United States showed up, the rules changed. Narrator (Kore): That is the next episode's darkness, and we will not rush it. Narrator (Kore): For

now, keep one more of his sentences in your pocket, because it is going to come back again and again in this series. Education, he writes, Quote (Charon, flat documentary): is one of the Anglo settlers' most important weapons. Narrator (Kore): The way a place teaches its children to feel at home in what was done to make it. Narrator (Kore): The Ohlone did not vanish. Their descendants, including the Muwekma Ohlone Tribe, are here today, and many trace their families through a very specific kind of document: the baptismal registers of the Spanish missions at San Francisco and Santa Clara. Which means that, for the people who were here first, the paper trail of their own ancestry begins in a book written by the people who came to change them. Narrator (Kore): Let's open that book.

03. What a Mission Was

[CUE: singing-bowl chapter chime] [MUSIC: aria from 0s, ducked] [AMBIENCE: church.mp3]

Narrator (Kore): What a Mission Was. Narrator (Kore): January 12, 1777. On the bank of the Guadalupe River, near an Ohlone village the Spanish wrote down as So-co-is-u-ka, laurel wood, Franciscan friars plant a cross and put up a shelter for worship. They name the place Misión Santa Clara de Thámien, after Saint Clare of Assisi and after the local people, the Thámien. It is the eighth mission in Alta California, and the first one named for a woman. Narrator (Kore): But it almost was not built there. Remember the tall redwood on San Francisquito Creek? In 1774, Father Palóu put up a cross beside it and marked the spot for a mission. Two years later, in March 1776, the commander Juan Bautista de Anza and Father Pedro Font rode out to inspect it. Font measured the tree. Then they looked at the creek, decided its flow was too unreliable to water mission fields through a dry summer, and moved on. The mission went about fifteen miles to the southeast instead, to Santa Clara. Narrator (Kore): So the very first plan for a big institution on that creek was a mission. More than a century later, someone would try again, with a railroad fortune instead of a royal charter. Santa Clara itself kept moving. Floods, fire, and an earthquake forced six rebuilds of the church. Narrator (Kore): Every person who came into a mission was written into a book. A baptismal book. A marriage book. A burial book. The line gave a Christian name, the parents, the village, a godparent, and the minister's signature. That is the shape of the form, not one person's entry. The books themselves were kept with real care, which is why historians can count what happened, and why Ohlone families today can trace their ancestors through them. Narrator (Kore): So what was a mission? For generations, California fourth-graders built them out of sugar cubes. The state's history framework now tells teachers to stop. Quote (Charon, flat documentary): Building missions from sugar cubes or popsicle sticks does not help students understand the period and is offensive to many. Narrator (Kore): Let me teach this one properly, because a California mission was four institutions in one. Narrator (Kore): One: a church. Its stated purpose was conversion, to bring Native people to Catholic Christianity. A baptized Native person was called a neófito, a neophyte, a new believer. An unbaptized one was a gentil, a gentile. Narrator (Kore): Two: an economy. A mission was a farm, a ranch, a workshop, and a warehouse. California was months by sea from Mexico, and supply ships were small, so the friars

needed the neophytes to grow the wheat, tend the cattle, weave the cloth, tan the leather, and render the tallow. By December 31, 1832, at the peak of the system, the missions together held about one hundred fifty-one thousand head of cattle and nearly one hundred thirty-eight thousand sheep. And at their height they controlled roughly one-sixth of all the land in Alta California. Narrator (Kore): Three: a military partner. Every mission sat in the district of a presidio, a fort with a small garrison of soldiers. The soldiers protected the friars, and they also went out to bring back people who left. Narrator (Kore): Four: a labor system. This is the part the sugar-cube models leave out. Once baptized, a neophyte was expected to live at the mission, under the friars' rules, and could not simply walk away. People who left without permission were treated as runaways, and soldiers were sent to bring them back. Santa Clara University's own Ohlone heritage project puts it plainly. Native people from dozens of communities were brought to live together at the mission, and, Quote (Charon, flat documentary): laboring without pay, these Native Californians built the mission structures and provided the agricultural and industrial workforce to support Spain's colony in California. Narrator (Kore): A teacher's note on a hard word. Was it slavery? Historians argue about the label. Neophytes were not bought and sold, and some came, at least at first, for food, trade goods, or safety. What is not in dispute is that they could not leave freely, they were not paid, and they were physically punished. The sailor Richard Henry Dana, whom you will meet in a few minutes, saw the system as it was ending and wrote that the neophytes were Quote (Charon, flat documentary): virtually slaves. Narrator (Kore): Now jump back ten years, because there is a handoff here that almost nobody knows. Narrator (Kore): The friars who built the California missions were Franciscans. But the mission chain did not start with them. On the peninsula of Baja California, below today's border, the Jesuits had built a chain of missions. Then, in 1767, King Charles the Third of Spain expelled the entire Jesuit order from his empire. The man sent to carry out the expulsion in Baja was the new governor, Gaspar de Portolá. The Jesuits gathered at Loreto and left for exile in February 1768. A month later, the king's inspector general, José de Gálvez, handed their missions to the Franciscans, under a friar named Junípero Serra. Narrator (Kore): That is the Jesuit-to-Franciscan handoff. And a year after it, Gálvez sent Portolá and Serra north, by land and by sea, to occupy San Diego and Monterey before the Russians or the British could. Narrator (Kore): Here is the twist. In 1851, after the Americans arrived, the new Catholic bishop of California transferred the struggling parish at Mission Santa Clara from the Franciscans to the Jesuits. They opened a school there called Santa Clara College. Today it is Santa Clara University, and it is the only university campus in California built on the site of one of the twenty-one missions. The order Spain threw out in 1767 came back eighty-four years later and inherited the mission. Narrator (Kore): From 1769 through December 31, 1832, the friars across all the California missions recorded eighty-seven thousand seven hundred eighty-seven baptisms. They also recorded sixty-three thousand seven hundred eighty-nine deaths. Put those side by side. For every ten people baptized, about seven burials. The death rate for children was especially high. At Mission San Gabriel, one missionary reported that three out of four children died before the age of two. Measles epidemics in 1806 and 1828 swept through the crowded compounds. The missions kept needing new converts because the people already there kept dying. Narrator (Kore): Santa Clara was the busiest of all. It recorded more baptisms, more marriages, and more burials than any

other mission in the chain: eight thousand five hundred thirty-six baptisms and six thousand eight hundred nine burials. Its Native population peaked at about one thousand four hundred sixty-four in 1827. Narrator (Kore): Let me give you a comparison you can feel. Fourteen hundred sixty-four people is smaller than one year's incoming first-year class at Stanford today. In the fall of 2025 the university enrolled one thousand eight hundred sixty-six new first-years, the class of 2029. Now imagine a class that size living in one compound, working the fields without pay, with a burial book that, over the life of the mission, filled up almost as fast as the baptismal book. Narrator (Kore): Junípero Serra was born Miguel José Serra on November 24, 1713, in Petra, on Mallorca. He held a chair of philosophy at the university in Palma, gave it up in 1749, and was president of the California missions from 1769 until his death on August 28, 1784, at Mission San Carlos Borromeo in Carmel, when only nine of the twenty-one missions existed. He is buried there. He walked that coast on a leg injured years earlier, on the road up from Veracruz. Narrator (Kore): Hold the name San Carlos. On that saint's feast day, the soldiers on Sweeney Ridge saw the bay. The mission where Serra is buried carries the same saint. A ship of that name is about to sail in. Narrator (Kore): His reputation has two halves, and both go on the air. On September 23, 2015, Pope Francis canonized him in Washington, D.C., and praised him as a witness to the joy of the Gospel in these lands. Three years later, on September 13, 2018, Stanford University announced it would take his name off its main campus mall, a dormitory, and an academic building, saying the mission system inflicted great harm and violence on Native Americans. Serra Mall became Jane Stanford Way, effective October 7, 2019. But Stanford kept one small Serra Street, so that, in the committee's words, history would not be erased from the campus. Narrator (Kore): His successor, Fermín Francisco de Lasuén, founded nine more missions between 1786 and 1798, including La Purísima in 1787. There is a Lasuen Street at Stanford. We will walk it in the final episode. The mission was also a ledger: the first written record, for thousands of people, of their own names.

04. The Royal Road

[CUE: singing-bowl chapter chime] [MUSIC: borodin from 0s, ducked] [AMBIENCE: horses.mp3, surf.mp3]

Narrator (Kore): The Royal Road. Narrator (Kore): A rider leaves Mission Santa Clara heading north with a pouch of letters. The path he follows was a footpath long before he was born, an Ohlone trail along the edge of the hills, above the marshes. He passes the creek with the tall redwood. He reaches Mission San Francisco de Asís, founded in 1776, about forty-five miles up the peninsula. From there, letters go on by ship, or south by horse to Monterey, the capital, and from Monterey down the coast to San Diego, and eventually to Mexico City. Narrator (Kore): The man who created this series calls this road the Silk Road of the West. The metaphor is his, not a historian's term, so test it. What it captures is a single corridor for friars, orders, tools, and very slow news. By the time all twenty-one missions existed, they stretched from San Diego, founded in 1769, to Sonoma, founded in 1823, about six hundred miles, roughly the distance from London to Milan. What it gets wrong is the cargo. The Silk Road moved luxury goods between empires. Spanish California's long-distance freight came by sea, on supply

ships from San Blas, in Mexico, and later on foreign traders. The road was for people, animals, letters, and orders. Narrator (Kore): And here is the bigger correction, a real one. The record shows there was no single El Camino Real in Spanish times. Camino Real means royal road. There were many caminos reales, a general term for roads used under the Crown's authority, and most of them followed older Native trade routes. The Autry Museum of the American West says it bluntly. El Camino Real, as a single road lined with missions, each a day's ride from the next, is an American construct of the twentieth century. Narrator (Kore): Who constructed it? Boosters. Around 1900, Californians fell in love with the ruins of the missions, which had been crumbling for decades, and began restoring them and selling the romance of the Spanish days to tourists. In 1906, the first of the famous cast-iron bell markers was installed outside the Old Plaza Church in Los Angeles. It was designed by a woman named Harrie Forbes, who signed herself Mrs. A. S. C. Forbes. Each bell carried the words El Camino Real and two dates: 1769 and 1906. There is no known family tie between her and the Boston Forbes family you will meet later in this series. Do not let the name do that work. Narrator (Kore): Think about those two dates on one bell. The year the friars arrived, and the year the boosters arrived. The marker commemorates both, and only one of them is about the eighteenth century. Narrator (Kore): The name for this is an invented tradition, the term Eric Hobsbawm and Terence Ranger gave, in 1983, to customs that feel ancient and were built fairly recently, to lend a new order an old legitimacy. The mission bell on the highway is one. It will matter later, because the Stanfords built their university in a style that echoes the missions, red tile and long arcades, on purpose. That is episode four. Narrator (Kore): Drive the peninsula today and the corridor is still the main street. From Santa Clara toward San Francisco it is State Route 82, and the signs still say El Camino Real. Interstate 280, up in the hills, is the Junipero Serra Freeway. You can cross the Spanish century without leaving the car. Narrator (Kore): Now the sea, because the sea is what the road metaphor leaves out. Jump six years from the camp on the creek. Narrator (Kore): [HARD] August 5, 1775. Night. A small Spanish supply ship called the San Carlos, a packet built at the naval base of San Blas, feels her way through a strait that is, at its widest, only about three miles across, and often hidden in fog. Ships on the Manila trade had been passing this coast for two hundred years. They knew Point Reyes. They knew the Farallon Islands, thirty miles out. They had not known there was a bay behind the cliffs. The men on Sweeney Ridge had seen the water from land. Nobody had sailed into it. Narrator (Kore): The San Carlos was the first ship that did. Her commander, Lieutenant Juan Manuel de Ayala, had been shot in the right foot in April, by a loaded pistol his predecessor left in the cabin. He wrote that he could not do the work in person. His pilot, José de Cañizares, took the launch and drew the first working map of the bay, south to what is now Suisun. Ayala arrived the night of August fifth, stayed forty-four days, and called it one of the best harbors on the coast. They anchored by an island he named Isla de los Ángeles. The cove is Ayala Cove. The island is Angel Island. Narrator (Kore): On September 18, rudder damaged, they ran down to Monterey and arrived the next day. California Historical Landmark 236, at Aquatic Park, Beach and Larkin, marks the entrance. Without that chart, the presidio and the mission of 1776 would have been a guess. Narrator (Kore): And the name keeps moving. On the feast of Saint Charles, in 1769, the land party saw the bay. In 1770 Serra's headquarters, Mission San Carlos, was founded, and moved to Carmel, where he

is buried. In 1775 the ship of that name was the first European keel in the water. The town of San Carlos, down the peninsula, says on its own website that the origin is lore, not a document. Three candidates: the ship, King Charles the Third, or that feast day. The record confirms all three existed. It does not say which one the developers meant. Narrator (Kore): The road's one hard fact was its slowness. Mexico City was months away, so whoever controlled the landings and the stamps held everyone else. In a few minutes, that becomes money.

05. Saints on the Map

[CUE: singing-bowl chapter chime] [MUSIC: scottish3 from 0s, ducked] [AMBIENCE: wind.mp3, horses.mp3]

Narrator (Kore): The Story of Our Streets. Saints on the Map. Narrator (Kore): Almost nobody knows the stories behind the names they drive past. In California the map is a family portrait, and the oldest layer is Catholic and Spanish. Narrator (Kore): Start with the big one. San Francisco is Saint Francis of Assisi, the founder of the Franciscan order, the friars who built the missions. Mission San Francisco de Asís was founded in 1776. Today almost everyone calls it Mission Dolores, after a nearby creek and lagoon the Anza expedition named for Our Lady of Sorrows, Nuestra Señora de los Dolores. And the town by the bay was not called San Francisco at first. It was Yerba Buena, good herb, for a wild mint that grew on the hills. Narrator (Kore): On January 30, 1847, the American alcalde, navy lieutenant Washington Bartlett, ordered the name San Francisco used in every official paper, because Yerba Buena was unknown beyond the district. Mariano Vallejo had wanted a rival town named Francisca, for his wife. Bartlett got there first. Vallejo used her second name. The town is Benicia. Narrator (Kore): Santa Clara is Saint Clare. San José is Saint Joseph. Sacramento is the sacrament. Saints' names run down the coast like beads on a rosary. Narrator (Kore): Then the explorers. Portola Valley, in the hills above Stanford, and Portola Drive in San Francisco, are named for Gaspar de Portolá. The story goes that he discovered the bay. The record shows a soldier carrying out Gálvez's plan, who founded Monterey on June 3, 1770, sailed away on July 9, and never came back. He later governed Puebla, in Mexico. Interstate 380, toward the airport, used to be the Portola Freeway, because Sweeney Ridge is just south of it. The state renamed it for Quentin L. Kopp. The ridge stayed. The sign moved. Narrator (Kore): A quarter-millennium later, Harris writes, Portola Valley was the richest town per capita in the richest country in the world. Sand Hill Road, between the campus and those hills, is where he says the Valley's investment money sits. Page Mill Road, west into the redwoods, began as William Page's haul road from a shingle mill. Both belong to later episodes. Narrator (Kore): Then the friars. In San Mateo there is Junípero Serra High School, a Catholic boys' school, and two of the most famous athletes in American history went there. The baseball home-run king Barry Bonds, class of 1982, and the quarterback Tom Brady, class of 1995. Every time a broadcaster said Serra High, they were saying the name of an eighteenth-century friar from Petra, on Mallorca. San Mateo County's Junipero Serra Park, just up the hill, sits on land that used to be a cattle ranch. We are about to say the ranch's name. Narrator (Kore): And then the strange one. Fly into San Francisco International Airport, and you

land on what used to be a Mexican cattle ranch called Rancho Buri Buri. The name has nothing to do with Spanish. It is traced to an Ohlone village recorded as Urebure, of the Yelamu people, whose families were brought into Mission Dolores. The rancho covered fourteen thousand six hundred thirty-nine acres. That is about fifty-nine square kilometers, almost exactly the land area of the island of Manhattan. Today that ground is San Bruno, Millbrae, South San Francisco, and part of Burlingame. There is still a Buri Buri Park. A Native village name survives because a Mexican ranch borrowed it, and it is now under the runways of one of the world's busiest airports. Narrator (Kore): Saint, soldier, ship, village, freeway. The paper changed. The names stayed, and they do not all mean what the signs imply.

06. Silver and Hides

[CUE: singing-bowl chapter chime] [MUSIC: brahms3_3 from 0s, ducked] [AMBIENCE: surf.mp3, wind.mp3]

Narrator (Kore): Silver and Hides. Narrator (Kore): Richard Henry Dana, writing about Monterey in 1835. LETTER (Puck: Richard Henry Dana, about twenty, a Harvard student turned common sailor, sardonic, sea-wind in the voice): "They have no circulating medium but silver and hides, which the sailors call California bank notes." Narrator (Kore): He is the first of a type this series keeps meeting: the Eastern gentleman who comes west and writes it down. Dana was a Boston boy at Harvard. Measles damaged his eyes so badly he could not read, and the family could not afford Europe, so in August 1834 he signed on as a common sailor aboard the brig Pilgrim, around Cape Horn. He was home in September 1836, finished Harvard, and in 1840 published Two Years Before the Mast. For a generation back east, it was the picture of California. The Pilgrim was there to collect cowhides. Narrator (Kore): The missions and the ranches ran enormous herds. There was no railroad and no bank, and beef could not reach a market, so the skins were the crop. Boston houses sent ships around South America with manufactured goods and swapped them for hides, which went home to become shoes. Narrator (Kore): The price of a hide was fixed by custom at two dollars. Josiah Royce, born in a California mining town in 1855, writing in 1886, says the traders took them at that uniform value. A hide was money. Rancheros paid their debts in hides. Narrator (Kore): Let's convert that, and let's do the method once, because this series will use it every time. We are using the consumer price index published by the Federal Reserve Bank of Minneapolis, the long table that runs back to 1800. The 2026 index on that table is 1004.8. The index for 1835 is 31. Divide one by the other and two dollars in 1835 is roughly sixty-five dollars in 2026 grocery money. Before 1913 these figures are estimates, so the honest word is roughly. Imagine walking into a store with a stiff, folded cowhide under your arm worth about sixty-five dollars, and paying with it. Narrator (Kore): Dana's work was carrying those hides. At San Diego his company had a hide-house built to hold forty thousand of them, and the crew would go in and stare at it, calculating how long it would take to fill. On May 8, 1836, the ship he sailed home on had finally loaded, in his words, our forty thousand hides, thirty thousand horns, along with barrels of otter and beaver skins. Narrator (Kore): Forty thousand hides at two dollars is eighty thousand dollars. By the same index, roughly two point six million

dollars today. One ship. One voyage. Narrator (Kore): Now here is the part that makes Dana's book useful as economics, not just adventure. He noticed the prices going the other way. LETTER (Puck: Richard Henry Dana, the young sailor, dry and exact): "Their hides, too, which they value at two dollars in money, they give for something which costs seventy-five cents in Boston; and buy shoes, like as not made of their own hides, and which have been carried twice around Cape Horn, at three or four dollars." Narrator (Kore): And things sold, he wrote, Quote (Charon, flat documentary): at an advance of nearly three hundred per cent upon the Boston prices. Narrator (Kore): A cow is skinned in California. The hide becomes a shoe in Boston. The shoe sails back, and the person who raised the cow pays several times the Boston price. That is chokepoint economics. One way in, one way out, and whoever holds the gate sets the price. Narrator (Kore): In Mexican California, there was one official custom-house, at Monterey. Every trading ship had to enter its cargo there and pay duties before it could trade anywhere on the coast. There were very few safe harbors. Dana counts San Francisco, Monterey, and San Diego as the only ports safe from the winter southeasters. And there were very few ships. If you lived inland, you bought what the ships brought, at the ships' price, or you went without. Narrator (Kore): Dana captured how isolating that was in one line about Santa Barbara, where the ship anchored three miles offshore and the town was nearly a mile farther inland. So far as the crew were concerned, he wrote, the town Quote (Charon, flat documentary): might almost as well have been in the middle of the Cordilleras. Narrator (Kore): The Cordilleras: the Andes. Remember the gate. In 1835 it is a custom-house and an anchorage. Later it will look like other things. Narrator (Kore): What was life like for the Californios on the other side of that trade? Royce collected the memories of people who knew them. The white population, he wrote, was made up mainly of Spanish and Mexican colonists Quote (Charon, flat documentary): whose chief industry was raising cattle for the hides and tallow, and whose private lives were free, careless, and on the whole, as this world goes, moderately charming and innocent. Narrator (Kore): They were, he says, Quote (Charon, flat documentary): gay and jovial, full of good fellowship and hospitality. Narrator (Kore): They paid the Yankee traders a little slowly, but generally paid in the end, in the currency of the country, that is, in hides. Narrator (Kore): Notice who is missing from that charming picture. The Native vaqueros and laborers who did most of the work of raising those cattle. Royce's own sentence, in the same passage, says it. Quote (Charon, flat documentary): The mission Indians had in large part disappeared. Narrator (Kore): Keep both halves in view. That is the honest way to hear the old romance of the ranchos. Narrator (Kore): In December 1835, Dana's ship sailed into San Francisco Bay. Past the Presidio on its cliff, there was a crumbling mission, a few ranches, and a newly begun settlement, mostly of Yankee Californians, called Yerba Buena. A Russian brig from Sitka, in Russian Alaska, was anchored there, loading tallow and grain. And Dana, a sailor who had never heard of a gold rush, wrote this. LETTER (Puck: Richard Henry Dana, young, looking at the bay): "If California ever becomes a prosperous country, this bay will be the centre of its prosperity." Narrator (Kore): One Eastern beat, then back on land. In 1839 Alexander Forbes, a Scottish merchant in Tepic, Mexico, published in London what is generally called the first full English book about California: a rich, thinly governed province. No family tie has been shown to the Boston Forbes. The outside world was already leaning in.

07. Secularization

[CUE: singing-bowl chapter chime] [MUSIC: mozart40_2 from 0s, ducked] [AMBIENCE: fs_42139.mp3]

Narrator (Kore): Secularization. Narrator (Kore): Now jump back from Dana's voyage to 1821. Mexico wins its independence from Spain, and with it, the far-off province of Alta California becomes Mexican. Narrator (Kore): The new Mexican republic had a problem with the missions. They were run by Spanish-born friars loyal to a church still tied to Spain, and they controlled the best land in California, along with most of its labor. Mexican liberals believed in separating church power from the state. Ambitious soldiers and settlers in California wanted that land. And some officials genuinely believed the neophytes ought to be free citizens. Narrator (Kore): The word is secularization, from the Latin saeculum, the world, as opposed to the cloister. A secular priest serves a parish under a bishop. A regular priest, like a Franciscan, lives under a religious order. To secularize a mission meant, on paper, to turn it into an ordinary parish and to release its lands, herds, and people. Narrator (Kore): It came in steps. On July 25, 1826, California's first Mexican-born governor, José María de Echeandía, issued a proclamation of emancipation that let qualified neophytes in the southern districts leave mission rule and become Mexican citizens. Five years later, in 1831, about eighteen thousand six hundred eighty-three Native people were still under mission control, compared with four thousand three hundred forty-two soldiers, settlers, and what the count called other classes. Four to one. Narrator (Kore): Then, on August 17, 1833, the Mexican Congress passed the decree for the secularization of the missions of California. And on August 9, 1834, Governor José Figueroa issued the rules for how it would actually work. The promise, in substance, was this. Each head of a family, and everyone over twenty, would receive a lot of mission land, not more than four hundred varas square and not less than a hundred. A vara is a Spanish yard, about thirty-three inches, so that plot came to somewhere between about seven and twenty-eight acres. Half the livestock, shared out. Half the tools, the seed, and the movable goods, or less. Narrator (Kore): On paper, it is one of the most sweeping land redistributions in North American history. Narrator (Kore): On paper. Narrator (Kore): Here is a second concept, and it is the key to this whole episode. Trust versus title. During the mission era, the land was supposed to be held in trust by the friars, for the benefit of the Native people, until they were judged ready to manage it themselves. Santa Clara University's Ohlone heritage project puts it this way. The lands Quote (Charon, flat documentary): were supposed to be held in trust by the Franciscans on behalf of Native people. Narrator (Kore): A trust is a promise. A title is a deed with your name on it. And when the moment came to turn the trust into titles, the titles mostly went to someone else. Narrator (Kore): Who ran the handover? Secular officials called administradores, administrators. Dana, who was on the coast as it happened, had a sailor's contempt for them. The law, he wrote, declared all the Indians free and independent rancheros. The change in their condition was, as might be supposed, only nominal. They were virtually slaves, as much as they ever were. And the missions' great possessions were given over to be preyed upon by Quote (Charon, flat documentary): the harpies of the civil power, Narrator (Kore): the administrators, who usually ended, in a few years, by making themselves fortunes.

Narrator (Kore): At Santa Clara, secularization began in 1836. In 1832, one thousand one hundred twenty-five Native people lived at the mission. By 1839, two hundred ninety-one. A fall of about seventy-four percent in seven years. Robert Jackson, working through the baptism and burial registers, found that deaths alone do not explain it. Hundreds left, or were pulled onto the new private ranches. The mission's own census, signed December 31, 1839, by Father Jesús María Vásquez del Mercado, is still in the archives. Narrator (Kore): Some Ohlone people did get land, and their stories deserve to be told by name. Santa Clara issued four land grants to Ohlone men who had entered the mission as children and asked for their own ancestral village sites back, places recorded as Ulistac and Posolmi. Four is the most of any Bay Area mission. But as Santa Clara's own historians note, those grants were only a small fraction of the total acreage the mission system controlled. In the Bay Area, nearly all other lands were granted to colonial elites during the 1830s and 1840s. Narrator (Kore): To understand how fast the land moved, take one mission in the south. In 1845, Governor Pío Pico sold Mission San Juan Capistrano, the whole thing, to his brother-in-law, John Forster, for seven hundred ten dollars. On the same index, that is roughly twenty-five thousand dollars today. An entire mission. Narrator (Kore): [SILENCE] So. A law that freed people on paper, a set of rules that promised them land, and an administration that mostly handed that land to someone else. Trust became title. Just not their title. Narrator (Kore): Where did the land go? Into a new kind of property. Let's go back to the creek with the tall redwood, and meet the people who owned it next.

08. The Rancho

[CUE: singing-bowl chapter chime] [MUSIC: eroica2 from 240s, ducked] [AMBIENCE: horses.mp3, wind.mp3]

Narrator (Kore): The Rancho. Narrator (Kore): 1839. In Monterey, the governor of Alta California is a Californio named Juan Bautista Alvarado, born in Monterey in 1809, so about thirty, a man who had come to power in 1836 in a local revolt against a governor sent from Mexico City. Now he grants a former soldier, Antonio Buelna, a ranch on the southwest side of San Francisquito Creek. Rancho San Francisquito. One thousand four hundred seventy-one acres. Narrator (Kore): That piece of ground, today, is the northern part of the Stanford University campus. Narrator (Kore): Here is how you got a rancho, because the process becomes a disaster in the next episode. You petitioned the governor. If you could not write, a neighbor wrote it. You attached a diseño, the drawing, a map with no surveyor behind it: a creek bend, a big oak, a hill. The governor asked the local officials and the mission whether the land was free and whether you owned cattle. If they said yes, he issued a título, a title. Then a local official walked the bounds with you and put you in juridical possession. The oak was the corner. The neighbors were the survey. Narrator (Kore): What is missing is a precise line. A grant might say two leagues, more or less. Under Mexican law a governor could grant as little as a third of an acre or as much as eleven square leagues, about fifty thousand acres. Two leagues was most common. In a country with plenty of land and few people, more or less was fine. Your neighbors knew where the oak was. Narrator (Kore): Two ways of owning land are about to collide. Mexican ranchos worked on memory, on neighbors, on the oak tree.

The American system that was coming worked on measured lines, filed documents, and courts. When they met, the oak tree lost. Hold that thought until the hook. Narrator (Kore): Fourteen hundred seventy-one acres is almost six square kilometers, nearly three times the country of Monaco. Narrator (Kore): And Stanford today holds eight thousand three hundred fifty-three acres, in seven cities and counties, from Portola Valley to Palo Alto to the unincorporated hills. So Buelna's whole rancho is less than a fifth of what the university now owns. Where did the rest come from? Narrator (Kore): The southern part of the campus sits on a second, bigger rancho. Rancho Rincón de San Francisquito, the corner of the San Francisquito. Eight thousand four hundred eighteen acres, granted by Alvarado in 1841 to a man named José Peña. Narrator (Kore): Peña was an old artilleryman from the Presidio of San Francisco who had grazed cattle on this ground with the mission's permission since 1822, went away to Mexico, came back, and in 1841 was teaching school in the little settlement at Santa Clara. In 1847 he sold almost the whole rancho to two brothers, Secundino and Teodoro Robles. Narrator (Kore): And here is a detail you could not invent, and should not romanticize. Back in 1824, Ohlone people had shown Secundino Robles where they dug a red pigment rock south of San José. It was cinnabar, the ore of mercury, quicksilver, and by 1845 that deposit was the New Almaden mine, which would become one of the most valuable mines in North America. The brothers were paid in cash and a share. Mercury is what miners use to pull gold out of crushed rock. Hold on to that, too. The people who knew the pigment did not get the mine. The men who were shown the place did. Narrator (Kore): The Robles brothers did not hold the land. By the late 1850s much of it had gone to lawyers' fees and debts. That pattern, land eaten by the cost of proving you own it, is the next episode. Remember their names when we get there. Narrator (Kore): The campus does not sit on the Soto family's grant. That was a neighboring ranch, the Rinconada del Arroyo de San Francisquito, about two thousand two hundred twenty-nine acres, given in 1835 to Rafael Soto, whose father had come north with the Anza expedition. That land is now, roughly, downtown Palo Alto, and the tall redwood stands at its northwest corner. The campus itself sits on Buelna's rancho in the north and Peña's in the south. Narrator (Kore): And one more fact, because the tree keeps teasing us. The Stanfords put El Palo Alto on their university seal. But the tree stands on the east side of the railroad line, and a land map made for Leland Stanford shows it was never on land he owned. The university's symbol was always just across the fence. Narrator (Kore): So by 1846, the ground under the future campus was divided among Mexican rancheros, held by grants, diseños, and memory. Native people who had worked these fields for the mission were now working them for the rancheros, or gone. And the province was about to change hands one more time, faster than anyone could have imagined.

09. Nine Days

[CUE: singing-bowl chapter chime] [MUSIC: brahms1_1 from 0s, ducked] [AMBIENCE: crowd.mp3, wind.mp3]

Narrator (Kore): Nine Days. Narrator (Kore): June 14, 1846. Armed American settlers ride into Sonoma and take Colonel Mariano Vallejo prisoner in his own house. They paint a bear, in berry juice, on a flour sack. Malcolm Harris, quoting the period, calls it

something that they called a bear. The California Republic lasts twenty-five days. On July 9, Navy lieutenant Joseph Revere raises the Stars and Stripes. Congress had declared war on May 13. Sonoma had not heard. Narrator (Kore): The pioneer John Bidwell, who watched it, called it a pretense, the Texas game. General John Frisbie later told a Bancroft interviewer that the real object was San Francisco Bay. Ulysses S. Grant, who fought the war as a young officer, wrote in his memoirs that it was Quote (Charon, flat documentary): one of the most unjust ever waged by a stronger against a weaker nation. Narrator (Kore): The war ended in Mexico City, with Mexico defeated and its capital occupied. Now I am going to give you two scenes, nine days apart, two thousand miles apart, and I want you to picture them side by side. Narrator (Kore): Scene one. Monday, January 24, 1848. Coloma, on the South Fork of the American River. James Marshall, a carpenter from New Jersey, is building a sawmill for John Sutter, a Swiss immigrant whose Mexican grant is eleven square leagues, the legal maximum, nearly fifty thousand acres, bigger than Liechtenstein. Marshall has been running water through the mill's tail-race to scour it deeper. In the morning he walks the channel and sees something shining in the gravel. Royce warned that the tale is simple in its main incident and tangled in its details. The date the histories use is January 24. The speeches later put in Marshall's mouth do not agree. The flake is the part that holds. Narrator (Kore): Scene two. Wednesday, February 2, 1848. The town of Guadalupe Hidalgo, just outside Mexico City. An American diplomat named Nicholas Trist sits down with three Mexican commissioners, Luis Gonzaga Cuevas, Bernardo Couto, and Miguel Atristain, and signs a peace treaty. Narrator (Kore): Trist is not even supposed to be there. President James K. Polk had recalled him months earlier. Trist ignored the order. On December 4, 1847, he wrote to his wife. LETTER (Puck: Nicholas Trist, a diplomat writing privately to his wife, decided and a little defiant): "Knowing it to be the very last chance and impressed with the dreadful consequences to our country which cannot fail to attend the loss of that chance, I decided today at noon to attempt to make a treaty; the decision is altogether my own." Narrator (Kore): Article Five of the treaty draws the new border. Article Twelve, in the English of the signed text: Quote (Charon, flat documentary): In consideration of the extension acquired by the boundaries of the United States, as defined in the fifth article of the present treaty, the Government of the United States engages to pay to that of the Mexican Republic the sum of fifteen million of dollars. Narrator (Kore): [SILENCE] [HARD] Nine days. The gold was found on January 24. The treaty that handed California to the United States was signed on February 2. And nobody at that table knew. The news from Coloma took weeks just to reach San Francisco. The first printed report appeared in a small San Francisco paper, The Californian, on March 15. The President of the United States did not officially confirm the gold to Congress until December 5, 1848, ten months after the discovery. Narrator (Kore): When people say California happened in five days, this is the coincidence they mean. It was nine. Mexico sold the land without knowing what was in it. The United States bought it the same way. Narrator (Kore): And the speed does not stop. Same treaty, very different clocks, and we will teach the machinery next time. California, two years and seven months, treaty to statehood on September 9, 1850, with no organized territorial stage at all. Nevada, sixteen years and nine months, October 31, 1864. Utah, almost forty-eight years, to January 4, 1896. New Mexico, January 6, 1912. Arizona, February 14, 1912, about sixty-four years. California reached statehood roughly twenty-

five times faster than its neighbors to the southeast. Narrator (Kore): The treaty transferred more than five hundred twenty-five thousand square miles, fifty-five percent of Mexico, on the order of France, Spain, and Germany put together. It became California, Nevada, Utah, most of Arizona and New Mexico, and parts of Colorado and Wyoming. Half a country, on a paper signed by a man the president had already fired. Narrator (Kore): The price was fifteen million dollars, plus up to three and a quarter million in debts the United States agreed to assume. The Louisiana Purchase, forty-five years earlier, had carried the same sticker price, for a larger piece of the continent. This one came with a harbor Dana had already named, and, though nobody at the table knew it, a mill race full of gold. Narrator (Kore): The ledger detail is what the Senate did on March 10, 1848. It struck Article Ten, the article that had guaranteed Mexican land grants. The treaty still promised to respect the property of Mexicans who stayed. The explicit guarantee for the diseños, the oak trees, and the more or less, was gone. Narrator (Kore): Mexico asked what that meant. On May 26, 1848, in the city of Querétaro, before the ratifications were exchanged, American commissioners Nathan Clifford and Ambrose Sevier sat down with Mexico's foreign minister, Luis de la Rosa, and signed an explanation. The Protocol of Querétaro. On the deleted article, the American government said it Quote (Charon, flat documentary): did not in any way intend to annul the grants of lands made by Mexico in the ceded territories. Narrator (Kore): Those grants, the protocol says, preserve the legal value they may possess, and the grantees may cause their legitimate titles to be acknowledged before the American tribunals. Legitimate, it adds, meant legitimate under Mexican law in California up to May 13, 1846. A promise that you could prove your paper in an American court. Remember the verb. Prove. It is about to become a bill. Narrator (Kore): And now, the synchronicity. On March 29, 1848, nineteen days after the Senate vote, a man died in his house in New York City. His name was John Jacob Astor. A German-born immigrant who had made a fortune in furs and then in Manhattan real estate, he was the richest man in America, leaving an estate commonly estimated at about twenty million dollars. Narrator (Kore): In consumer prices, roughly seven hundred seventy million dollars today. Grocery prices understate a fortune like that. Twenty million dollars was a far larger share of a smaller country than a shopping basket can show. His will left four hundred thousand dollars for a library that later became part of the New York Public Library. Thirty-seven years earlier, in 1811, his Pacific Fur Company had built Fort Astoria at the mouth of the Columbia, the first American settlement on the Pacific. He had looked west and tried to own it by trade. In the spring he died, the West's own fortune was lying in a mill race, waiting. Narrator (Kore): Old Eastern money passing. New Western money about to be born. That same spring, the first Chinese residents of San Francisco were arriving, servants brought from China by an American businessman. Within two years, many more. That is the next episode.

10. Prove It

[CUE: singing-bowl chapter chime] [MUSIC: brahms4_2 from 300s, ducked] [AMBIENCE: wind.mp3, hum.mp3]

Narrator (Kore): Prove It. Narrator (Kore): Let's go back to the tall stick. Narrator (Kore): By the end of 1848, the redwood on San Francisquito Creek stands on the corner of the Soto family's rancho. Across the creek, Buelna's widow, María Concepción Valencia, holds Rancho San Francisquito. To the south, the Robles brothers hold the corner of the San Francisquito. The mission at Santa Clara is a struggling parish. Its fields, which the neophytes were promised, belong to rancheros. The Ohlone families who built that mission are scattered, working for others, or gone. The Native population of California, about three hundred thousand in 1769, is roughly half that. Narrator (Kore): Everyone who holds land here holds it by paper. A grant signed by a Mexican governor, a diseño with an oak tree for a corner, and the memory of neighbors. Narrator (Kore): And the new owners of California are about to ask a question the old system was never built to answer. Narrator (Kore): In 1851, Congress will pass a law that requires every rancho owner in California to prove, in an American court, in English, with American lawyers paid by the hour, that their land is really theirs. More than eight hundred claims will be filed. Most of them will be confirmed. Confirmation will not be the same thing as keeping the ranch. Buelna's widow will file her claim in 1853. She will not receive her final patent from the United States until 1868, fifteen years later. By then, squatters will have settled on the land, and a wealthy San Franciscan named George Gordon will have bought out both the squatters and her family. Most rancheros will win their cases. Many of them will lose their land anyway, to lawyers' fees, to debts, to squatters, to time. Narrator (Kore): And two thousand miles east, in Port Washington, a lake town in Wisconsin, a young lawyer from New York is trying to build a practice and not getting far. His brothers leave for the gold fields ahead of him. In 1852 a fire burns out his office. He follows them. In the mining country he opens a store, and the men digging for gold buy shovels and flour from him, because the shovel is a surer business than the gravel. He does not know that yet. The listener does. Narrator (Kore): [SILENCE] [HARD] His name is Leland Stanford. Narrator (Kore): Next time on Cardinal: The Toll.

Cardinal — Episode 2: The Toll (audio script)

Cardinal — Episode 2: The Toll (audio script)

Source: outline plus Ambrose research, Oct 18, 2026. Narrator = Kore. Documentary and official quotes = Charon. Judah's letters = Puck. E. B. Crocker = Orus. Hopkins = Algieba. Huntington = Algenib. Stanford's inaugural = Iapetus. Lincoln on the porch = Alnilam. Newspapers = Sadachbia. Volpe = Rasalgethi. [HARD] = music gated out. [SILENCE] = 2 s of silence before a reveal. Pronunciation notes and the fact-check log are in the review file, not narrated. Money is Minneapolis Fed CPI, said as "roughly."

01. Done

[CUE: singing-bowl chapter chime] [MUSIC: hebrides from 30s, ducked] [AMBIENCE: crowd.mp3, wind.mp3]

Narrator (Kore): May 10, 1869. A dry flat in the Utah Territory, north of the Great Salt Lake, a place called Promontory Summit. Water has to be hauled in. There is no band and no stage. Two locomotives face each other across a gap of a couple of rail lengths. One is the Central Pacific's Jupiter. The other is the Union Pacific's Number 119. Narrator (Kore): The ceremony is two days late. Leland Stanford, governor of California turned railroad president, is the only one of his four Sacramento partners who came. In his hand is a silver-headed maul. In the tie beneath him, polished California laurel, a golden spike sits in a hole already drilled so the precious metal will not be scarred. Reporters have been telling the country that the blow will be heard farther than any sound a mortal man has made. A telegraph operator named W. N. Shilling beats a warning down the wire to every office that is asking when. Quote (Charon, flat documentary): To everybody. Keep quiet. When the last spike is driven at Promontory Point, we will say done. Do not break the circuit, but watch for the signals of the blows of the hammer. Narrator (Kore): The engineers, Booth and Bradford, pull the whistles. A veteran in the crowd said later they all yelled like to bust. Stanford's speech runs less than three minutes. No one at the center of the flat says the name of a worker. A squad of Chinese men in clean blue frocks has already carried one rail into the gap, set it, and stepped back. An Irish squad has carried the other. The laurel tie is in place. The spike is seated in its hole, so the gold will not have to take a real blow. Narrator (Kore): Stanford swings. Narrator (Kore): [SILENCE] [HARD] He misses. The maul hits the rail. Narrator (Kore): [HARD] It does not matter. The operator closes the circuit anyway. The wire goes out across the continent. Quote (Charon, flat documentary): done Narrator (Kore): The officials will say that the two railroads together have laid one thousand seven hundred seventy-six miles of track. As in the year of independence. A nation that had proclaimed itself in 1776 was, ninety-three years later, announcing that it had nailed itself together, and it wanted the number to rhyme. Narrator (Kore): Hold the miss. The ceremony is a machine for making a myth, and the myth is that a spike finished a country. The rail was already graded. The holes were pre-drilled. The man with the maul missed, and the telegraph told the truth the organizers wanted, which was not quite the truth of the morning. When the dignitaries leave, other workers will pull the gold and the laurel out, put ordinary iron and ordinary wood back in, and drive the spikes that actually hold. A newspaper will notice. The photograph of the day will not quite show them. Narrator (Kore): We will come back to that flat. First we go backward twenty-one years, to a place that is not yet a state, where the money is not yet in the railroad because there is no railroad. There is gold in a mill race, a treaty on a table, and a passage so hard that the passage itself is the business. Narrator (Kore): This is Cardinal. Episode two: The Toll.

02. The Fastest State

[CUE: singing-bowl chapter chime] [MUSIC: brahms4_2 from 0s, ducked] [AMBIENCE: horses.mp3]

Narrator (Kore): The Fastest State. Narrator (Kore): September 1, 1849. Monterey. Forty-eight delegates meet in Colton Hall, a two-story schoolhouse of local stone above the old Spanish presidio, to decide what California is. The United States has owned the country, on paper, since the Treaty of Guadalupe Hidalgo the year before. It has not organized it. There is no territorial governor sent from Washington. There is a military man, a flood of gold miners, and a question. Narrator (Kore): You already heard the clock. Two years and seven months from that treaty to statehood, about twenty-five times faster than New Mexico and Arizona. A territory is the waiting room. Congress governs it. The president appoints the governor. The people do not elect senators, and they do not vote for president. A state writes a constitution, elects its own officers, and gets two seats in the Senate no matter how large or small it is. Most of the West sat in that waiting room for decades. California walked past it. Narrator (Kore): On September 5 the delegates vote on whether to ask Congress for a territorial government or to write a state constitution and demand admission. The clerk's count is twenty-eight to eight, for a state. They are not asking permission to wait. They are writing the document and daring Washington to refuse it. The constitution they sign on October 13 bans slavery. On November 13, 1849, the voters ratify it, twelve thousand sixty-one to eight hundred eleven, a landslide in a place that, a year earlier, had almost no American electorate at all. Millard Fillmore, who became president when Zachary Taylor died, signs the admission bill on September 9, 1850. Congress has spent the year bargaining. The men in Colton Hall did not. Narrator (Kore): Why California? Three forces, and they arrive together. Gold, which by the end of 1849 has dumped a population into a place that had almost no American government. The refusal, at Monterey, to wait. And the Compromise of 1850, the national bargain that let California in as a free state, with no slavery, and postponed the breakup of the Union for a decade. A free state on the Pacific was a prize both sides of that fight thought they understood. They did not yet understand what it would do to a railroad. Narrator (Kore): Remember the free state. In a few years the question of where to lay the tracks will be the same question as where slavery is allowed to grow. California's speed is not only a gold-rush story. It is the reason a transcontinental railroad becomes possible the moment the slave states walk out of Congress.

03. Panama

[CUE: singing-bowl chapter chime] [MUSIC: borodin from 0s, ducked] [AMBIENCE: surf.mp3, wind.mp3]

Narrator (Kore): Panama. Narrator (Kore): 1849. Collis Potter Huntington is twenty-eight, a hardware merchant from upstate New York, and he wants California. There is no railroad across the continent and no practical wagon road for a shopkeeper in a hurry. The sea route around Cape Horn is eighteen thousand miles. One army officer's ship, William Tecumseh Sherman's, takes two hundred two days. Stephen Ambrose, looking back at that isolation, writes that it must be doubted whether so desirable a place had

ever been so cut off. Narrator (Kore): Huntington takes the other way, the isthmus of Panama. His steerage ticket costs more than two hundred fifty dollars. On the price index we are using, the Minneapolis Fed's consumer prices, two hundred fifty dollars in 1849 is roughly ten thousand dollars today. The ship that should meet him on the Pacific side is not there. For two months he is stranded in the mud and the rain. He writes to his brother. LETTER (Algenib: Collis Huntington in 1849, young, writing from Panama to his brother, grim and plain): Death is carrying off the Americans most fearfully. There are eight to ten deaths a day. Narrator (Kore): Cholera, dysentery, malaria, yellow fever. Huntington does not die. He goes into business on the beach, selling medicines and food to men who cannot leave. He later said he made three thousand dollars at it. Roughly a hundred and twenty thousand dollars now, earned because other people were stuck. He remembered the walk this way. Quote (Charon, flat documentary): It was only twenty-four miles. I walked it. Narrator (Kore): He walked it at least twenty times, buying and selling, while the fever worked through the camps. The man who will later buy Congress, as his own letters will boast, learns his California fortune at a bottleneck. The gold is in the mountains. The price is on the path. Narrator (Kore): When a place is hard to reach, the money is not in the thing at the end of the path. It is in the passage. Shovels, flour, mule teams, a toll wagon road, and at last a railroad that the government pays you to build, and that you then own, so that everyone behind you has to pay you to move. Huntington does not know the last step yet. He knows the first. Do not dig. Sell to the people who are digging, and charge them for the crossing.

04. The Store

[CUE: singing-bowl chapter chime] [MUSIC: brahms3_3 from 0s, ducked] [AMBIENCE: horses.mp3]

Narrator (Kore): The Store. Narrator (Kore): Sacramento, in the early eighteen-fifties, is a levee town that floods, burns, and rebuilds, selling tools to men on their way to the gravel. Four shopkeepers find each other. Huntington and Mark Hopkins, the quiet partner with the ledger, are neighbors at 52 K Street, in the hardware trade. Charles Crocker is a big man, around two hundred fifty pounds, dry goods and a freighting business, who will become the foreman. Leland Stanford, the lawyer who burned out of a practice in Wisconsin, sells groceries. Shovels and flour. The miner's lottery is the customer's problem. The margin on the shovel is the store's. Narrator (Kore): June 1850, on the trail west, before Crocker is a railroad man. At the Green River crossing, another party moves its animals onto ground Crocker's men have picketed. Crocker drives the animals off. Quote (Charon, flat documentary): That ground is ours. Narrator (Kore): They threaten to pull his stakes. He pulls a pistol. Quote (Charon, flat documentary): You do, if you dare. Narrator (Kore): They back down. Ambrose records this as one of several times Crocker used the pistol, and he quotes Crocker's own rule. A man well assured of his own position, who shows a bold front, need not fear anybody. On that same journey Crocker found his horse-guards playing cards, called for the deck, burned it, and told them it would not do. They had to have their wits about them. The weapon the record gives him, on a riverbank in 1850, is the pistol. An ax handle will show up later, in another man's hand, used to discipline workers, not to hire them. Narrator

(Kore): The four merchants incorporate the Central Pacific Railroad on June 28, 1861. Their combined wealth, Ambrose says, is a hundred and fifty-nine thousand dollars. The subsidy they are about to chase pays sixteen thousand dollars a mile on flat land. Ten of those miles cost a hundred and sixty thousand. Their whole fortune, pooled, is about one flat mile short of ten miles of easy track. They are not rich enough to build a railroad. They are rich enough to hold the franchise while the United States builds it with them. A shopping-basket translation of that fortune would still understate it. In 1861 those dollars were a larger share of a smaller country than groceries can show. Narrator (Kore): Theodore Judah is the engineer who brings them the map. In the fall of 1860 a druggist in Dutch Flat, Daniel Strong, writes to him. Strong knows the old emigrant road over Donner Pass, the one abandoned after the Donner party, and he thinks a railroad can use it, because there the Sierra crests only once. Judah rides up with him. Anna Judah, later, remembered that the two men nearly got snowed in and had to break camp in the dark. Back in Strong's store, Judah stretched his paper on the counter and would not sleep until the figures were down. Then he told Strong he would survey the Donner route above every other. Narrator (Kore): The ridge is not gentle. At Cape Horn, above what was then called Illinoistown, the old wagon road hangs over a drop of about fifteen hundred feet, roughly the height of the Empire State Building. People who had stood there said a railroad was impossible. Judah said it was the line. Narrator (Kore): He tells the storekeepers, in his wife's later account, that they are tradesmen in Sacramento, that their property is there, and that if they will help him survey, he will make them the Central Pacific. With a congressional bill passed, he says, they will have the control of business that makes a fortune. If nothing more, they can own a wagon road from Dutch Flat over the mountains. They take both. Quietly they form the Dutch Flat and Donner Lake Wagon Road, a toll company on the line the railroad will follow. Crocker said later that none of them knew anything about railroad building. They were enterprising men, and they wanted the road to come to Sacramento. Narrator (Kore): The wagon road starts charging before the rails exist. In February 1866 Hopkins writes Huntington that snow stops the mountain work five or six months a year, and that getting freight over the pass means paying teamsters a dollar and a quarter a ton. Even then, he says, the men would rather stay on the old Pacific Road. More taverns. More waiting girls and barmaids. A familiar road. The partners already own a toll, and they are already learning that people pay for the path they know. Hopkins's same letter puts the construction in a builder's arithmetic. Reach the desert this season and it is a six-month job. Miss it, and it is eighteen months. On the flat, beyond the snow, a day of track is a day of government bonds. Narrator (Kore): Before any of them, there is a story the Stanford family liked to tell, and it has to be marked as a story. According to a biographical sketch in the Senate's memorial addresses after Leland's death, a Mr. Whitney spent a night at Josiah Stanford's farm, Elm Grove, on the road between Albany and Schenectady, when Leland was thirteen, and talked about a railroad to the Pacific. The boy never forgot it. Leland was born on March 9, 1824, so thirteen is about 1837. The record problem is the name. There were two Asa Whitneys. One made car wheels and worked on the Mohawk and Hudson Railroad. The other was a China merchant who came home convinced that the United States should be the road to Asia, went to China only in 1842, and began his public campaign in 1844, years after a thirteen-year-old Leland would have sat at that table. The memorial says only Mr. Whitney. Chang's history of the Chinese railroad

workers identifies the visitor as the China merchant. The dates argue against it. Treat Elm Grove as family legend, not as the night the Pacific railroad was handed to a boy. Narrator (Kore): What is not legend is the merchant's own book. In 1849 Asa Whitney, the promoter, published A Project for a Railroad to the Pacific. He wrote that the change of the route for the commerce with Asia had, since before the time of Solomon, changed the destinies of empires, and that its march had always been westward. Quote (Charon, flat documentary): Through us must be the route to Asia, and the change to our continent will be the last, the final change. Narrator (Kore): Senator Thomas Hart Benton, in the same season of prophecy, imagined Chinese workers building that West, and said they must talk together, and trade together, and marry together. Hold that sentence against the inaugural we are about to hear. The China dream and the China panic are the same decade. The storekeepers will get rich on both.

05. Three Signs

[CUE: singing-bowl chapter chime] [MUSIC: scottish3 from 0s, ducked] [AMBIENCE: wind.mp3, horses.mp3]

Narrator (Kore): Three Signs. Narrator (Kore): A sidebar, for anyone who walks these streets, and for anyone who has never seen them. Place names are arguments that won. Narrator (Kore): Union Square, in San Francisco, is a shopping square now, a column in the middle, Geary and Powell at the northeast corner. California Historical Landmark 623 records where the name came from. It originated in 1860, from public meetings held on that ground in support of the Union, in the year before the Civil War broke. Not a labor union. The federal Union, the one California's gold was helping to pay for, the one a Pacific railroad was supposed to pin to the coast. If you have stood in that square waiting for a cable car, you have stood in a war meeting. Narrator (Kore): Sanchez Street runs south through the Mission and Noe Valley. The old directory of street names, in the appendix to The Beginnings of San Francisco, says it was named for the family of José Antonio Sánchez, a soldier of Juan Bautista de Anza's company, the expedition of 1776. The airport you heard about last time sits on Rancho Buri Buri, and that grant carried a Sánchez name too. Street and ranch may be one soldier or a family of them. The sign does not sort it out. Narrator (Kore): Judah Street is the third. Same appendix. Named for Theodore D. Judah. The engineer who found the Donner route, sold it to four storekeepers, and then watched them use it in a way he came to hate. Most people who turn onto Judah are going to the beach, the N-Judah streetcar, the sunset. The name is a man who died at thirty-seven, on November 2, 1863, before the tracks met, having brought a franchise to a door and decided the men inside would beggar it. We will hear him say that. The street does not say it. Streets rarely do. Narrator (Kore): Three signs, one war. A square for the Union. A street for a Spanish soldier's family, easy to confuse with a land grant. A street for the engineer the merchants outgrew. The map looks neutral. It is a minute of the meeting.

06. Vigilance

[CUE: singing-bowl chapter chime] [MUSIC: tragic from 0s, ducked] [AMBIENCE: crowd.mp3]

Narrator (Kore): Vigilance. Narrator (Kore): May 14, 1856. The corner of Montgomery and Washington, San Francisco. James King of William, editor of the Evening Bulletin, is shot by James P. Casey, a city supervisor whose criminal record King has been printing. King dies on May 20. On May 22 the Committee of Vigilance hangs Casey, and with him a gambler named Charles Cora, from the windows of its headquarters at 41 Sacramento Street. The building has sandbags piled against the walls. People call it Fort Gunnybags. Narrator (Kore): This is the second committee. The first, in 1851, was merchants and vigilantes hanging men in a gold-rush city that did not yet have a working court. The 1856 committee is larger, more military, and more sure that it is the government. Its members turn out under arms. A fort on a commercial street. California has been a state for less than six years. Executions after a newspaper editor dies. Call it private governance. When the official state is new, slow, or captured, the people with warehouses decide they will be the police and the court. Sometimes they are punishing a murderer. Sometimes they are deciding who counts as the public. Narrator (Kore): The state already has a habit of building institutions beside the law and then asking the law to bless them. The vigilance committee is one. The railroad company, chartered by the state and subsidized by Congress, will be the lasting one. The committee hangs two men and disbands. The company stays, and it charges rent on the only modern road out.

07. The Governor

[CUE: singing-bowl chapter chime] [MUSIC: aria from 0s, ducked] [AMBIENCE: crowd.mp3, wind.mp3]

Narrator (Kore): The Governor. Narrator (Kore): September 4, 1861. Leland Stanford is elected governor of California. The Democrats are split, and a Republican who was a grocer a few years ago wins with a plurality. He takes the oath on January 10, 1862, in the middle of the Great Flood. Sacramento is a lake. The new governor comes to the oath by boat, through the streets, and the capitol waiting for him is only temporary. Narrator (Kore): He serves one term, to December 10, 1863. For that whole term he is also president of the Central Pacific, an office he will hold from 1861 to 1890. The house he works from is the governor's office and the railroad's office. The man who signs state paper for the railroad is the railroad's president. His partners will call him lazy and vain. He will insist, for the rest of his life, on being called Governor. He is good at being the face. Narrator (Kore): In the assembly chamber he reads his inaugural from a text, and he tells the state what kind of settler he wants. Gordon Chang records the words. SPEECH (Iapetus: a young Leland Stanford, governor in 1862, reading an inaugural from a prepared text): The settlement among us of an inferior race is to be discouraged. Narrator (Kore): He says that Asia, with her numberless millions, sends to our shores the dregs of her population. He calls the Chinese a degraded and distinct people, a deleterious influence upon what he calls the superior race. The chamber applauds. These remain, Chang notes, his best-known words about the Chinese. On the stump, the line was shorter. The dregs of Asia. That degraded race. The inaugural is one

occasion. The campaign is another. They agree with each other. Narrator (Kore): The same year, in the governor's house, Jane Stanford is seriously ill with a lung infection. Western doctors do not help her. The family's cook, Moy Jin Kee, brings a doctor from Sacramento's Chinese quarter, Yee Fung Cheung, who treats her with herbs. She recovers. Jane becomes so attached to Jin Kee's teenage brother, Jin Mun, that she talks of adopting him. When the boy leaves their service she gives him a gold ring engraved with his name. He wears it for seventy years. Narrator (Kore): Public speech, private kitchen. The governor who wants the Chinese discouraged owes his wife's recovery to a Chinese physician, and his company is about to decide it cannot lay the track without Chinese crews. California's gold and Nevada's silver are helping pay for the war just begun. There is loose talk that the state might follow the South out of the Union. The railroad directors, one of them the governor, will have none of secession. They want the coast held, and they want the subsidy that holding it will justify. Narrator (Kore): Washington is about to give it to them, because the men who had blocked a northern railroad are no longer in the room.

08. The Porch

[CUE: singing-bowl chapter chime] [MUSIC: mozart40_2 from 0s, ducked] [AMBIENCE: wind.mp3]

Narrator (Kore): The Porch. Narrator (Kore): August 13, 1859. A hot day in Council Bluffs, Iowa, on the east bank of the Missouri, looking at Nebraska. Abraham Lincoln is in town. He is not yet the nominee. He is a full-time politician, a year past the debates with Stephen Douglas, trying to become the Republican candidate for president. That evening he speaks at Concert Hall. The crowd is large. The local paper, the Nonpareil, has told the town to go and hear old Abe. Narrator (Kore): He did not take the easy way in. At St. Joseph, Missouri, he could have ridden the only railroad across the state and gone home to Illinois. Instead he boarded a stern-wheel steamboat and came nearly two hundred miles up the Missouri to this bluff. The morning before the speech, he walked to the top with the Puseys, the family he was staying with. Lewis and Clark had stood on that same bluff in 1804. Lincoln looked at the river and the opposite shore and told his friend Pusey that not one, but many railroads, would center here. Narrator (Kore): The next day a group of citizens gathers on the big porch of the Pacific House, a hotel, to hear him answer questions. When he finishes, the crowd thins. W. H. M. Pusey recognizes a young man still standing there. Grenville Mellen Dodge is twenty-eight, a railroad engineer. Pusey points him out and says Dodge knows more about railroads than any two men in the country. Narrator (Kore): Lincoln's head comes around. He studies Dodge for a moment, and goes over to him. He sits, crosses his long legs, swings a foot, puts his hand on the younger man's forearm, and goes straight at it. SPEECH (Alnilam: Abraham Lincoln, 1859, intent, the campaign voice dropped): Dodge, what's the best route for a Pacific railroad to the West? Narrator (Kore): Dodge answers at once. From this town, out the Platte Valley. Lincoln asks why. Dodge says the forty-second parallel is the most practical and economic route, which makes Council Bluffs the logical place to begin, because of the railroads already building west from Chicago to this point. Lincoln keeps asking. Years later Dodge will describe the conversation in a

soldier's phrase. Quote (Charon, flat documentary): He shelled my woods completely and got all the information I'd collected. Narrator (Kore): There is a money angle under the geography, and it is small, and it is real. Lincoln is about to accept seventeen lots in Council Bluffs as collateral on a loan of three thousand five hundred dollars to a fellow lawyer, Norman Judd. The lots are in the Riddle Tract, on the floodplain, the same tract where Dodge is buying. Three thousand five hundred dollars in 1859 is roughly a hundred and thirty thousand dollars now. The future president is also looking at his collateral. He is, in that year, a politician and a railroad lawyer running for president. Narrator (Kore): The railroad was already his subject. At twenty-three, in his first campaign, for the Illinois legislature, Lincoln told voters that no other improvement could equal the utility of the railroad. He finished eighth of thirteen. He lost, and he kept the subject. In a tax case for the Illinois Central he billed two thousand dollars. The railroad answered that this was as much as Daniel Webster himself would have charged. Lincoln re-billed five thousand, sued, and won. Ambrose calls him, on a scholar's verdict, one of the foremost railroad lawyers in the West. In 1857 he defended the first railroad bridge across the Mississippi, the Rock Island bridge, after a steamboat called the Effie Afton hit it and burned. Lincoln went to the river and measured the currents himself. He argued for travel from East to West, whose demands, he said, are not less important than those of the river. The jury hung. A hung jury was a win for the bridge. The boats did not get to own the crossing. Richard Rayner, in a later life of the associates, quotes him in that case on the manifest destiny of the people to move westward. The destiny was a legal argument about a bridge. Narrator (Kore): The Iowa railroad men do not forget the porch. At the 1860 convention they help nominate him, and they get John Kasson to write the railroad plank of the Republican platform. It says that a railroad to the Pacific Ocean is imperatively demanded by the interests of the whole country, and that the federal government ought to render immediate and efficient aid in its construction. On the train east to his inauguration, Lincoln takes a turn at driving the locomotive. Dodge, in the audience for the address, writes to his wife that old Abe has delivered the greatest speech of the age. It is backbone all over. Then Dodge adds the practical line. It looks as though we can get all our measures through. Narrator (Kore): The route he gave Lincoln on that porch was never only a survey. The question sat next to slavery in Lincoln's mind, and it was part of the question of slavery's future in the Republic. A northern railroad and a southern railroad were two different countries.

09. The Elephant

[CUE: singing-bowl chapter chime] [MUSIC: brahms1_1 from 0s, ducked] [AMBIENCE: telegraph.mp3]

Narrator (Kore): The Elephant. Narrator (Kore): Through the eighteen-fifties the line could not be chosen, because choosing it chose slavery's direction. Jefferson Davis, secretary of war, sent out the Pacific railroad surveys of 1853 and wanted the thirty-second parallel, from New Orleans to San Diego. No free-state politician was ready to charter or fund a road that would help slavery expand. Free-soil men wanted Chicago, or St. Louis, or Minneapolis. No slave-state politician was willing to give the road to them. The United States paid ten million dollars in the Gadsden Purchase for the strip that

made Davis's southern route possible on American soil, and the Army experimented with camels for the desert. The camels are a detail. The blockade is the fact. Narrator (Kore): Judah put it in a pamphlet in 1857, reaching for the Bible a year before Lincoln's House Divided speech made the verse famous. The Pacific railroad, he wrote, is a house divided against itself. It cannot be done until the route is defined, and if the route is defined, the opposing interest is powerful enough to defeat it. In January 1859 he wrote that President Buchanan favored the extreme southern route and would veto a bill for any other. In 1860 he reported that the debate on slavery, and on other matters, had left Congress no time to consider the Pacific railroad bill. Narrator (Kore): Then the Deep South leaves the Union, in the spring of 1861, and its members leave Congress. The votes that had blocked every northern route are no longer in the chamber. Chang's history says it in one turn. A divided Congress could not agree. Secession let Washington authorize a northern route the South had opposed. The historian Robert Russell adds the cold institutional point. Because the Southerners were gone, Congress could pass a Pacific railway law several years earlier than it otherwise would have. Dodge's account of Lincoln's motive is military and political at once. The road had to be built not only as a military necessity, but as a means of holding the Pacific Coast to the Union. California's gold was paying for the war. The railroad was how you kept California. Narrator (Kore): Judah goes back to Washington and, at his wife Anna's suggestion, turns a room in the Capitol into what he calls the Pacific Railroad Museum. Maps, surveys, and Anna's paintings of the Sierra on the walls. Members come to look. Many of them ask the obvious question. How do you get across those mountains? Aaron Sargent, a California congressman and Judah's ally, speaks for the bill on January 21, 1862. He calls it a military necessity. He says trade with Japan and China will jump, that California's loyalty will be assured, that no foreign army will dare attack the coast. Judah's complaint is that Sargent spoke before an empty hall. Judah himself is everywhere the bill is. He serves as secretary of the Senate's Pacific railroad committee and as clerk of the House subcommittee, with the privilege of the floor in both houses. The engineer is also the staff. Narrator (Kore): Thaddeus Stevens of Pennsylvania, who owns a foundry, writes in a rule that the iron be American-made. He says the Southerners will be back someday, with the same arrogant, insolent dictation which the North has cringed to for twenty years, forbidding any road that does not run along the southern border. Justin Morrill of Vermont grumbles that the nation can hardly afford both guns and railroads. The House passes the bill on May 6, 1862, seventy-nine to forty-nine. The Senate passes it on June 20, thirty-five to five. The House agrees to the Senate's changes. Narrator (Kore): Lincoln signs it on July 1, 1862, the same day as Malvern Hill, the last battle of the Seven Days on the Virginia Peninsula. Ambrose's account gives McClellan's army about a thousand dead and three thousand wounded in that fight, and a country depressed by it. In Washington, a pen. In Virginia, a battle the Union wins and does not feel like a victory. The law that results is, in Ambrose's phrase, complicated to an almost incomprehensible degree. The shape of it is simpler. Narrator (Kore): Two corporations build toward each other. The Union Pacific west from the Missouri. The Central Pacific east from Sacramento. Each gets a right of way two hundred feet on either side of the track. Each gets land, in a checkerboard. A section is one square mile, six hundred forty acres. The grant is five alternate sections per mile on each side of the line. Alternate means every other square. Ten sections, sixty-four

hundred acres, for each mile of track. The squares the government keeps, in between, become valuable because the railroad is there. That argument, about who gets the rise in the land, will have a name later. Not this hour. Narrator (Kore): And the government loans its credit, in bonds. Sixteen thousand dollars a mile on flat ground. Thirty-two thousand in the foothills. Forty-eight thousand in the mountains. A thirty-year loan at six percent. A loan, not a gift. The bonds are a debt the companies are supposed to repay. In practice the debt will outlive the builders and land on their heirs, and the land grant is the part that behaves like a gift. A mountain mile pays triple a flat mile. In today's prices that mountain mile is roughly a million six hundred thousand dollars. The Central Pacific has to build fifty miles within two years, and finish its line by July 1, 1876, a few days before the nation's hundredth birthday. It will beat that deadline by seven years. So it matters, enormously, where someone says the mountains begin. Narrator (Kore): From 1861 to 1864, prices in the East jump, because the Union is printing paper money, the greenbacks. California's economy still runs on gold coin. Sacramento's dollar and Washington's dollar are not the same dollar in those years. Narrator (Kore): Judah wires his partners in Sacramento. Quote (Charon, flat documentary): [HARD] We have drawn the elephant. Now let us see if we can harness him. Narrator (Kore): An elephant, in the American phrase of that century, is a prize you have won and may not be able to feed. They have the franchise. The Sierra is still there. Narrator (Kore): The next day, July 2, 1862, Lincoln signs a second land law. Justin Morrill's act, the one he wanted even while he doubted the country could afford guns and railroads both. It grants each loyal state thirty thousand acres of federal land for every senator and representative it has, to endow at least one college of agriculture and the mechanic arts. The count follows the census of 1860. New York had thirty-one representatives and two senators, thirty-three seats. Thirty-three times thirty thousand is nine hundred ninety thousand acres. Stanford University's campus today is eight thousand three hundred fifty-three acres. New York's land-grant pile was on the order of a hundred and twenty of those campuses. Cornell University, chartered in 1865, is the school New York built with that idea. Stanford is not a land-grant college. It will be founded later, privately, by a railroad man, on a fortune this week's other law helps create. Same week. Same president. Public land turned into a college by a statute, and public land turned into a railroad subsidy. Do not mix them up. The confusion flatters the railroad. Narrator (Kore): In the spring of 1863 the war and the railroad sit in the same room again. General Grant orders Grenville Dodge to Washington. Dodge is alarmed. He thinks he is about to be called to account. Lincoln wants to talk about the Missouri River, about towns fifty miles up and down the bank that all want to be the place the Union Pacific starts, even as Robert E. Lee is preparing to march into Pennsylvania. Lincoln remembers the porch in 1859. Dodge's verdict is short. He found Mr. Lincoln well posted. Narrator (Kore): A year later the companies come back for more. Lincoln signs the amendment on July 2, 1864. The land grant doubles, to ten alternate sections on each side, about twelve thousand eight hundred acres a mile. Set that against the campus again. One mile of finished track can now earn land equal to about one and a half Stanford campuses. The companies may issue their own first-mortgage bonds, equal to the government bonds, which pushes the United States into second place in line if the railroad fails. The government will advance bonds after twenty miles of track instead of forty. Representative Elihu Washburne described the galleries, when it passed, as a tempest of the wildest disorder, lobbyists

male and female. The Union Pacific's man, Thomas Durant, had a lobbyist, Joseph P. Stewart, spreading two hundred fifty thousand dollars in Union Pacific bonds, twenty thousand of it to Charles T. Sherman, the general's brother.

10. Moving the Mountains

[CUE: singing-bowl chapter chime] [MUSIC: eroica2 from 240s, ducked] [AMBIENCE: wind.mp3, horses.mp3]

Narrator (Kore): Moving the Mountains. Narrator (Kore): The triple rate for mountain miles is an invitation, and Charles Crocker accepts it. He drives the state geologist, Josiah Whitney, out to Arcade Creek, about seven miles east of Sacramento. This is not Asa Whitney the promoter, and not the car-wheel Whitney. This Whitney maps mountains for a living. The ground at the creek is flat. What Crocker shows him is a fan of reddish earth washed down from the hills. Whitney agrees that the Sierra Nevada, for the purposes of the subsidy, begins here. On September 29, 1862, Governor Stanford telegraphs Washington about evidence to fix the western base of the Sierra Nevada. Aaron Sargent, no longer in Congress but still in the capital, carries the geologist's report to Lincoln. Lincoln says it seems about right. Narrator (Kore): The mountains move about fifteen miles west, on paper. Foothill miles pay thirty-two thousand dollars. Mountain miles pay forty-eight thousand. The difference is sixteen thousand a mile. Sixteen thousand times fifteen miles is two hundred forty thousand dollars in extra bonds. In 1863 prices, roughly six and a half million dollars now. The United States has been talked into paying mountain rates for a fan of red dirt on a valley floor. Narrator (Kore): Sargent told it as a joke at Lincoln's expense, and at his own. Quote (Charon, flat documentary): [SILENCE] Here you see, my pertinacity and Abraham's faith moved mountains. Narrator (Kore): Another telling puts the joke in Lincoln's mouth. Here is a case, that version says, in which Abraham's faith has moved mountains. Sargent is the man we can quote. Lincoln is the man who agreed to the map. Faith, in the joke, means a signature. The mountains did not move. The category did. Narrator (Kore): Economists call this rent-seeking: money earned by bending a rule, rather than by making something people need. The track still has to be laid. The extra two hundred forty thousand dollars is not payment for a steeper grade. It is payment for a definition. Judah refuses to sign the affidavit. He tells Strong he cannot, because the foothills do not begin here. He writes to Anna that he has brought them a franchise and laid it at their door. LETTER (Puck: Theodore Judah, an idealistic engineer in his thirties, writing bitterly to his wife Anna): Rightly used it gives them unlimited credit throughout the world, and they would beggar it. Narrator (Kore): He also tells her he cannot make these men appreciate the elephant on their shoulders. They will not do what he says they must. There will be trouble in Congress if they go on this way. He cannot find the money to stop them. He is out. On November 2, 1863, Theodore Judah dies. He is thirty-seven. Narrator (Kore): A buggy, a creek, a geologist, a president who agrees, and a subsidy that follows the words instead of the slope.

11. The Patent and the Furnace

[CUE: singing-bowl chapter chime] [MUSIC: schubert2 from 0s, ducked] [AMBIENCE: church.mp3, telegraph.mp3]

Narrator (Kore): The Patent and the Furnace. Narrator (Kore): March 18, 1865. The war is not over. Lee has not surrendered. In Washington, Abraham Lincoln signs patents that convey Mission San Juan Capistrano and Mission San Luis Rey to Bishop Joseph Sadoc Alemany of California. The patents confirm a decree the Land Commission issued in 1855. Alemany had filed the church's claim in 1853. A later Supreme Court case, Foster against Mora, measures the San Juan Capistrano parcel, the buildings and the gardens, at forty-six and seventy-four hundredths of an acre. Narrator (Kore): Lincoln did not return all the missions. He signed these two, on this date. Other mission parcels move on other days. San Diego's patent is May 23, 1862. San Gabriel's is November 19, 1859. La Purísima waits until 1874. A press release that says one document gave them all back is wrong. What Lincoln did was narrower and more interesting. Narrator (Kore): Last episode, a Mexican governor sold the buildings at San Juan Capistrano for seven hundred ten dollars. The American patent is how the church gets its parcel back, through the same federal title machine that is confirming rancho grants, and that, in this same decade, is giving railroads millions of acres. One office, three kinds of land. A mission church. A Mexican ranch. A checkerboard along a track. If you can get your paper in front of the United States and survive the lawyers, the United States will draw the line. The railroad is simply the applicant that arrives with a statute written for it. Narrator (Kore): And the men who hold that statute decide they should also hold the contract to build. The Big Four award the construction to Charles Crocker and Company, a firm they own. Ambrose calls it an almost identical device to the Crédit Mobilier. In the East, Thomas Durant and a promoter named George Francis Train buy a sleepy Pennsylvania corporation, the Pennsylvania Fiscal Agency, and rename it the Crédit Mobilier of America. The Union Pacific pays that company, which the Union Pacific's own directors own, to build the Union Pacific. The profit is in the building, whether or not a ticket is ever sold. Among the insiders is Congressman Oakes Ames of Massachusetts. His family made the shovels that Huntington, Hopkins, and Crocker had sold to miners. The store and the subsidy have the same supplier. The investigation that will make Crédit Mobilier a scandal is still years away. Narrator (Kore): The books go later, while Hopkins is still alive. In February 1873, Representative Jeremiah Wilson's committee puts Huntington on the stand and asks for the construction accounts. Rayner writes that Hopkins carried fifteen leather volumes down to the basement furnace. Huntington came back and told the committee the records were lost. LETTER (Algenib: Collis Huntington, dry, before a congressional committee): My partner Hopkins is a peculiar man. He considered the paper no longer worth saving. Narrator (Kore): Hopkins dies in March 1878, five years after that fire. For now the tracks are still climbing, and the men on them are about to stop.

12. The Summit

[CUE: singing-bowl chapter chime] [MUSIC: brahms4_2 from 300s, ducked] [AMBIENCE: wind.mp3]

Narrator (Kore): The Summit. Narrator (Kore): The winter of 1866 into 1867. Snow so deep, and so bright, that men go snow-blind. E. B. Crocker, Charles's brother and the company's lawyer, writes to Huntington that they have had to buy up all the cheap goggles in the market. Think of that, he says. What an item in railroad building. Narrator (Kore): Inside the summit tunnel the work is a few inches at a time. Crews of three stand at the granite face, one man holding the drill, two swinging sledges of fourteen to eighteen pounds. They drill, they pack the hole with black powder, they run, they wait for the blast, and they walk back into the smoke and do it again. Ambrose's account of that facing is six to twelve inches in a day, with shifts going around the clock. A month of that work moves the mountain less than the length of a room. Narrator (Kore): On Christmas Day, 1866, the Dutch Flat Enquirer reports that a gang of Chinese workers was covered by a snow slide and that four or five died before they could be dug out. Near the summit tunnel, an avalanche carried away some twenty Chinese, whose bodies were found after the spring thaw. How many died on the line, the company did not publish. The historian Thomas Chinn's judgment is that the loss of life was heavy. In June 1870 a train will pass through Sacramento carrying, by one newspaper report, the bones of perhaps twelve hundred Chinese workers, boxed to be sent home. Narrator (Kore): James Harvey Strobridge is the construction boss. Irish, over six feet, thirty-seven when he takes the work. At Bloomer Cut a black-powder blast costs him his right eye. The Chinese who learn English call him the one-eyed bossy man. Foremen clap a hand over an eye to make a crew think he is coming. He says that men generally earn their money when they work for him. When there is trouble among the Chinese crews, Ambrose writes, Strobridge can pick out the leaders with a glance. He confronts them, usually with an ax handle, and they give way. The handle is Strobridge's. Crocker is the man who hired him. Crocker, for his part, later claimed he quarreled with Strobridge about the way he spoke to the men. They are human creatures, Crocker said. Strobridge answered that you have got to do it, and that they are about as near brutes as they can get. Crocker said he found out that it was true. Put that next to the pistol at Green River. The bold front, turned on his own workforce. Narrator (Kore): Strobridge had not wanted them hired. His line, in Ambrose, is plain. I will not boss Chinese. He said they could not do the work. Then he did boss them, because the white crews were leaving for the mines and the mountain was not going to wait. The story goes that Crocker answered him with the Great Wall of China. The payroll is plainer, and earlier. Chinese crews under contractors including Hung Wah and Ah Toy are already on the Central Pacific's books in January and February of 1864. The usual telling puts the decision a year later. Hung Wah is a labor contractor. At the end, Strobridge will seat his Chinese foreman, most likely Hung Wah, at the head of a dinner table, and a paper will run the scene under the heading Chinese Laborers at Table. The honor is private. The pay packets are the public fact. Strobridge's wife, Hanna Maria, lives in a passenger car fitted out as a house at the end of track, with an awning and a caged canary by the door. The men call her the heroine of the Central Pacific. Ambrose calls her the only white woman who saw the work through from beginning to end. The canary is real. It does not soften the ax handle. Narrator (Kore): By Chang's account the company needs a workforce it can barely imagine, and it prints thousands of copies of a woodcut flyer to circulate in California and in China. No copy survives. Crocker admits to Huntington that no one on the company side has translated the flyer back into English. The men who believe they are

buying labor cannot read their own advertisement. Chinese workers become about ninety percent of the Central Pacific's labor. Chang calls them, for those years, by far the largest single workforce in American industry to that date. Narrator (Kore): June 24, 1867. A Monday, near the solstice, the middle of the building season. Three thousand Chinese workers put down their tools at once, from Cisco to Truckee, nearly thirty miles of mountain. The demand, as the company understands it, is parity with white workers. Forty dollars a month instead of thirty-five, and a day of ten hours instead of eleven. Five dollars, and an hour. The strike is not a demand to get rich. It is a demand to be paid like the other men on the same rock. A newspaper called it the greatest strike the country had known. Narrator (Kore): Mark Hopkins writes to Huntington with the fear of a bookkeeper. If they succeed in this demand, then they control. E. B. Crocker writes the most honest line. The truth is they are getting smart. His remedy is to inundate California and Nevada with laborers. Freedmen, Chinese, Japanese, all kinds. Charles Crocker goes up the mountain. The strike leaders tell him, in words he later reports as theirs, eight hours a day is good for a white man, all the same good for a Chinaman. In that telling they ask for eight hours, not the ten the company wire had named. The number shifts with the teller. The pay does not. He answers in a pidgin of his own making. Hear it as his rendering, not as their language. Quote (Charon, flat documentary): [HARD] John Chinaman no make laws for me. I make laws for Chinaman. You sell for thirty-five dollars a month, me buy. You sell for forty dollars and eight hours a day, me no buy. Narrator (Kore): Then the food stops. Crocker will claim he cut it off. The labor agents stop supplying provisions. E. B. Crocker writes that none of them went near the strikers for a week. They did not want to exhibit anxiety. After eight days most of the men go back to work. Histories that follow Charles Crocker's version call it a defeat. Chang reads the partners' own letters and sees men who were shaken, and he notes that pay for skilled Chinese workers rises over the following months. What the principals want next is not a negotiation. It is a flood. LETTER (Orus: E. B. Crocker, matter-of-fact, in a business letter): [HARD] We want a hundred thousand chinamen here, so as to bring the price of labor down. Narrator (Kore): Huntington goes further. It would be all the better for us and the state, he writes, if half a million came over in 1868. The governor who had called them the dregs now loves the supply. The economist's word for a market with one big buyer is monopsony. The Central Pacific is that buyer on the mountain. A strike is the workers' attempt to be a seller with a price. Cutting off the food is a lockout. Wanting a hundred thousand more men is the plan to make the price fall. On August 3, 1867, Chinese crews hole through the summit tunnel and let the air in.

13. Ten Miles

[CUE: singing-bowl chapter chime] [MUSIC: brahms1_1 from 420s, ducked] [AMBIENCE: crowd.mp3, wind.mp3]

Narrator (Kore): Ten Miles. Narrator (Kore): April 28, 1869. Seven in the morning, a whistle in the Utah desert. The Central Pacific's crews set out to lay more track in a day than anyone has laid. Crocker has bet Durant ten thousand dollars that they can do ten miles. Ambrose says Durant, reportedly, was sure they could not, and took the bet. Ten

thousand dollars in 1869 is roughly a quarter of a million dollars now. A day's spectacle, at the price of a small fortune. Narrator (Kore): By seven that evening they have laid ten miles and fifty-six feet, up the slope toward Promontory, bending rails around curves by hand. Almost a mile an hour, for twelve hours. The celebratory accounts name eight Irish rail handlers. They do not name a single Chinese worker. Several thousand men did the day. The story kept eight names. A few days later some of those men, perhaps including Hung Wah, climb a bluff near Monument Rock with a brown-glazed jug of rice wine that has come from the Pearl River delta. They can see the line, and beyond it the Great Salt Lake. They drink, leave the jug, and go back to camp. Archaeologists find it on the outcrop a century and a half later. Chang writes that the bottle, though mute, had something to say. It is the celebration that was not on the program. Narrator (Kore): You have already heard May 10. Stanford misses, and the wire says done, and the mileage is read out as seventeen seventy-six. Here is who the picture loses. Andrew J. Russell's photograph, East and West Shaking Hands, two engines nose to nose, becomes the image of the railroad. Chang is careful. There is no evidence that Russell arranged the crowd to keep Chinese workers out of it. Look closer, and a ghostly figure with his back to the camera, blurred because he was moving, stands in the middle of the frame. From his clothes, his hat, and his stature, he is Chinese. He is in the most famous picture of the road he helped build, and he is a smear. Narrator (Kore): The ceremony was late for a reason that should not be smoothed over. Union Pacific workers, owed wages, detached Thomas Durant's private car, set a guard on it, and held him until he wired for the money and paid them. The vice president of the road, a hostage to his own unpaid crews. The builders, again, were paid last, when they were paid. Narrator (Kore): In 1969, at the centennial, a cabinet secretary named John Volpe asks the crowd, SPEECH (Rasalgethi: John Volpe, a 1960s cabinet secretary at the centennial, proud, public-address cadence): Who else but Americans could drill tunnels in mountains thirty feet deep in snow? Narrator (Kore): The question answers itself by leaving people out. The golden spike survives. It sits, today, in the art museum at Stanford. The laurel tie burns in the San Francisco fire of 1906. Narrator (Kore): Doctors were already selling California's climate before the rails met. In 1852 James Blake wrote that there was not a day in the year when the mind or the body was weakened by heat or numbed by cold. J. Praslow, a physician in San Francisco from 1849, wrote that the climate was so healthy that illness was rarely found. The crowds come later. From 1840 to 1900, one estimate puts illness behind about a quarter of the new arrivals, and the rush itself runs from 1870, after the trains, when a state report will call California the sanatorium of the world. The railroad will carry a different passenger soon enough. The coughing rich. A boy. A sick young man whose son will matter to a valley of companies. This hour ends on freight. Narrator (Kore): [SILENCE] [HARD] The road is finished. Now they own it. Everyone who wants to move a bushel of wheat has to pay. The toll is the point of the passage. It always was, from a beach in Panama, to a shovel in Sacramento, to a definition of where a mountain begins, to a spike that missed and a telegram that did not. Narrator (Kore): Next time on Cardinal: The Octopus.

Cardinal Podcast Ep. 3: The Octopus (audio script)

Cardinal Podcast Ep. 3: The Octopus (audio script)

Episode 3 of 8. The Octopus (1869–1884). Markup matches the pipeline in "The Spike in the Museum": Narrator = Kore. Documents and flat quotations = Charon. Newspapers = Sadachbia. Letters and speeches use the voice names below. Bracketed cue lines are not narrated. Narration target 8,500–9,000 words.

01. The Wheat Field

[CUE: singing-bowl chapter chime] [MUSIC: tragic from 0s, ducked] [AMBIENCE: wind.mp3, horses.mp3] [MOTIF: railroad stinger]

Narrator (Kore): The Octopus. Narrator (Kore): This is Cardinal. Episode three. Narrator (Kore): May eleventh, 1880. A Tuesday morning in a wheat field, two and a half miles north of a new town called Hanford, in the San Joaquin Valley of California. The valley is the long flat trough that runs down the middle of the state, between the coast mountains and the Sierra Nevada. In spring the wheat stands high enough to hide a man on a horse. Narrator (Kore): In Hanford that morning, families had come in for a picnic. The children, at least, expected to have fun. The adults had heard who got off the train a few hours after midnight. His name was Alonzo W. Poole. He was a United States marshal, and he was carrying eviction notices. Narrator (Kore): Poole rode out of town in a wagon with a Southern Pacific land man beside him. Two neighbors who were buying the disputed ground from the railroad met them in a buggy. Their names were Mills Hartt and Walter Crow. In the buggy, Richard Rayner writes, they had two shotguns, a Spencer rifle, and three pistols. Hartt had been bragging in town about what he would do to settlers who would not move. Crow was known as one of the best shots in California. He had knocked wild geese out of the air with that rifle. Narrator (Kore): They stopped at a homestead where nobody was home, put the family's belongings in the road, and lined shotgun shells up on the porch. Then they went on to a place held by Henry Brewer and John Storer, part of which the railroad had sold to Walter Crow. It was about half past ten. Narrator (Kore): At the picnic, word arrived. Riders came out through the wheat and the alfalfa. Poole later said he saw forty or fifty. Other witnesses said about fifteen. Testimony cannot settle the number of men on horseback. A legend will. Narrator (Kore): Poole knew the neighborhood, and the neighborhood liked him. A settler named James Patterson asked him to wait until the court appeals were heard. Poole said he was doing his job. Someone told him to give up his pistol. He would not. Then a nervous horse struck him and knocked him down. He was the most reliable

witness, and he was in the dust when the shooting started. Nobody agrees who fired first. In less than a minute, twenty or thirty shots. Walter Crow did most of the killing. James Harris. Iver Knutson, forty-nine, a father of nine, a man Crow knew. Daniel Kelly. Archibald McGregor, armed with a penknife. John Henderson. Hartt was hit too, and took more than a day to die. Narrator (Kore): There was a pause. Smoke over the wheat. Crow backed the others off with a pistol and said, one at a time. The marshal told him to take a horse and go to the timber. Crow would not. He walked into the wheat he had bought from the railroad, the field he had told his friends he would harvest that year. A mile and a half on, armed settlers met him. He raised his pistol. He was shot in the back, with his own rifle, according to one report, and he fell dead in a ditch. NEWS (Sadachbia: the Visalia Times, May 15, 1880, read like a period town crier, flat): "As has long been anticipated, the conflict." Narrator (Kore): That is how the Visalia Times opened its account four days later. The paper named seven dead: five settlers, and two men who had come to take the land. An eighth man, Edwin Haymaker, had only a scalp wound that morning. He cleaned it at the well. He developed pneumonia, and he was dead within the month. Richard Rayner, who wrote the history of the railroad partners, counts eight if you count Haymaker. The next legend of the American West, the gunfight at the O.K. Corral in Tombstone, a year and a half after this, killed three. The movies kept Tombstone. The wheat field is the one that tells you how California actually worked. Narrator (Kore): Here is the fight under the fight. This was not empty land, and it was not a picnic that went wrong for no reason. In 1877 the Southern Pacific finally received its federal patents to land in this part of the valley, along a branch line toward the Mussel Slough of the Kings River. Settlers were already on it. Many of them had brought water to ground the county histories call barren, and had made a crop. Railroad agents then announced they would sell at prices from seventeen dollars an acre to forty. Narrator (Kore): Hold the price up to the company's own advertisement. The company itself had advertised the land at two dollars and fifty cents an acre and, in the company's own weasel word, upwards. Seventeen dollars is almost seven times that advertised floor. Forty dollars is sixteen times it. The difference is not the dirt. The difference is who gets paid for the work. The Pacific Railway Act of 1862, the law that gave the railroad its land in the first place, had named a different ceiling: grant land still unsold three years after the road was finished was to be open to settlers at not more than one dollar and twenty-five cents an acre, paid to the company. Narrator (Kore): A San Francisco newspaperman had just written the most famous economics book of the century to explain the rise in the land. Henry George published Progress and Poverty in 1879. His argument was that a railroad, a ditch, a town, all of that improvement, raises the value of the land around it, and the rise flows to whoever holds the title, not to the people who did the digging. He did not need a wheat field to draw the picture. He already had the railroad. Quote (Charon, flat documentary): A railroad company approaches a small town as a highwayman approaches his victim. Narrator (Kore): George wrote. Quote (Charon, flat documentary): The threat, "If you do not accede to our terms we will leave your town two or three miles to one side!" is as efficacious as the "Stand and deliver," when backed by a cocked pistol. Narrator (Kore): Build through the town, or build past it and let it die. After the road is built, charge what the town cannot refuse. The settlers at Mussel Slough had improved the ground. The railroad held the patent. The patent won. In the end the courts sided with the company. Some families

left. Some paid the higher price. The Southern Pacific kept the land. Narrator (Kore): The story goes that this was honest farmers against a monster. The record is meaner. Some of the settlers had fought for the Confederacy. Their league drilled like a militia. Masked men burned a homestead, and each side blamed the other. Crow knew the men he killed. Rayner thinks nobody rode out with a settled plan to start a slaughter, and that drawn guns made one likely. Both things are true. The railroad wrote the price. The men in the field pulled the triggers. Narrator (Kore): As has long been anticipated, the conflict. Violence had become weather. A toll road had finished its tracks and started acting like a government. It set prices. It bought newspapers and lawmakers. When the public got angry, it found a scapegoat who could not vote. And when a court finally wrote the story down, a caption did more work than the ruling. Come up the hill and meet the men who send the bill.

02. Nob Hill

[CUE: singing-bowl chapter chime] [MUSIC: borodin from 0s, ducked] [AMBIENCE: horses.mp3, crowd.mp3] [MOTIF: railroad stinger]

Narrator (Kore): Nob Hill. Narrator (Kore): Jump back four years, to 1876, and up a steep street in San Francisco. The hill did not yet have its fame. It had a view, wind, and sand. The men who had just finished a railroad decided the view was the point. Narrator (Kore): Leland Stanford built first, at California and Powell. The house was finished in 1876. Richard Rayner describes a mansion of more than fifty rooms, on about two acres, inside a wall thirty feet high. The front doors were mahogany and rosewood, seven inches thick. A glass dome over the hall sat seventy feet up. The carved framework of the conservatory cost thirty thousand dollars. A custom library ladder, three rungs, cost a hundred and fifty. Rayner puts the whole production at two million dollars in the money of the eighteen seventies. Narrator (Kore): Two million dollars. In grocery dollars, at 1876 prices, using the Minneapolis Fed's price index and its estimate for 2026, two million then is roughly sixty-three million now. Narrator (Kore): In 1899 an economist named Thorstein Veblen gave this a name: conspicuous consumption, spending whose job is to be seen. The mansions came first. The theory arrived to describe them. Stanford's house was not a machine for living. It was a machine for being looked at, from the bay and from the offices of men who paid the railroad's rates. Narrator (Kore): Getting up the hill was its own advertisement. Cable cars already climbed Clay Street. That line, Andrew Hallidie's, had started in 1873, and it stopped two blocks short of Stanford's door. So Stanford and his railroad friends built another one, up California Street itself, past the new palaces. It opened on April tenth, 1878. It is still running. If you have stood on a San Francisco cable car and felt the floor bite as the grip takes the cable, you are on a vehicle a railroad president built so the public could reach his front steps, and so he would not have to walk the last two blocks. Narrator (Kore): Mark Hopkins let his wife talk him into a house a block up the hill, at California and Mason. He was the quiet partner, the ledger. Rayner says that even after he was a multimillionaire he kept a small cottage, kept no carriage, and took no vacations. The palace went up anyway. He never lived in it. We will come to the night he died. Narrator (Kore): Charles Crocker wanted a whole block. California, Sacramento, Taylor, and

Jones. One lot on it belonged to a man named Nicholas Yung, an undertaker, comfortable but not rich, who liked the view and would not sell at Crocker's price. The offers climbed. Six thousand, then more, and at some point Yung is said to have named twelve thousand. Crocker stopped bargaining and built a fence. Narrator (Kore): The story goes that the fence was thirty feet high, and that Henry George called it a veritable coffin. The record is plainer and a little different. Richard Rayner, citing the press, says a wooden fence forty feet high, three sides of Yung's house, so the sun barely got in. A later history of the hill says the same, forty feet. Some retellings shrink it to thirty. Call it four stories of boards, on three sides of a family's windows. The San Francisco Examiner was writing about it by August of 1878, under the name the city gave it: a spite fence. Narrator (Kore): The cost, in the newspaper accounts that survive in later write-ups, was about three thousand dollars. In 1878 grocery money that is roughly a hundred thousand dollars. Crocker could lose that sum the way another man loses a cufflink. What he bought was a lesson. The price of refusing the railroad was darkness at noon. Narrator (Kore): The story goes that Yung answered by putting a coffin on his roof, skull and crossbones turned toward Crocker's windows. What a Chronicle report preserves is the threat: a coffin above the fence, as an advertisement for the undertaking business and a reminder of mortality. Threat or object, the picture stuck. The man who buried the dead, boxed in by the railroad. Narrator (Kore): Yung moved the house to another lot he owned, on Broderick Street. He would not sell the empty ground. He died in 1880. The fence outlived him. The heirs kept the feud for years. The wind worried the boards. Eventually the lot was sold, and the fence came down, and Crocker, by then dead himself, got the block in the only way that counts to an estate: on paper. Narrator (Kore): Stand on that pavement now and the houses are gone. The fire after the earthquake of 1906 took the hill in one night. What replaced them is the city's memory, pretending to be an address. The Stanford Court Hotel sits at 905 California, on the footprint of Stanford's mansion. The Mark Hopkins Hotel sits on Hopkins's corner, and the bar at the top looks down on the same bay the dome looked down on. Huntington Park is the lot of the house that had been David Colton's, where Collis Huntington later lived. His widow gave the ground to the city for a park in 1915. Grace Cathedral rises on ground the Crocker family turned over to the Episcopal church after the fire. The palaces lasted thirty years. The names are still on the hotel awnings.

03. The Story of Our Streets

[CUE: singing-bowl chapter chime] [MUSIC: brahms3_3 from 0s, ducked] [AMBIENCE: wind.mp3]

Narrator (Kore): The Story of Our Streets. Narrator (Kore): This series keeps a short ledger of names, because California wrote its history on signposts and then forgot the ink. Nob Hill is one of those names. Narrator (Kore): Nob is not a saint. The Spanish named towns for Francis, Clare, Joseph, and the rest, and those names are still the mission chain. Nob is short for nabob. And nabob is an English mangling of a word from India, nawab, a deputy of the Mughal emperor, a man who governed in someone else's name and got rich doing it. The British brought the insult home for merchants who made a fortune in the East and spent it loudly in London. San Francisco picked it up for

a different east: Sacramento grocers and Nevada silver men who hauled their money up one steep ridge and built houses the rest of the city had to crane its neck to see. Narrator (Kore): So the hill is a joke that froze. Catholic California named places for heaven. Railroad California named a hill for showing off. You can still ride the California Street cable car, the line Stanford opened in 1878, step off at a hotel named for his bookkeeper, and look across a park named for the partner who stayed in New York. Then go west through Golden Gate Park to a museum that still carries the name de Young. Narrator (Kore): Michael Henry de Young, M. H. de Young, was the brother who lived. We are about to meet the brother who did not. In 1894 Michael organized a fair in Golden Gate Park, the California Midwinter International Exposition, a world's fair staged in a California winter to tell the rest of the country the sun still worked. The art building was left standing when the fair came down. The museum that grew out of it carries his name. The de Young. If you have been in that building, you were in a room descended from a newspaper family that spent the eighteen seventies as a political weapon. The marble does not mention the weapon. That is what a name is for. Narrator (Kore): Across the bay, the name is Hearst. Hearst Avenue runs along the north side of the University of California at Berkeley. The Hearst Memorial Mining Building is there because Phoebe Apperson Hearst paid for a great deal of that campus. She was George Hearst's widow, and from 1897 until she died in 1919 she was the first woman on the university's board of regents. George had bought the ranch at San Simeon, on the coast, in 1865. The castle tourists drive up to see is his son's later building, on the father's ground. Mine, newspaper, ranch, campus, castle. One family, five ways of making a name feel like geography. Narrator (Kore): Spanish California put saints on the land and made the church the landlord. Railroad California put nabobs on a hill and made a corporation the landlord. It built houses like altars and bought papers like pulpits. When a boy died, it would build a university that looked like a mission. That building is the next episode. The papers come first.

04. Friends

[CUE: singing-bowl chapter chime] [MUSIC: brahms3_3 from 0s, ducked] [AMBIENCE: telegraph.mp3] [MOTIF: railroad stinger]

Narrator (Kore): Friends. Narrator (Kore): The partners did not particularly like each other, and they liked Stanford least. Friendship, in the Gilded Age, was not a feeling. Richard White, in his history of the transcontinental railroads, treats it as an arrangement. Measures in California lived or died on who would stand with whom when the vote was called. Narrator (Kore): Some of the arrangements were with Congress, and we can hear them, because a man named David Colton saved his mail. Colton was the partners' man in San Francisco, a junior associate who wrote constantly to Collis Huntington in New York. Huntington wrote back in his own voice, dry, busy, sure of the price of things. Between the middle of the eighteen seventies and Colton's death in 1878, the letters piled up. In 1883, during a lawsuit by Colton's widow, they became public. They were reprinted in a pamphlet with a title that does not bother to whisper: How Members of Congress Are Bribed. Narrator (Kore): Read Huntington in his own words. November thirtieth, 1874, on the Congress about to sit. LETTER (Algenib:

Collis Huntington, private, dry, amused): "I think this coming session of Congress will be composed of the hungriest set of men that ever got together." Narrator (Kore): January seventeenth, 1876. A rival, Tom Scott, was pushing a bill to help his own Pacific line, the Texas and Pacific. Huntington took the night train to Washington. LETTER (Algenib: Collis Huntington, private, matter of fact): "It cost money to fix things so that I would know his bill would not pass. I believe that with $200,000 I can pass our bill." Narrator (Kore): And the sentence he used whenever he wanted to sound clean. January twenty-ninth, 1876, and again that March. LETTER (Algenib: Collis Huntington, self-satisfied, private): "I keep on high ground." Narrator (Kore): The second time he finished the thought. He kept on high ground so that no investigation could hurt them. Narrator (Kore): Two hundred thousand dollars, in a letter dated January seventeenth, 1876. In grocery dollars that is roughly six million three hundred thousand now, as the cost of one bill. Narrator (Kore): Lobbying is arguing for a bill in public, with your name on the argument. Bribery is paying an official to use the office as a lever. Huntington's letters live in the gap. He does not write that he handed a senator an envelope. He writes that two senators were switched this week and can be switched back with the proper arrangements. The high ground is not innocence. It is a method. Pay in ways that look like friendship, so that if a committee asks, the paper smells like business. Narrator (Kore): White's point is that nineteenth-century Americans were not naive about this. They were shocked less by a bought newspaper or a greedy congressman than by what it added up to: a republic whose decisions were for sale. The word they used was corruption, and they meant it the old way, as rot in the structure, not as one man's bad manners. Narrator (Kore): This was not a California invention. While Huntington priced a bill, Tom Scott of the Pennsylvania Railroad was asking Congress to back the Texas and Pacific, a southern route to the coast. The Colton letters are in large part Huntington fighting Scott. Farther north, Jay Gould was gathering lines. Charles Perkins of the Chicago, Burlington and Quincy said the Union Pacific's life depended on high rates more than any other company's. John Murray Forbes, whose Boston family had made its first fortune in the China trade, said Gould had as many aliases as a London professional thief. Stanford's face was Western. The technique was national. Narrator (Kore): Colton himself did not live to see the letters read aloud. On the evening of October eighth, 1878, a carriage stopped outside his house on Nob Hill. He was unconscious inside it. The company said a horse had thrown him. He died two days later. The rumor in the street was a stabbing. The truth of that night stayed dark. What the letters did, five years on, was turn the private voice of the railroad into a public document. Huntington's defense, for the rest of his life, was sarcasm and a bad memory.

05. The Bought Press

[CUE: singing-bowl chapter chime] [MUSIC: hebrides from 30s, ducked] [AMBIENCE: telegraph.mp3]

Narrator (Kore): The Bought Press. Narrator (Kore): Friendship with a newspaper worked the same way as friendship with a senator. You did not always buy the printing press. You bought the pipe the news came through. Narrator (Kore): In the eighteen seventies the Associated Press was that pipe. A paper that was not in the association did

not get the national wire, which meant it did not get the country, not in time to print it. The wire was a chokepoint, the same shape as a port or a mountain pass. Whoever stood at the gate decided which stories moved. Narrator (Kore): Huntington understood the gate. He wrote to Hopkins that if the Associated Press agent in Washington were seen to, a great deal of hostile copy could be kept back. His lobbyist, Richard Franchot, once got that Washington reporter onto the Central Pacific payroll. It worked until the Associated Press editor, James Simonton, noticed that the tone of the stories had changed, and suspected why. Narrator (Kore): In 1875 the partners tried something larger. They wanted an exclusive Associated Press membership for the San Francisco Chronicle, a paper they then had influence over, and they offered double the going rate. Double. The point of paying double was not generosity. It was to cut two other papers, the Bulletin and the Call, off the wire. Papers the railroad hated. The attempt failed. The instinct is the lesson. A news monopoly is worth more than a fair price, because a news monopoly decides what a fair price sounds like in print. Narrator (Kore): Jump to 1886, and a memo prepared for the president of the Union Pacific, which ran down the New York papers one at a time, the way a quartermaster runs down boots. One sheet existed to blackmail. One would print what it was handed. And the Star, the memo said, did as Huntington directs. White copies the verdict in a sentence that belongs at the center of this story. Quote (Charon, flat documentary): Nineteenth-century Americans were not shocked by the corruption of the press. Narrator (Kore): It was how the trade worked. What frightened people was the use. A railroad that can raise a rate and then rent the description of the rate is not only a carrier. It is an editor. Narrator (Kore): There is a cartoon that puts the whole arrangement in one palm. February nineteenth, 1881, in the San Francisco Wasp. The title is "In the Hollow of His Hand." Crocker and Huntington stand on the railroads and hold Stanford up, and Stanford holds the state railroad commission in his hand, a commissioner small enough to close his fingers on. The commission was new. Voters had adopted a fresh state constitution in 1879, over the railroad's opposition, and that constitution created a railroad commission, three elected men whose job was to watch the rates. Two years later a humor magazine drew the watchdogs sitting in the palm of the man they were supposed to watch. Narrator (Kore): Lawyers now call this regulatory capture. It sounds like jargon. It is a simple motion. A public creates a guard to stand between a monopoly and its customers. The monopoly then hires the guard, or elects him, or tires him out, until the guard's reports describe the monopoly's wishes as the public's safety. You do not have to repeal the law. You inhabit it. The Wasp did not have the phrase. It had the picture. Stanford's hand, and inside it, the commission.

06. The Sandlot

[CUE: singing-bowl chapter chime] [MUSIC: tragic from 0s, ducked] [AMBIENCE: crowd.mp3]

Narrator (Kore): The Sandlot. Narrator (Kore): July, 1877. The country was in the long depression that had started in 1873. In San Francisco, unemployed men gathered on a vacant lot near the city hall, a sandlot, and the meeting that was called against the railroads turned, fast, against the Chinese. The historian Malcolm Harris, who grew up

in Palo Alto, follows the crowd up the hill. Narrator (Kore): They marched on Leland Stanford's mansion. The man at the front of the Workingmen's Party of California was Denis Kearney, an Irish-born drayman, a man who hauled freight for a living, and a speaker who had found the sentence that would move a crowd. SPEECH (Rasalgethi: Denis Kearney, a drayman on a sandlot in 1877, hoarse, addressing a crowd): "I will give the Central Pacific just three months to discharge their Chinamen, and if that is not done, Stanford and his crowd will have to take the consequences." Narrator (Kore): Three months. That is the number in the threat. Now put a census next to it, even though the count comes three years later, in 1880, because the people Kearney was pointing at were not a rumor. The 1880 census counted 75,132 Chinese people in California, in a state of about 865,000. Nearly one Californian in twelve. And of all the Chinese people in the United States, about 105,000, seven in ten lived in that one state. Narrator (Kore): Kearney's slogan, the one painted on banners, was four words. The Chinese must go. Harris's judgment of the man is short. Kearney was better at being a racist than at being a labor leader. It was easier to make Chinese workers pay for the depression than to make Stanford pay. Stanford had Kearney arrested. Then Stanford did what the very rich have often done when a crowd learns the address. He left town. The family already had a farm south of the city, bought in 1876, about six hundred and fifty acres, a place called Mayfield Grange. The crowd had marched on a mansion. The owner was becoming a country gentleman. The farm would grow, and one day it would have a different name. Narrator (Kore): Scapegoating is not just meanness. It is a political technology. The anger in the sandlot was not imaginary. Wages were down. The railroad really was a monopoly. A man who hauled goods for a living had reason to hate a company that set the price of moving goods. Kearney took that anger and nailed it to a people who could not vote, who had built the railroad, and who were still in the company's workforce because the company wanted them there. In his 1862 inaugural, read from a text, he said Asia sent the dregs of her population, and he called the Chinese a degraded and distinct people. On the stump the line was shorter. The dregs of Asia. The inaugural is one occasion. The campaign is another. By the time the tracks were finished he wanted more of them, because they kept the wage down. In 1877 his neighbors wanted them gone, and he was happy to let the neighbors do the shouting. Narrator (Kore): The workers and the owner were not on the same side of the door. The crowd was outside the mansion. The men the crowd wanted fired were the railroad's labor, paid by the month, living in camps the mansion never had to see. A monopoly can survive a riot if the riot aims at its workforce instead of its rates. That is what the sandlot gave the Big Four. A villain who was not them. Narrator (Kore): The party Kearney fronted did not stay in the street. It got men elected. It helped write the streak of anti-Chinese clauses into the new state constitution of 1879. And it produced, in San Francisco, a mayor the railroad's favorite paper tried to kill.

07. The Chronicle

[CUE: singing-bowl chapter chime] [MUSIC: brahms1_1 from 0s, ducked] [AMBIENCE: crowd.mp3]

Narrator (Kore): The Chronicle. Narrator (Kore): The San Francisco Chronicle began in January of 1865 as a theatrical handbill, the Daily Dramatic Chronicle, started by two brothers, Charles and Michael Henry de Young. Charles was the editor who treated scandal as a product. By the late eighteen seventies the handbill had become a daily, and for a time the railroad had a hand on it. That grip did not make the paper gentle. It made the paper a faction. Narrator (Kore): In 1879 a Baptist minister named Isaac Kalloch ran for mayor as the Workingmen's candidate. The Chronicle went after him personally. Kalloch answered from the pulpit and, by the paper's own later account, said things about Charles de Young's mother that a man of that temper would not let stand. In August of 1879 Charles de Young shot Kalloch. The minister lived. The sympathy of the shooting helped elect him. Narrator (Kore): Learn the dates, because the city learned them. On the evening of April twenty-third, 1880, Kalloch's son, Milton Kalloch, walked into the Chronicle office and shot Charles de Young dead at his desk. The New York Times ran the wire the next day, plain as a bill of lading. Charles de Young killed. Mayor Kalloch's son. The Chronicle office. Half past seven. Narrator (Kore): The son was tried and acquitted. One juror's reported view was that he would have done the same. Michael de Young, the surviving brother, took the paper. He would run it for decades, build a fair, and leave his name on a museum. Charles left a statue and a warning. Narrator (Kore): A newspaper can be a public record. It can also be a political weapon owned by a side. De Young did not shoot Kalloch because of a disagreement about grain rates. He shot him because the paper and the candidate had become personal enemies, and in that San Francisco the personal was how politics was conducted. The railroad did not order the shot. It did not have to. It had spent years teaching the city that a paper was a piece of property, to be captured, starved of wire service, or aimed. Once that is normal, editors start carrying the same tools as the men they cover. Narrator (Kore): Put the spring of 1880 on one calendar. April twenty-third, a son kills an editor in a newspaper office on the bay. May eleventh, eighteen days later, seven men die in a wheat field, and an eighth is dying of a scalp wound he does not yet know will kill him. Same state. Same railroad sitting under both stories, not as the finger on every trigger, but as the condition both fights were fought inside. A city and a valley that had come to expect the conflict. Narrator (Kore): White's line about the press belongs here too. People were not shocked. The Chronicle's own history tells the shootings as office lore. A museum in the park carries the surviving brother's name. The abnormal, repeated, becomes a plaque.

08. Yuma

[CUE: singing-bowl chapter chime] [MUSIC: hebrides from 30s, ducked] [AMBIENCE: wind.mp3] [MOTIF: railroad stinger]

Narrator (Kore): Yuma. Narrator (Kore): Jump back two years. The night of March twenty-eighth, 1878. Some notices date the death the twenty-ninth. The scene, in Rayner's telling, is a private railroad car standing on a siding at Yuma, on the Colorado River, where California meets Arizona. Narrator (Kore): Mark Hopkins was sick. He had not wanted the trip. Huntington wanted eyes on the ground the Southern Pacific was reaching for, out past the river, and Hopkins knew that Huntington's schemes usually

had something behind them. He got off his sickbed, into his own company's car, and went to look. He went to sleep in the car. He did not wake up. Narrator (Kore): He was the partner the others could stand. The bookkeeper. The man who had answered panic, in the building years, with as few words as possible. He had let a palace rise on Nob Hill that he would never occupy. He died in the one house he actually used, a car on a company track, at the edge of the map. LETTER (Algenib: Collis Huntington, writing after Hopkins's death, briefly softer): "My old friend Mr. Hopkins has gone over the river. His death has made me very sad. I was very fond of Mr. Hopkins. We had been friends from the day of our first meeting, and partners in business, as you know, for many years." Narrator (Kore): Huntington did not come west for the funeral. He was, Rayner notes, too busy. He added that there might be other men as good as Hopkins, but that he would not live long enough to form such a friendship again. Read that twice. The affection is real. So is the arithmetic. A partnership is a machine that expects a man to be replaced, and the man who ran the machine from New York was not going to miss a vote in order to stand by a grave. Narrator (Kore): This is succession, the unromantic kind. Four men had owned the Central Pacific in equal shares and let a construction company they also owned pay them for building it. By the spring of 1878 one of the four was a car on a siding. Within months Colton, the fifth man they had let partway in, was dead on Nob Hill in a story nobody could finish. The survivors inherited the dead men's leverage, and they used it on the widows. Colton's widow, Ellen, was walked through a story in which her husband had been an embezzler, and she was pressed to give the shares back. Grief was not a pause in the business. It was an opening in the cap table. Narrator (Kore): Hopkins's house on the hill outlived him as an art school, then as a fire, then as a hotel with his name in gold on the roof. The siding at Yuma did not get a plaque. The railroad kept the car.

09. Birth of the Octopus

[CUE: singing-bowl chapter chime] [MUSIC: borodin from 0s, ducked] [AMBIENCE: telegraph.mp3, crowd.mp3] [MOTIF: railroad stinger]

Narrator (Kore): Birth of the Octopus. Narrator (Kore): In 1882 California elected an antimonopoly governor, George Stoneman, a Democrat, and a legislature that looked, on paper, like the railroad's worst day. The new constitution already promised a commission and a harder hand on rates. Stoneman wanted taxes, including back taxes, and a bill to outlaw the special contracts by which the railroad charged one shipper a quiet price and another shipper a punishment. In 1884 he called a special session of the legislature to do railroad business and nothing else. Narrator (Kore): Count the votes he thought he had. A Democratic majority of sixty-one to nineteen in the assembly. Thirty-two to eight in the senate. The bills sailed through the assembly. In the senate, bill after bill ended in a tie, and the lieutenant governor, a Democrat, broke the tie against them. Nine of ten Democratic senators from San Francisco voted the railroad's way. They were the lambs of a man the city called the Blind Boss. Narrator (Kore): Christopher Buckley. White uses the nickname the city used, and he cites the historians who wrote Buckley's career up as the Blind Boss of San Francisco. Buckley had publicly supported the antimonopoly bills. Then his senators killed them. White's explanation is not that

Buckley loved Stanford. It is that a fight between a railroad and a reform wave was an opportunity only a fool would waste. The railroad could taste defeat, or it could sweeten Buckley's pot. Nineteenth-century slang for the dollar was simoleons. Buckley liked them. Narrator (Kore): The Pacific Railway Commission, looking back, estimated that the Southern Pacific spent about seven hundred thousand dollars on politics that year. That year is 1884, the special session. In grocery money, at 1884 prices, it is roughly twenty-six million dollars now. It did not all go to one envelope. White describes a stew: railroad money, kickbacks, payments from anyone who needed a franchise or a contract, protection, the ordinary boodle of a city boss, mixed until a senator could not have told you which dollar had started in a freight tariff. That mixing is the method. Once the money is in the stew, every dollar looks like every other dollar. Narrator (Kore): White's sentence for what came out of the pot is the sentence this episode is named for. Quote (Charon, flat documentary): The political marriage of Leland Stanford and Christopher Buckley gave birth to the Octopus. Narrator (Kore): The creature already had a body in the cartoons. In the Wasp, a drawing called The Curse of California put the railroad on the page as an octopus, tentacles on the farmers, the miners, the whole state. Ambrose Bierce, who edited the Wasp in these years, printed that kind of picture and wrote the captions in acid. The public had the image before White had the marriage certificate. What White adds is a date and a mechanism. Not a metaphor floating free. A boss, a senator, a special session, and a pile of money large enough to turn a sure majority into a tie. Narrator (Kore): Regulatory capture, again, but now you can watch the gears. The voters built a commission and elected a hostile governor. The railroad did not repeal the commission. It bought the senate votes that kept the commission's teeth in a drawer. The law stayed on the books, where it could be pointed at. The practice stayed in Buckley's pocket, where it could be used. A government that looks like it is governing the railroad, and a railroad that is governing the government. That is the octopus. Not the animal. The arrangement. Narrator (Kore): It did not last in every market. Within a year other transcontinental lines were reaching the coast, and a second track is the one thing an octopus in a rate table cannot survive. The political animal lasted longer. Between the middle eighties and the early nineties, White writes, the Octopus was quite real. In 1885 Stanford, with Buckley's help, bought his way into the United States Senate. Senators were still chosen by state legislatures, not by the voters. A boss who owned a city's delegation owned a piece of the Senate. Narrator (Kore): Frank Norris published The Octopus in 1901. It is a novel, not a history. In it the railroad raises its tariff to whatever, in the novel's phrase, all the traffic would bear, and its men have a sentence ready. Quote (Charon, flat documentary): The railroad could not be operated at a legitimate profit. Narrator (Kore): Legitimate does the moral work. A rate extracted from a farmer with no other road is redescribed as the minimum a decent company needs in order to exist. Norris's narrator sees tentacles of steel clutching the soil. Fiction, and late, and still the name the public kept. Charge to the point where the customer breaks, and call the breaking point a principle.

10. Exclusion

[CUE: singing-bowl chapter chime] [MUSIC: tragic from 0s, ducked] [AMBIENCE: wind.mp3]

Narrator (Kore): Exclusion. Narrator (Kore): The sandlot got its federal law. It took five years from Kearney's three-month threat, and it arrived wearing the language of a treaty. Narrator (Kore): In 1879 Congress passed a bill to cut Chinese immigration hard, down toward fifteen passengers on a ship. President Rutherford B. Hayes vetoed it on the first of March, 1879. His reason was a treaty. The United States had promised China, in the Burlingame Treaty, a freer movement of people. Hayes did not veto because California's workers had changed his heart. He vetoed because he did not want to tear up a diplomatic bargain in public. Then he sent a minister to renegotiate the bargain. The new treaty let the United States restrict Chinese immigration, so long as the restriction was reasonable. Reasonable is a lawyer's word. It means whatever the next Congress can pass and the next president will sign. Narrator (Kore): In 1882 Congress tried for a twenty-year ban on Chinese laborers. President Chester A. Arthur vetoed that one too, as more than the treaty allowed. Congress came back with ten years. On May sixth, 1882, Arthur signed it. The Chinese Exclusion Act. National Archives historians call it the first significant federal law restricting immigration, and the first time federal law barred an ethnic working group because Congress claimed that group endangered the good order of certain places. Narrator (Kore): Be precise, because the precision is the point. This was not the first unfriendly statute. The Page Act, in 1875, had already made it much harder for Chinese women to enter the country, under the stated target of prostitution and forced labor. The 1882 law was wider. It shut out Chinese laborers, skilled and unskilled, for ten years. It made Chinese people ineligible for citizenship. It required papers of the Chinese who were allowed to come and go, merchants, teachers, students, travelers. A workforce that had been recruited, by this same railroad, with handbills the railroad's own officers could not read, was now a people the federal government would not admit and would not naturalize. Narrator (Kore): The sequel is how you know a panic has become a policy. In 1892 the Geary Act extended the ban and ordered Chinese residents to carry a certificate of residence or face deportation. In 1902 Congress stretched it again, including to Hawaii and the Philippines. The exclusion laws stayed until 1943, when China was a wartime ally. Congress repealed them and left a quota of a hundred and five people a year. Sixty-one years. A man who had laid rail in the Sierra in 1867 was old, or buried, before the statute that answered him came off the books. Narrator (Kore): Put the census back beside the statute. Seventy-five thousand Chinese people in California. One in twelve of the state. The men who spoke for the state described an invasion. The count describes a minority, concentrated because the jobs were concentrated, on the railroad, in the fields, in the laundries and the factories of one coast. Exclusion did not break the octopus. It fed it a story. The monopoly remained. The people who had built it were barred at the door. Narrator (Kore): Stanford's own record is the cleanest exhibit. As governor he had called them a degraded people. As railroad president he had wanted them by the hundred thousand, to bring the wage down. As a senator, years after this law, he would sit in a body that had made the ban normal. The inconsistency is not a mystery and it is not a private paradox. It is a sequence. Use a people when you need their hands. Describe them as a danger when their hands make your neighbors angry. Sign nothing yourself if other men will sign it for you.

11. Bierce and Hearst

[CUE: singing-bowl chapter chime] [MUSIC: mozart40_2 from 0s, ducked] [AMBIENCE: fs_42139.mp3]

Narrator (Kore): Bierce and Hearst. Narrator (Kore): On April second, 1881, the Wasp printed a poem by Ambrose Bierce. He had the Mussel Slough dead in mind, and he aimed the last stanza at a man who had not been in the field. SPOKEN (Puck: Ambrose Bierce, sardonic, reading his own verse, a little tired): "These mounds are green where lie the brave who fought and fell at Hanford. O, point me out, I pray, the grave of Leland Stanford." Narrator (Kore): He had a nickname ready. Stealand Landford. Later, in the Examiner, he called Huntington's head a dromedary, and in another column gave him one leg in the grave, one arm in the treasury, and one eye on the police. When Huntington told a committee he could not recall, Bierce wrote that he took his hands out of his pockets long enough to be sworn. Narrator (Kore): The poem is 1881. The marriage of Stanford and Buckley is still ahead. So is the strangest consequence of Stanford's fight with his own partner. Narrator (Kore): George Hearst was a miner who had been lucky on a scale that looks like fiction. The Comstock, the Homestake, the Anaconda, claims that became the backbone of American metal. He wanted a Senate seat. In those years the California legislature still handed the seat over, and a rich man campaigned by feeding legislators. Rayner writes that Hearst got there in 1886 with the support of Leland Stanford, who chipped in because Hearst's opponent was Aaron Sargent, a man in Huntington's orbit. The official record is a little more precise than the verb elected. Hearst was appointed to a short vacancy in March of 1886, and then chosen for a full term that began on March fourth, 1887. He died in office in 1891. Narrator (Kore): He owned a newspaper, the San Francisco Examiner. He had picked it up in 1880, a weak Democratic daily, third in the city. On the day his full Senate term began, March fourth, 1887, the masthead changed. W. R. Hearst, proprietor. William Randolph Hearst was in his middle twenties. He had looked at Joseph Pulitzer's New York World and decided the Examiner should shout. Bigger headlines. Fires, adultery, violent crime, and preferably all three in one story. He hired, at high salaries, a staff of famous writers. Bierce was the first of them. Narrator (Kore): Watch the loop. Stanford helps a miner into the Senate to spite Huntington. The miner hands his son a newspaper. The son hires the poet who had already asked to be shown Stanford's grave, and points him at Huntington. Stanford's feud financed, at one remove, the most widely read attack on his partner. He did not mean to. A weapon you help load does not ask which enemy you had in mind. The avenue, the mining building, the castle, the masthead: downstream of a Senate seat a railroad man supported because he hated his partner. The street sign does not say so.

12. The Caption

[CUE: singing-bowl chapter chime] [MUSIC: mozart40_2 from 0s, ducked] [AMBIENCE: fs_42139.mp3]

Narrator (Kore): The Caption. Narrator (Kore): Now jump forward. Past a death we have not reached yet. Past the founding of a university. To a law book, and a paragraph at the top of a case that is not the case. Narrator (Kore): California tried to tax the

railroad's property without letting the company deduct its mortgages the way an ordinary landowner could. The Southern Pacific refused to pay, and turned the refusal into test cases that ran for years. Counties spent time and money they did not have. The judge who heard the cases on the federal circuit in California was Stephen Field, a justice of the United States Supreme Court, riding circuit, which justices still did. The legal scholar Adam Winkler puts the relationship in capital letters. Leland Stanford of the Southern Pacific had a close relationship with Justice Stephen Field. Narrator (Kore): When the railroad decided to bring the test cases, Field recommended a lawyer, a law professor named John Norton Pomeroy, who admired the way Field wrote opinions. Recommending a lawyer to a friend is not, by itself, a crime. Field then did the thing modern judicial ethics would forbid. He presided. He did not step aside. And during the litigation, Winkler writes, he shared with the railroad's lawyers confidential memoranda about the case written by other justices. Narrator (Kore): In the circuit opinions, Field said corporations were persons protected by the Fourteenth Amendment. The amendment had been ratified in 1868 for a different person: the formerly enslaved, and anyone a state might try to crush. Its promise was that no state shall deny to any person within its jurisdiction the equal protection of the laws. Field's move was to slide the railroad into the word person. Treating the company's property differently from a widow's property was, he wrote, the very essence of tyranny. He offered a comparison meant to shock. A tax law that changed its rules according to whether the owner was a white man or a black man would be an outrage. So, he said, was a tax law that changed its rules because the owner was a corporation. Narrator (Kore): The amendment was written because states were denying rights to black Americans. Field used that sentence to protect a railroad's balance sheet. The people it was written for would spend the next half century losing most of the cases they brought under it. The railroad was just getting started. Narrator (Kore): In January of 1886 the Southern Pacific's case, Santa Clara County versus Southern Pacific Railroad, reached the Supreme Court itself. The question on the table was Field's question. Does the Fourteenth Amendment cover corporations? Before the argument, Chief Justice Morrison Waite told the lawyers the Court did not want to hear about it. When the decision came down, the Court ruled for the railroad on the narrowest ground available. The county had included the value of the fences along the tracks in the tax bill. California law did not allow that. Because the case could be decided on that ground, the opinion said it was not necessary to consider any other questions, including the grave constitutional questions decided below. Narrator (Kore): Read that again, because the next man counted on you not to. The Supreme Court did not hold, in 1886, that corporations are persons under the Fourteenth Amendment. It said it did not need to reach the question. It decided a fence. Narrator (Kore): Then the reporter went to work. His name was J. C. Bancroft Davis. His job was to write the syllabus and the headnotes, the summary at the front of a published opinion, so a busy lawyer can see what was decided without reading the decision. Headnotes are not the law. They are a caption. Davis's caption said this. Quote (Charon, flat documentary): The defendant Corporations are persons within the intent of the clause in section one of the Fourteenth Amendment to the Constitution of the United States, which forbids a state to deny to any person within its jurisdiction the equal protection of the laws. Narrator (Kore): Above the opinion, he printed a remark he attributed to the Chief Justice, from before the argument. Quote (Charon, flat documentary): The Court

does not wish to hear argument on the question whether the provision in the Fourteenth Amendment which forbids a state to deny to any person within its jurisdiction the equal protection of the laws applies to these corporations. We are all of the opinion that it does. Narrator (Kore): Davis checked with Waite before he published. Waite's reply survived. LETTER (Charon: Chief Justice Morrison Waite, a terse private note): "I leave it with you to determine whether anything need be said about it in the report inasmuch as we avoided the constitutional question in the decision." Narrator (Kore): [SILENCE] [HARD] Davis printed it anyway. He had once been president of a railroad himself, the Newburgh and New York Railway. Winkler does not think the caption was a sleepy mistake. A reader who stopped at the headnote would walk away believing the Court had held that corporations were people. The Court had not. Narrator (Kore): Three years later the caption became a citation. In 1889 a railroad, the Minneapolis and St. Louis, killed three hogs in Iowa and was ordered to pay twenty-four dollars in damages. Twenty-four dollars. In grocery money of 1889, roughly nine hundred dollars today. The company spent tens of thousands of dollars taking the case to the Supreme Court. The money was cheap if the prize was a stronger shield against every future fence law, livestock law, and tax. Field wrote the opinion. He upheld the Iowa statute. Along the way he wrote that corporations are persons under the Fourteenth Amendment, and then he added one sentence. Quote (Charon, flat documentary): It was so held in Santa Clara County versus Southern Pacific Railroad. Narrator (Kore): [HARD] It had not been so held. A caption had been promoted into a precedent. Field had written, three years earlier, an angry note that the Court had dodged the constitutional question. He had not forgotten. He cited the dodge as if it were a decision. Narrator (Kore): Three words. A holding is the rule a court must use to reach its result. Dicta are remarks a judge makes on the way by, which do not control the case. A headnote is a reporter's summary, not a vote. Santa Clara's holding was a fence. The corporate sentence was a headnote, plus a remark the Chief Justice left the reporter free to omit, plus Field later treating that headnote as if the Court had voted it. A strange idea becomes a normal one by repetition, in the book every lawyer keeps on the desk. Narrator (Kore): In 1912 a lawyer named Charles Wallace Collins, who had served as a librarian for Congress and for the Supreme Court, counted every Fourteenth Amendment case since 1868. Six hundred and four. Twenty-eight involved African Americans, and those plaintiffs usually lost. Three hundred and twelve, more than half, involved corporations, which used the same sentences against laws on wages, zoning, and child labor. Collins called the rebuilt amendment the Magna Charta of accumulated and organized capital. Narrator (Kore): If those cases were a year, the cases about black Americans would run out before the end of January. The corporate cases would last until November. Narrator (Kore): Mussel Slough made the papers for a week. The headnote is still in the bound volume. A later court can narrow it. It cannot pretend it was never printed. The railroad did not need to win every case. It needed the word person to be available the next time a state tried to tax a fence.

13. A Telegram from Florence

[CUE: singing-bowl chapter chime] [MUSIC: scottish3 from 0s, ducked] [AMBIENCE: wind.mp3] [MOTIF: railroad stinger]

Narrator (Kore): A Telegram from Florence. Narrator (Kore): Come back. The caption is two years in the future. The boy is in Italy now. Narrator (Kore): March, 1884. Leland Stanford is a railroad president and a former governor. The Senate seat is still a year away. The house on Nob Hill has a dome seventy feet high. He and his wife have one child. The boy was born on May fourteenth, 1868. He is fifteen. He is in Florence. David Starr Jordan will call the illness Roman fever. University histories call it typhoid. Narrator (Kore): On March thirteenth the fever kills him, two months short of his sixteenth birthday. His mother will keep his room as he left it. His father has spent twenty years turning a franchise into a government. Narrator (Kore): Grief does not launder a fortune. It can change what the fortune is for. The next hour is the farm. Narrator (Kore): [SILENCE] [HARD] In a hotel in Florence, a fifteen-year-old boy has a fever. The octopus has an heir. The heir will be made of stone, and scholarships, and a clause that says forever.

Cardinal Podcast Ep. 4: The Boy and the Farm (audio script)

Cardinal Podcast Ep. 4: The Boy and the Farm (audio script)

Episode 4 of 8. The Boy and the Farm (1884–1905). Markup matches the pipeline in "The Spike in the Museum": Narrator = Kore. Documents and flat quotations = Charon. Newspapers = Sadachbia. Letters and speeches use the voice names below. Bracketed cue lines are not narrated. Narration target 8,500–9,000 words.

01. The Verandah

[CUE: singing-bowl chapter chime] [MUSIC: hebrides from 0s, ducked] [AMBIENCE: surf.mp3, fs_42139.mp3] [MOTIF: Jane, Goldberg Aria phrase]

Narrator (Kore): The Boy and the Farm. Narrator (Kore): This is Cardinal. Episode four. Narrator (Kore): February twenty-eighth, 1905. A Tuesday, on the verandah of the Moana Hotel, at Waikiki, in the Territory of Hawaii. Jane Lathrop Stanford is seventy-six. She was born on August second, 1828. She is the widow of a railroad president, the surviving founder of a university, and she has spent the day being happy. Narrator (Kore): A carriage took her slowly along the windward coast, because she wanted the ride to last. Her maid, May Hunt, remembered that she chattered and sang. Around one o'clock they picnicked in a grove. Bertha Berner, her secretary, later told the Hawaiian Star what the hotel had packed: sandwiches of graham bread, Swiss cheese, tongue and lettuce, hard-boiled eggs, gingerbread, oranges, chocolate candy, and cold coffee. Jane ate heartily of everything but the eggs. They were back at the hotel by four. Narrator

(Kore): The Moana had opened on the beach in 1901. She checked in on February twenty-first. Room 120. Berner and May Hunt shared 122. Narrator (Kore): At half past six the three women went down to dinner. On the verandah Jane told a guest named Mrs. Grinbaum about the pleasant day. Dinner was a bowl of soup. It took ten minutes. She sat outside until eight, or a quarter past, and then went up with Berner and asked her to set out the medicine. Narrator (Kore): Berner put a cascara capsule on the bed stand. Cascara sagrada was a laxative sold over the counter, and the capsules of that era often carried a little strychnine, on the theory that the poison hurried the herb. Then Berner took a small bottle of bicarbonate of soda from a medicine basket. The bottle was two to two and a half inches tall, with a mouth wide enough for a teaspoon. She asked May Hunt for a spoon, measured the half teaspoon Jane wanted, and left it beside the capsule. DRAMA (Sulafat: Jane Stanford, tired, ordinary, in a hotel room, take 1 of 3): "Give it time to get out of my stomach." Narrator (Kore): That is what she said she would do with a glass of water, before the medicine. She urged Berner to take some of the soda too. Berner declined. The two women left around nine. Jane called May Hunt back. She was having trouble with the lock. Hunt showed her, and went to bed. Narrator (Kore): Berner thought her employer fell asleep without taking the dose, and woke later and swallowed it. Hours after that, the hallway filled with a voice. A guest across the hall, Adam Heunisch, opened his door and found her in her doorway, arms thrown up. DRAMA (Sulafat: Jane Stanford, in pain, gripping a door casing, take 2 of 3): [HARD] "Oh I am so ill. Get me a doctor." Narrator (Kore): Heunisch ran for the elevator. Dr. Francis Humphris lived in the hotel. Berner told the elevator boy the doctor should not even stop to dress. May Hunt tried to get Jane back to the bed. Jane would not go. She held the casing. She said a spasm had thrown her out of bed. She said she had been poisoned again, and she asked Humphris for a stomach pump. When he asked how she knew, she said she had been poisoned before, and she turned to Berner. Berner said the earlier bottle, Poland Spring water, had held arsenic, and then corrected herself. Strychnine. Narrator (Kore): The pump did not come in time. Her jaws set. Humphris felt the muscles lock. He took the glass from her hand. Berner watched the last spasm. DRAMA (Sulafat: Jane Stanford, a broken prayer, jaw stiffening, take 3 of 3): [HARD] "Oh God, forgive me my sins." DRAMA (Sulafat: Jane Stanford, the same breath, almost no voice left): "Is my soul prepared to meet my dear ones?" Narrator (Kore): And then the sentence Richard White, who wrote the history of this death, says she rarely if ever spoke, because she had spent twenty years calling death a passage into a higher life. DRAMA (Sulafat: Jane Stanford, the last words, nearly a whisper): [SILENCE] [HARD] "This a horrible death to die." Narrator (Kore): [SILENCE] The body went rigid. The soles of the feet turned inward toward each other, insteps arched, toes pointing forward, knees wide. The eyeballs protruded. The pupils dilated. The jaws fixed. The fingers contracted, and the thumbs dug into the palms. Breathing stopped. From the moment the doctor entered the room to the last spasm, ten minutes. Narrator (Kore): The inquest is three weeks away. The reason any of this is a university story is twenty-one years away, in Florence, where a boy of fifteen died and his parents decided the fortune would outlive them. Come back to the bedroom across the hall.

02. Florence

[CUE: singing-bowl chapter chime] [MUSIC: aase from 0s, ducked] [AMBIENCE: wind.mp3] [MOTIF: Jane, Goldberg Aria phrase]

Narrator (Kore): Florence. Narrator (Kore): You have already stood in that hotel. March thirteenth, 1884. A boy of fifteen. What this hour keeps is the room across the hall, and the sentence that came out of the night. Narrator (Kore): His bedroom in the house on Nob Hill sat across the hall from his mother's. Jane turned the tutor's room next door into a study where she worked with Bertha Berner. She kept the bedroom as it was when he died. She described what she did there. DRAMA (Sulafat: Jane Stanford, alone, kneeling, a private habit, not a speech): "into dear Leland's room and kneel down and tell all my sorrows to the blessed Jesus. I feel my darling boy and my dear husband are pleading with me." Narrator (Kore): In 1893, nearly a decade later, curators from the university museum measured every piece of furniture and where it stood, and moved the boy's things into the museum. They told her they had arranged the room just as it was in her city home. Narrator (Kore): The sentence the tours love has two wordings, and they are not the same sentence. Jordan writes that after a long and dreary night the father woke with these words on his lips. SPEECH (Iapetus: Leland Stanford, as David Starr Jordan remembered a sentence from years later, quiet, certain): "The children of California shall be my children." Narrator (Kore): Malcolm Harris, in his history of Palo Alto, prints a different pronoun. He has Leland telling Jane, the children of California shall be our children, when they decided to build on the farm. Jordan says the father told him the story in 1891. Mark it as a remembered sentence. My children makes the university a father's replacement son. Our children makes it a couple's. Narrator (Kore): Herbert Nash, the boy's tutor, wrote a sketch that put Jane at the center: young ladies and the conventions of society had no attractions for the boy, but toward his mother he would display the most attentive gallantry. Jane made Nash the university librarian. The sketch became the life that was not to be questioned. The next room is where that legend meets a medium, a preacher, and a ruined president.

03. The Séance and the Preacher

[CUE: singing-bowl chapter chime] [MUSIC: tragic from 0s, ducked] [AMBIENCE: fs_413852.mp3] [MOTIF: Jane, Goldberg Aria phrase]

Narrator (Kore): The Séance and the Preacher. Narrator (Kore): In the fall of 1891, after the university had opened, a story went around that the school had been founded under spiritualist influence, and that a medium named Maud Lord Drake had been the guiding hand. In 1892 Leland and Jane sat down with Jordan and dictated a statement they wanted kept. Jordan prints it. The father made his will, looking to the endowment of the university, in Paris, on April twenty-fourth, 1884. The mother made her will the same way, and copies were sent to America. Maud Lord Drake was unknown to them until they met her at a séance with the Grants in October 1884. About that time, the statement says, Mrs. Drake was detected in fraud. Narrator (Kore): Jane added a sentence. DRAMA (Sulafat: Jane Stanford, dictating, firm, correcting a rumor): "No spiritualistic influence affected the decision. Mrs. Drake had no more to do with it than a babe unborn." Narrator (Kore): No record survives of what was said in that parlor. Jordan

adds that both Stanfords were, for a time, deeply interested in a spiritualism that seemed to offer a demonstrable life after death. With the Grants they attended several séances in Washington. He says they never got, through mediums, any evidence they found convincing. The wills come first, in Paris, six weeks after the boy died. The Drake claim comes after, and they spent a page denying it. Narrator (Kore): The Grants are in this story for a more ordinary reason. Patronage. The word means the trade in favors inside a circle that does not have to call them favors. A railroad pass. A pulpit. Later, an embassy. On June second, 1884, from New York, Ulysses S. Grant wrote Leland Stanford. Before the failure of the firm Grant and Ward, Grant's youngest son, J. R. Grant, had sent his wife, child, and nurse to California and had expected to follow. His ready money had been in the firm. LETTER (Rasalgethi: Ulysses S. Grant, an old general asking a favor, plain, a little ashamed): "He deposited with the firm on the 2nd of May $86,000." Narrator (Kore): Grant asked whether it was too much, under those circumstances, to request a pass from New York out and back, for the son and the family. He told Stanford to refuse if it was inconvenient. Eighty-six thousand dollars, in 1884, wiped out in a Wall Street partnership that carried a president's name. In grocery dollars, using the Minneapolis Fed's price index and its 2026 estimate, that deposit is roughly three million two hundred thousand now. It had been a fortune. The circle held. Narrator (Kore): Five years later Frederick Grant, the general's older son, writes Stanford about his appointment as American minister to Austria. A pass, then a post. The office is public. The ask is private. Narrator (Kore): The pulpit was louder. The Reverend John Philip Newman had been the Grants' pastor in Washington and then in New York. In 1884 he was between jobs. The Stanfords brought him west to preach the boy's memorial. White writes that they reportedly promised him ten thousand dollars, equal to his yearly salary. The payment is not in an account anyone has produced. Narrator (Kore): The boy traveled west in a funeral car draped in black. Publicity claimed the greatest floral displays on the West Coast, supposedly twenty thousand dollars, roughly seven hundred forty thousand now. White marks that figure as a claim, not a bill. Narrator (Kore): White says that if it were not for the parents' grief, the praise of a rich dead boy the preacher had never known would have been risible. The sermon drew charges of blasphemy for comparing the boy to Christ, and a church fight over whether a minister who was not an Episcopalian should have been in an Episcopal service. A dead child, a borrowed Christ, a fee the size of a president's year. The university was decided in April, in Paris. The performance was in the winter. Grief was real. So was the stage.

04. The Founding Grant

[CUE: singing-bowl chapter chime] [MUSIC: brahms4_2 from 0s, ducked] [AMBIENCE: telegraph.mp3] [MOTIF: railroad stinger]

Narrator (Kore): The Founding Grant. Narrator (Kore): On March ninth, 1885, the legislature passed an enabling act so a private deed of trust could found a university. The title begins, an act to advance learning, the arts and sciences, and to promote the public welfare. It was written as a general law. It had one institution in mind. Narrator (Kore): On November eleventh, 1885, Leland Stanford and Jane Lathrop Stanford signed the grant. By then he was a United States senator. Senators were still chosen by legislatures,

and he had taken the seat that March. The deed says they desire to found, on their estate known as the Palo Alto Farm, in San Mateo and Santa Clara counties, a university for both sexes. Colleges, schools, museums, galleries, everything necessary to a university of high degree. They convey the land to a board of trustees and to their successors forever. Narrator (Kore): Read two clauses, because they are the machine. Quote (Charon, flat documentary): "the principal thereof shall forever remain intact, and that the rents, issues, and profits thereof shall be devoted to the foundation and maintenance of the University hereby founded and endowed" Narrator (Kore): And, inside the rights the founders kept for themselves while they lived: Quote (Charon, flat documentary): "but this reservation does not include the right or power to sell or encumber any of the real property granted." Narrator (Kore): A trust is property held by people who do not own it for themselves. The trustees spend the income on the purpose in the deed. The principal stays. Lawyers call a clause that keeps ruling after the giver is dead a dead hand. The same lock can freeze a mistake. Jane will later try to cap the women at a university the deed had opened to both sexes. The clause that stopped a sale is the clause that stopped her from turning the school into a boys' college. Narrator (Kore): One name on the trustee list is a callback. Stephen J. Field, a justice of the Supreme Court of the United States. The last episode sat in his courtroom, where a railroad fence case grew a caption that treated corporations as persons. The same man is now on the board of the railroad president's university. The deed does not blush. Narrator (Kore): The 1885 grant does not forbid tuition. It lets the trustees decide. The promise that California students would not pay comes later, in a constitutional amendment, with an escape we will reach. What the founders told Jordan was a fortune. The ultimate endowment, they said, would be about thirty million dollars. Jordan had fought to get Indiana's budget to fifty thousand. He was offered thirty times that, someday. Narrator (Kore): Someday was not a deposit. Before Leland died he gave three ranches, eighty-five thousand acres. Vina never turned a profit, and the vines died of phylloxera. Gridley's grain was in decline. The Palo Alto stables usually lost seven thousand dollars a year, most of Jordan's ten-thousand-dollar salary. George Crothers said the ranches brought nothing but taxes. White says taxes did exceed the revenue, and the papers prohibited a sale. The cornerstone went down on May fourteenth, 1887, the day the boy would have turned nineteen. The doors opened on October first, 1891.

05. A Supermission

[CUE: singing-bowl chapter chime] [MUSIC: borodin from 0s, ducked] [AMBIENCE: wind.mp3, horses.mp3] [MOTIF: railroad stinger]

Narrator (Kore): A Supermission. Narrator (Kore): Frederick Law Olmsted, the landscape architect who had made Central Park, was hired to site the campus. He wanted the buildings in the foothills. The Stanfords wanted flat land. They were building a monument you could march through. Olmsted suggested, and Charles A. Coolidge designed, the low buildings of the inner quadrangle, Encina Hall for the men, and a long drive from the railroad station planted in Mexican fan palms and Canary date palms. Palm Drive. The successor firm of the architect H. H. Richardson, Shepley, Rutan and Coolidge, carried the buildings. Flat land. A quad. An arch. A church at the center.

Narrator (Kore): Gray Brechin, in his history of imperial San Francisco, describes the result. A symmetrical composition of sandstone, continuous arcades, a Romanesque chapel loosely modeled on Trinity Church in Boston, the whole roofed in red tile, because the Stanfords wanted the buildings to resemble California's missions. Brechin's word for the place is a supermission. At the entrance stood a triumphal arch. The frieze was called The Progress of Civilization. The donors put themselves in it, on horseback, ahead of a locomotive entering the Sierra. Jane suggested the outline of that story and gave herself and her husband featured roles. Augustus Saint-Gaudens, the most famous sculptor in the country, turned her suggestions into the program. White adds a measurement. The arch was San Jose sandstone, more than a hundred feet tall, second in size only to the Arc de Triomphe, a Babylonian eruption, he writes, in the middle of the mission-style quiet. Narrator (Kore): Leland wrote to Jordan a sentence Brechin quotes with brackets, meaning the biographer is supplying a word so the grammar holds. Our graduates are to be missionaries to spread correct ideas of civilization. Hold that next to the sentence from Florence. The children of California shall be my children. And hold both next to the missions the quad was dressed up to recall. Narrator (Kore): The missions sat on crown land the friars held, in the official story, for the people they baptized. Secularization in the eighteen thirties scattered that claim. The Stanford grant says the real property may not be sold or encumbered. Adobe and red tile became sandstone and red tile. Under the missions, Native labor. Under the quad, a railroad fortune graded in large part by Chinese workers paid less than the white crews, which the Central Pacific's own managers admitted under oath. Narrator (Kore): A mission was a discipline. The university, on the face of the grant, was an invitation. Both sexes. Come if you can pass the exam. Hoover, in a few minutes, is the price of that invitation: two suits and a bicycle. Narrator (Kore): Today most of the twenty-one missions are Catholic parishes. La Purísima and Sonoma are state historic parks. The farm is still a private university on land the grant would not let anyone sell. The families who passed through the baptismal books at Mission Dolores and Mission Santa Clara are the Muwekma Ohlone. The campus sits in their watershed.

06. Two Widows

[CUE: singing-bowl chapter chime] [MUSIC: schubert2 from 0s, ducked] [AMBIENCE: wind.mp3, horses.mp3] [MOTIF: Jane, Goldberg Aria phrase]

Narrator (Kore): Two Widows. Narrator (Kore): A few miles from the farm, in the same Santa Clara Valley, another widow was building. The tourist story has the two of them backward. Narrator (Kore): Sarah Lockwood Pardee Winchester signed her name Sallie. She was heir to a rifle. Her husband, William Wirt Winchester, was the son of Oliver Winchester, the New Haven man who turned a shirt business into the Winchester repeating rifle. William died of tuberculosis in New Haven on March seventh, 1881. In 1886 Sarah bought an eight-room farmhouse near San Jose and named it Llanada Villa. Mary Jo Ignoffo, the historian who went through the house's papers, is the source for that purchase. The Stanfords laid their cornerstone on May fourteenth, 1887, the birthday the boy did not reach. Two grieving households, both rich, started monuments about fifteen miles apart, inside roughly a year of each other. The timing is a fact. A

shared motive is not. There is no evidence Jane Stanford and Sarah Winchester ever met, or wrote to each other. Narrator (Kore): The house on Winchester Boulevard, 525 South Winchester, is famous for a ghost. The story goes that a Boston medium told Sarah the rifles had stacked up so many dead that the spirits would haunt her unless the hammers never stopped. Stairs to nowhere. A window in a floor. The record does not show that medium. The name later books give him, Adam Coons, has no trace in her lifetime. It first appears in 1967, in Susy Smith's Prominent American Ghosts. Her companion, Henrietta Servers, said she was not superstitious. She had been raised Baptist, and after 1908 she sat in St. Paul's, an Episcopal church in Burlingame. Call the medium a legend. The broken stairs have a plainer cause. The 1906 earthquake damaged the house, and she largely stopped building. A carpenter added the number thirteens after she was dead. The first printed mention of them is 1929. Narrator (Kore): The séances the documents describe are Jane's. White puts them in the Nob Hill mansion, with the Newmans, friends of the Grants, trying to reach the boy. Mrs. Newman was the chief medium. The boy did not appear. A different child did, a spirit the Newmans called Ensign, and the medium reported that heaven rejoiced in the decision to dedicate the fortune to God and the church. That is a séance claim. It is not a deed. The wills were already signed in Paris. Berner later wrote that after Australia, where the dead did not answer, Jane never again attended a séance. The documented spiritualist in this valley is the railroad widow. Narrator (Kore): What Jane said about Yale is an insult, and it is on the page. While they were bringing the boy's body home, the Stanfords looked at Cornell, Harvard, the Massachusetts Institute of Technology, and Yale. She said graduates of Harvard and Yale were of benefit to no one, including themselves. Her husband, she said, employed DRAMA (Sulafat: Jane Stanford, dry, dismissive, remembering the men on her husband's payroll): "six of them as car-conductors on the Market Street Line now." Narrator (Kore): There is no evidence that ties Leland or Jane Stanford to Skull and Bones. In 1871 Oliver Winchester gave Yale the land for an observatory. The building went up in 1882. Narrator (Kore): The person who connects the two widows is not a medium. He is a lawyer in San Jose, Samuel Franklin Leib. He was a Stanford trustee. In 1899, afraid a constitutional campaign might fail and pull her gift down with it, Jane deeded the entire university endowment, twenty-five million dollars, with no reservations and no trust clauses, to Leib alone. The idea was that if a court ruled the gifts illegal, Leib could deed them back. Twenty-five million dollars in 1899 is, in grocery dollars at about forty to one, roughly a billion dollars now. George Crothers later called it a strange procedure. If Leib had died, or acted on the deed, the university might not have recovered the money. The same Samuel Leib was Sarah Winchester's adviser. He handled her property, her investments, her taxes, and her will, and he settled her estate after she died. History San Jose holds the Samuel Leib Papers, a Sarah Winchester series running from 1865 to 1922. One man in San Jose held the paperwork of the railroad fortune and the paperwork of the rifle fortune. The link is a filing cabinet. Narrator (Kore): In 1909, four years after Jane is in the mausoleum, Sarah sent three hundred thousand dollars, from funds she had entrusted to Yale, to endow the William Wirt Winchester Hospital in New Haven, for the treatment of tuberculosis. It is the disease that killed her husband. The money had come west with the widow. It went back east to fight the illness. Three hundred thousand dollars is thirty years of the salary the Stanfords paid their president. The cure she paid for was a hospital in the city she had

left. Narrator (Kore): On January twenty-third, 1905, nine days after the strychnine in the Poland Spring bottle, Jane went to the Hotel Vendome in San Jose to rest, and to see the trustees. Same town as Llanada Villa. Same town as Leib's office. White's account of the visit is old friends at the hotel, and a fight with her president over the budget. It is not a meeting with Sarah Winchester.

07. The Story of Our Streets

[CUE: singing-bowl chapter chime] [MUSIC: brahms3_3 from 0s, ducked] [AMBIENCE: wind.mp3]

Narrator (Kore): The Story of Our Streets. Narrator (Kore): Jordan, in the same memoir, explains the names on the roads in front of the quad. The road the quadrangle faced, he writes, we called Serra Avenue, for the Franciscan, Junípero Serra. Pronounce him hoo-NEE-peh-ro SEH-rah. Serra founded the first of the California missions. Lasuen Street, Jordan says, is for Firmin Lasuen, Serra's successor, the contained man after the impulsive one. The accent is Lasuén, lah-SWEN. He carried the chain farther, including La Purísima. Other roads took Cabrillo, Portolá, Crespi. The founders wanted the mission era in the directory. They got it. Narrator (Kore): On September thirteenth, 2018, the trustees renamed Serra Mall, the Serra dormitory, and Serra House, and kept Serra Street, the ordinary road toward El Camino Real. A dormitory you are assigned to is not a side street. Keeping one Serra, the university said, stopped the history from being rubbed off the map. Jane Stanford Way took effect on October seventh, 2019. The official address changed from 450 Serra Mall to 450 Jane Stanford Way. Narrator (Kore): In October 2020 they took David Starr Jordan's name off Jordan Hall, the psychology building, Building 420, and moved the statue of Louis Agassiz off the facade. The reason was eugenics. In 1906, the year after Jane died, Jordan became the first chair of the Committee on Eugenics of the American Breeders' Association. Eugenics is the attempt to breed human beings the way the farm bred horses, and to call it science. Serra Street stays so the mission is not pretended away. Jordan's name comes off the lab so the breeding project is not a compliment.

08. Opening Day

[CUE: singing-bowl chapter chime] [MUSIC: scottish3 from 0s, ducked] [AMBIENCE: crowd.mp3, horses.mp3] [MOTIF: railroad stinger]

Narrator (Kore): Opening Day. Narrator (Kore): October first, 1891. The inner quad was still being finished the morning of the ceremony. The university's public history counts five hundred fifty-five students in that first year, about a hundred and thirty of them women, roughly one in four. Richard White's history counts five hundred fifty-nine, mainly from California, which made it the largest college in the far west. Four students, two books. Believe the scale, not the last digit. Jordan spoke to the pioneer class. SPEECH (Schedar: David Starr Jordan, opening day, October 1, 1891, a young president, hopeful and public): "It is hallowed by no traditions; it is hampered by none.

Its finger posts all point forward." Narrator (Kore): Finger posts are signposts. A school with no traditions has not yet had time to lie about itself. Narrator (Kore): Herbert Hoover was an orphan at nine. His father, a Quaker blacksmith in Iowa, died, and then his mother. Relatives passed him along until he ended in Oregon with an uncle who did his duty without much tenderness. Narrator (Kore): A visiting engineer told him mining engineers were scarce and well paid. The East Coast schools were beyond his means. A new college in California was not going to charge tuition. That was the entire plan. In the summer of 1891 he went to Portland and sat the entrance exams. He came up short. Joseph Swain, a Quaker who would teach mathematics at Stanford, gave the exams and watched him attack plane geometry. Quote (Charon, flat documentary): I observed that he put his teeth together with great decision and his whole face and posture showed his determination to pass the examination at any cost. Narrator (Kore): Swain admitted him on the condition that he bone up. On August twenty-ninth, 1891, a few weeks after he turned seventeen, Hoover stood at the Salem station with a ticket for San Francisco. Two suits in a satchel, and a bicycle. He boarded. At the edge of the campus he studied, failed English, and was waved through again on the promise of effort. He was given the first room assignment under the red tile of Encina Hall, and for the rest of his life he liked to say he had been Stanford's first student. There was no electricity and no hot water for weeks after classes began. Narrator (Kore): What the biography shows is the price of a free tuition. A boy can enter without a father and without a fee, and still arrive owning two suits. Free, in the grant, meant the college would not send a bill for instruction. It did not mean the train, the room, or the year of geometry he had missed.

09. A Legacy of Debt

[CUE: singing-bowl chapter chime] [MUSIC: eroica2 from 0s, ducked] [AMBIENCE: telegraph.mp3] [MOTIF: railroad stinger]

Narrator (Kore): A Legacy of Debt. Narrator (Kore): June twenty-first, 1893. Leland Stanford dies. The Panic of 1893 is already on, the start of the deepest depression the country had been through. The Southern Pacific, the basis of the fortune, is missing dividends and cannot pay full interest on its bonds. Within two years of the opening, White writes, the new college is in danger of collapsing. The will leaves the university two and a half million dollars in cash. It leaves about seven million, largely, to his relatives, not to Jane's. The bulk, whatever it is actually worth, goes to Jane, and the university lives on her largesse. Narrator (Kore): Two and a half million, in 1893 grocery dollars, is roughly ninety-three million now. Seven million is roughly two hundred sixty million. The ranches could not be sold. Narrator (Kore): Then the federal government sued. In 1894 it went after the estate for fifteen million dollars and froze the assets, including the bequest to the university. The money was the old government loans to the Central Pacific, the construction debt from the last episodes, come due. Here is the mechanism, because it is the whole Gilded Age in one lawsuit. The Central Pacific was a California corporation, and California law made stockholders personally liable for debts the company did not pay. Stanford and his partners had drained money out of the railroad into their own pockets, sold the hollowed-out Central Pacific to British investors, and had the Southern Pacific, which they still controlled, lease it back. The

British were afraid of that stockholder-liability law, so the partners kept the stock in their own names. In 1894 the government came to the names. Narrator (Kore): Fifteen million dollars. In 1894 grocery dollars that is roughly five hundred eighty million now. DRAMA (Sulafat: Jane Stanford, to a visitor, the boast gone out of her voice): "People think that Governor Stanford left me a very rich woman. I thought so myself, but it now seems that I was left a legacy of debt, trouble and worry." Narrator (Kore): She kept the doors open on the household allowance the court allowed her, on an insurance policy, and on the sale of belongings. She cut salaries ten percent. The faculty, White says, would have left if there had been anywhere to go in a depression. Jordan stayed and defended the Stanfords, because the alternative was a dead college. He called it the cause of higher education in America. The cause, that year, was a widow selling furniture. Narrator (Kore): Her lawyer, Russell Wilson, hired Joseph Choate to argue at the Supreme Court. In 1896 the Court ruled that the stockholders were not personally liable, even if California law had said so when the railroad was incorporated. The government would have to collect from the Central Pacific itself. Official versions treat that ruling as the end of the money trouble. White says it solved nothing. Just before the decision, Choate wrote Jane that a victory might put perhaps forty million dollars in her hands. White calls the guess a considerable exaggeration. His advice was to rewrite the will and to fix a maximum the university should be allowed to hold. Her public promise had been to pour the fortune in. Her lawyer was telling her to pour less. Narrator (Kore): Probate is a court supervising a dead person's property. The boy was ten years dead. The university was three years old. The fortune was a lawsuit with a palm drive.

10. The Ballot Box

[CUE: singing-bowl chapter chime] [MUSIC: brahms1_1 from 0s, ducked] [AMBIENCE: crowd.mp3] [MOTIF: railroad stinger]

Narrator (Kore): The Ballot Box. Narrator (Kore): On May thirty-first, 1899, Jane sold the Southern Pacific stock. The reported price was eleven million dollars. Russell Wilson advised her. George Crothers later called the sale one of the great disasters to befall the university. He said it was seven million dollars less than the Speyer banking house got when it sold the same stock, the next year, to E. H. Harriman. In grocery dollars of 1899, seven million is roughly two hundred eighty million now. Eleven million is roughly four hundred forty million. She took the smaller pile. Harriman, the man who already ran the Union Pacific, took the railroad her husband had helped build. That is the East Coast hand. It comes back in the newspapers the week she dies. Narrator (Kore): The same day as the sale, she signed what Crothers called the Great Deed, a huge gift of most of her estate back to the university, and also, in his phrase, a great but defective grant. A clause, inserted by Mountford Wilson, gave her the power to grant, bargain, sell, convey, or lease all the property she had just given. She kept so much control that Crothers thought the gift had, in effect, not happened. She treated it lightly enough, he said, to pull millions back out. Remember the dead hand. A forever clause can be walked around by a later deed that says the donor may sell after all. The lock and the key were in the same folder. Narrator (Kore): November sixth, 1900. The amendment passed 137,667 to 67,737, about two to one. It confirmed the trusts and let

the legislature exempt Stanford's lands and trust funds while they were used for education. California students were not to be charged tuition unless the legislature later allowed it. A later education code did allow it. The 1900 promise was real, and it was not eternal. A tax exemption is a public subsidy that does not look like a check. The yes votes were paying for a private university by not collecting from its land. Narrator (Kore): Five days later, on November eleventh, 1900, Jordan accepted a resignation Edward Ross had written in June. He had held it, hoping the situation would change. He accepted it, he said, with great reluctance, and he praised Ross as a teacher and as a man. The timing is the story. Ross and others said that if the dismissal had been news before the election, the amendment might have failed. After the vote, Ross called in the San Francisco reporters and blamed Jane. She disliked his politics. He had talked about coolie labor, the era's slur for Chinese workers, and about municipal ownership, the idea that a city should own its own utilities instead of a franchise company. On November fourteenth the Chronicle ran the headline. NEWS (Sadachbia: the San Francisco Chronicle, November 14, 1900, a morning paper enjoying itself): "Dr. Edward Ross Forced Out of Stanford University." Narrator (Kore): Jordan had not yet read Ross's statement when the reporters reached him, and he praised Ross on the record. The praise ran next to the accusation. Ross thanked his great chief. White calls it a Judas kiss. The thanks put the firing on Jane. Narrator (Kore): Academic freedom is the principle that a professor's job does not end when a donor dislikes the lecture. It was not a statute in 1900. It was an argument Ross forced into the papers by refusing to go quietly. He lost the job. He won the headline. The amendment was already in the constitution. Narrator (Kore): The same years, she tried to close a different door. As women became a larger share of the students, about two in five by 1899 in the university's own magazine, she moved to cap them. The grant had said a university for both sexes. Crothers stopped her from an instruction that would have told the trustees to break the grant, because breaking it risked the whole trust. He got her down to a maximum of five hundred women. In 1904 she still called coeducation unsettled. Five hundred is a number with a law under it. Both sexes was the founders' sentence. Five hundred was the widow's revision, and the lawyer's compromise, and it lasted a generation.

11. The Dowager

[CUE: singing-bowl chapter chime] [MUSIC: mozart40_2 from 0s, ducked] [AMBIENCE: telegraph.mp3, wind.mp3] [MOTIF: railroad stinger]

Narrator (Kore): The Dowager. Narrator (Kore): Richard White, telling the Ross firing, gives Jane Stanford a title. She played the dowager queen. It was her university. The dismissal was her due. Narrator (Kore): There is another dowager in this hour, and no letter connects her to Jane. Western reports said the Empress Dowager of China, Cixi, had the soul of a tiger in the body of a woman. Pronounce her tsz-shee. The tiger is the papers' word. It is not a court title. She had held the real power of the Qing throne since 1861. In 1898 she put the reforming emperor under house arrest and moved to stop foreign businessmen from treating her realm as a buffet. In the summer of 1900 she backed the Boxers, the martial, magical movement that meant to drive the foreigners out. Narrator (Kore): June 1900. Tianjin. Herbert Hoover is twenty-five, the orphan who

came to Encina with two suits and a bicycle, now a mining engineer in the foreign settlement. On June fourteenth a Boxer force, described as twenty-five thousand strong, hits the Chinese city and burns the French cathedral. About two thousand foreign soldiers hold the settlement, protecting seven hundred foreign civilians, three hundred of them women and children, and thousands of Chinese Christians. Hoover helps build barricades at the gates out of sacks of sugar and grain. Imperial troops sent to scatter the Boxers join them and turn their guns on the compound. He fights fires and carries food on his bicycle, chewing nuts without shucking them. His wife, Lou, will not wait in the basement. She runs her own errands with a .38 Mauser on her hip. Narrator (Kore): Rail and telegraph are cut. At Stanford, faculty and students read that some of their best alumni are probably dead. Quote (Charon, flat documentary): "It is almost certain that they have fallen victims to the savage ferocity of the Mongolian hosts, unless they managed to make their escape previous to the bloody massacre, which is not likely." Narrator (Kore): They had not. Whyte's account says the casualties inside the foreign compound were light. Narrator (Kore): The coal was the point. Hoover considered the Kaiping mines, northeast of Tianjin, the only successful mining venture in China by Western standards. Nine thousand men worked three collieries. His employer, Chang Yen-mao, directed them. Chinese law kept the mines with the emperor, and Chang could not sell them to foreigners. During the siege the Boxers stopped the coal, the share price fell, and Russian and Japanese troops were said to be planting flags on the pits. The foreigners in the settlement arrested Chang. They thought he was negotiating his way out. Whyte notes there was no firm evidence. They locked him up anyway. From that confinement he signed a power of attorney and told Gustav Detring to put the company under the British flag on the best terms he could get. Narrator (Kore): Hoover and Detring wrote the deal and backdated the papers, so the reorganization would look as if it had been agreed before the uprising. A million shares, one pound each. Existing Chinese shareholders would receive three hundred seventy-five thousand of them. What the London promoters would take for themselves was not written down. Hoover would not let the deed go straight to his boss, Charles Algernon Moreing. He had it made out to himself, as trustee. He had already told Moreing that without absolute control the game was not worth the candle. Narrator (Kore): Whyte calls it the biggest mining transaction in the history of China. Harris quotes the historian Ian Phimister: the previous Chinese owners were defrauded of their property.

12. The Church of One

[CUE: singing-bowl chapter chime] [MUSIC: aria from 0s, ducked] [AMBIENCE: church.mp3] [ERA: phono] [MOTIF: Jane, Goldberg Aria phrase]

Narrator (Kore): The Church of One. Narrator (Kore): Jane called Memorial Church her Koohinoor, her spelling of the Koh-i-noor, the mountain of light, a famous diamond. The facade reads, Erected to the Glory of God and the Loving Memory of My Husband Leland Stanford. It is nondenominational. She chose the quotations on the walls herself. She wanted Christianity with the dogma stripped off, a church any student could enter. White's verdict is that the result spoke mainly to a sect of one. Jane Stanford. Narrator (Kore): She painted a line she attributed to her husband: the best form of religion is trust

in God and a firm belief in the immortality of the soul. Jane did not advertise her spiritualism. White's sentence is that she did not hide it from anyone who knew where to look. Narrator (Kore): Her brother-in-law, Thomas Welton Stanford, in Australia, funded a fellowship in psychic research and gave the university his apports, objects he believed spirits had delivered into a séance. White notes that pieces of that gift are still in the collection.

13. The Catholic Turn

[CUE: singing-bowl chapter chime] [MUSIC: brahms3_3 from 40s, ducked] [AMBIENCE: church.mp3] [MOTIF: Jane, Goldberg Aria phrase]

Narrator (Kore): The Catholic Turn. Narrator (Kore): George Crothers worked for her without a salary, so she would not treat him as an employee she could dismiss at will. In 1902 he became a trustee, a seat some people had expected to go to Herbert Hoover. From 1899 to 1905 he was afraid she would hand the university away. Narrator (Kore): Archbishop Patrick Riordan had her ear. In 1899 he tried, and failed, to stop her signing the Great Deed. She thought about a boys' school, about giving the university to the Catholic Church, about giving the endowment to Columbia, which White calls a cruel dig at Jordan, because Leland had first offered the presidency to Columbia's Nicholas Murray Butler, and about dividing the fortune among her heirs. Narrator (Kore): By 1902 she was a catechist, taking instruction to enter the Roman Catholic Church. Crothers told her a hierarchical church, grown rich on the sacraments, would betray her hope of a purer Protestantism. She postponed the conversion. She did not drop the wish that Stanford would feel more like a Catholic college. She sent for Crothers in the middle of the night with stories of students in the shady lanes. The stories came back false. Narrator (Kore): She came to see the women students as a danger to the men's celibacy, and she thought the gossip arrived through her Catholic friends. She told Crothers it might be better if she resigned as surviving founder before she made some foolish change. She did not convert. Crothers answered the midnight summons and kept the deed intact. The church at the center of the quad stayed hers.

14. The Expanding Peoples

[CUE: singing-bowl chapter chime] [MUSIC: brahms1_1 from 0s, ducked] [AMBIENCE: crowd.mp3, telegraph.mp3] [MOTIF: railroad stinger]

Narrator (Kore): The Expanding Peoples. Narrator (Kore): February twenty-fifth, 1898. Washington. The Secretary of the Navy, John D. Long, has gone home for the afternoon. Theodore Roosevelt, the assistant secretary, is acting secretary for those hours. He cables Commodore George Dewey, in Hong Kong. The cable is marked secret and confidential. Keep full of coal. In the event of declaration of war Spain, Dewey's duty will be to see that the Spanish squadron does not leave the Asiatic coast, and then offensive operations in the Philippine Islands. The next day Long writes that the very devil seemed to possess Roosevelt yesterday afternoon. Long leaves the order standing.

Narrator (Kore): Roosevelt did not start this war, and he did not blow up a ship. The battleship Maine exploded in Havana harbor on February fifteenth, killing two hundred sixty-six men. Navy inquiries in 1898 and 1911 blamed a mine. A 1976 study led by Admiral Hyman Rickover concluded the likeliest cause was a coal-bunker fire next to a magazine. It is still debated. President William McKinley asked Congress for the war. Congress declared it, in April. The Navy's own historians say the story that Roosevelt engineered the war by himself has largely been refuted. A war plan from 1896 already pointed at Manila. Gray Brechin puts it plainly. The Navy Department had planned the seizure of the Philippines two years before Gridley was ready to fire. What Roosevelt did, on one afternoon, was aim the fleet and refuse to whisper. Narrator (Kore): May first, 1898. Dewey's squadron slides past the old fort at Corregidor into Manila Bay. He tells his captain, you may fire when you are ready, Gridley. Six hours of shelling sinks the Spanish fleet. Narrator (Kore): The peace is a purchase. The Treaty of Paris, signed December tenth, 1898. Spain cedes Puerto Rico and Guam, gives up Cuba, and the United States pays twenty million dollars for the Philippines. In 1898 grocery dollars that is roughly eight hundred million now. The Senate ratifies the treaty on February sixth, 1899, fifty-seven to twenty-seven, barely the two-thirds a treaty needs, and two days after the shooting has started in Manila. The money went to Spain. It did not go to the Filipinos. Narrator (Kore): Hawaii comes in the same summer, by joint resolution, the Newlands Resolution, on July seventh, 1898, while the war is still on. Francis G. Newlands, whose name is on it, was an heir by marriage to the Sharon silver fortune, and he laid out Burlingame and Hillsborough, the suburbs just up the peninsula from the farm. In October his brother-in-law, Hall McAllister Jr., writes that keeping the entire Philippine group would prove of incalculable benefit to San Francisco. A colony, argued from a peninsula law office. Narrator (Kore): The war after the treaty begins February fourth, 1899. Roosevelt, president by then, will declare it over on July fourth, 1902. Fighting continues after the speech. The State Department's figures: more than forty-two hundred American combatants dead, more than twenty thousand Filipino combatants, and as many as two hundred thousand Filipino civilians dead of violence, famine, and disease. At the peak, seventy-five thousand American troops are in the islands. Americans called it an insurrection. The word keeps the other side from having a war. Narrator (Kore): Both sides commit atrocities, and the government's own historians say so. American forces at times burned villages, reconcentrated civilians, and used torture. Filipino fighters tortured captured soldiers and terrorized civilians who cooperated with the Americans. One American method has a name, the water cure. On November twenty-seventh, 1900, Major Edwin F. Glenn used it on Tobeniano Ealdama, the town president of Igbaras, on Panay. Glenn was court-martialed, convicted, and given a one-month suspension and a fifty-dollar fine. General William Shafter told a Chicago paper it might be necessary to kill half the Filipinos so the remaining half could be advanced to a higher plane of life than their present semi-barbarous state. On September twenty-eighth, 1901, at Balangiga on Samar, townspeople attacked a company of the Ninth Infantry. Dozens of the soldiers died. The published counts of the American dead do not agree. About twenty-eight villagers were killed in the attack. Afterward, Brigadier General Jacob Smith told a major he wanted no prisoners. Kill and burn. The more you kill and burn, the more you please me. Asked the age limit, he said everything over ten, and he wanted Samar turned into a howling wilderness where even the birds could not

live. Major Littleton Waller partly countermanded him. We are not making war on women and children. Smith was court-martialed for conduct to the prejudice of good order, sentenced to be admonished, and Roosevelt retired him. Narrator (Kore): David Starr Jordan campaigned in public against annexing the Philippines. His reason was not a sermon on consent. Harris records the argument. The colonies were incapable of civilized self-government. They were not habitable by Anglo-Saxon people. The home cannot endure in the climate of the tropics. Anti-imperial, and racist. Both are in the record. Mark Twain, a vice president of the Anti-Imperialist League from 1901, asked whether we would go on conferring our civilization upon the peoples that sit in darkness. Narrator (Kore): May thirteenth, 1903. Union Square, San Francisco. Roosevelt dedicates the Dewey Monument, a Roman column of the kind the old empire built for victories at sea. The ships' names are cut in the base. Olympia, Dewey's flagship, is first. He tells the crowd that the proper place of Americans is with the great expanding peoples, with the people that dare to be great. The civilian dead are already in the ground.

15. Natural Causes

[CUE: singing-bowl chapter chime] [MUSIC: tragic from 30s, ducked] [AMBIENCE: surf.mp3, telegraph.mp3] [MOTIF: Jane, Goldberg Aria phrase]

Narrator (Kore): Natural Causes. Narrator (Kore): January fourteenth, 1905. At the Nob Hill mansion she drinks her bedtime glass of Poland Spring water, tastes something wrong, and makes herself throw up. A pharmacist finds a lethal dose of strychnine, the same poison sold as rat poison. White distinguishes the attacks. January was a massive dose. Honolulu, six weeks later, was a small precise amount of the pure alkaloid. She survives January. In February she leaves. Narrator (Kore): February fifteenth, 1905. The steamship Korea, San Francisco for Hawaii and Japan. Thirteen days before she dies. The dock is crowded, and there are military bands, because General Arthur MacArthur is aboard, commander of the Department of the Pacific, military governor of the Philippines from May 1900 to July 1901, eased out after he clashed with William Howard Taft. The passenger list spells Bertha Berner as Benner and lists Yokohama. On the voyage Jane talks freely about the poisoned Poland Spring water, to MacArthur and his wife, and to Charles Sidney Dole, a Chicago grain merchant. Nothing in the record shows that anyone later asked the general what she said. She has packed seventy-five thousand dollars in jewels, roughly two million eight hundred thousand in today's grocery dollars. She is wearing a small endowment, and she is running from a bottle. Honolulu, when she docks, is a territory whose governor, George Carter, Theodore Roosevelt appointed in 1903 after consulting, among others, David Starr Jordan. Narrator (Kore): February eighteenth, 1905. Three days after she sails, ten days before she dies. In the Blue Room of the White House, Liang Cheng, the Chinese minister, gives Roosevelt a portrait of Cixi. An American painter, Katharine Carl, made it for the St. Louis fair the year before. Roosevelt accepts it for the United States. White's name for Jane, in the Ross fight, was the dowager queen. On this afternoon the other dowager's face is a diplomatic gift, and the Palo Alto dowager is on a ship, talking about a bottle. Narrator (Kore): She docked looking, the phrase in the record is, well and

hearty. The chemists found strychnine in the soda and in the body. A trace from the cascara capsules could not, the physicians said, explain the death. The coroner's jury delivered its verdict on the evening of March ninth. The newspapers printed it the next day. Quote (Charon, flat documentary, a clerk reading an inquest): "Jane Lathrop Stanford came to her death at Honolulu, Island of Oahu, Territory of Hawaii, on the twenty-eighth day of February, A.D. 1905 from Strychnine poisoning, said strychnine having been introduced into a bottle of bicarbonate of soda with felonious intent by some person or persons to this Jury unknown and of the contents of which bottle Jane Lathrop Stanford had partaken." Narrator (Kore): An inquest can name a poison and refuse to name a hand. This one did both. Judge William F. Stanley, for the estate, sat next to the deputy and whispered questions. The day before Berner testified, it became public that she had inherited fifteen thousand dollars, roughly five hundred sixty thousand now. Ariel Lathrop called her a medium. There is still no transcript of her testimony. White's portrait of her is five adjectives: bold, flirtatious, charming, quick, and calculating. None of them is a conviction. The papers also chased a Chinese employee, Ah Wing. Access is not a verdict. White does not name a killer. Narrator (Kore): Hawaiian officials decided the murderer was in San Francisco, and that Honolulu would not pay for a trial. High Sheriff Henry told the press that Jordan, Timothy Hopkins, and the detectives had come to cover a crime, not to solve one. The night before the funeral, Jordan and Hopkins drew up a statement. Natural causes. Advanced age, unaccustomed exertion, a surfeit of unsuitable food, and the unusual exposure of the picnic, perhaps aggravated by strychnine in the capsule and a small amount in the bicarbonate, which they called a pharmacist's error. No one found a bicarbonate on the market that contained strychnine. They did not release the report they said held their evidence. Doctors recruited in San Francisco called it angina pectoris, a spasm of the heart. Dr. Wood, in Hawaii, called that theory simply rot. Narrator (Kore): The conflict of interest is the whole case. Jordan's university inherited if the death was ordinary and the will stood. A murder investigation embarrassed the estate, invited a fight over the will, and raised the older question of the January bottle, which had not been handed to the police cleanly. The man with the most to lose from the word murder was the man who issued the sentence natural causes. Harris, a century later, writes that it is no longer particularly controversial to suggest Jordan was part of a conspiracy, most likely with Bertha Berner, and that no one has produced a definitive proof. Suggest. Not prove. The chemists found strychnine. The president found a picnic. Narrator (Kore): The railroad found a newspaper. W. H. Mills, an old creature of the Associates, paid off a wide stretch of the California press, including the San Francisco Post, on behalf of the Southern Pacific, which by then was Harriman's. The Post denounced the murder stories and praised Jordan and Wilson for refusing them. Wilson and Jane's brother, White writes, had presumably asked for the favor, and Mills granted it. The Post then spoiled the favor, a little, by adding that private detectives are more often hired to hush a scandal than to solve one. Harriman's road did not need to invent Jane's death. It needed the word murder off the front page of a paper it already paid. Narrator (Kore): After the service the hymn was Nearer, My God, to Thee, and the choir sang the alma mater. The casket went into the steel chest. The next day the chest went into the sarcophagus, between husband and son, and was sealed. For twelve years the marble had waited: Jane

L. Stanford. Born in Mortality, August second, 1828. Passed to Immortality, and a blank. On March sixteenth a workman cut the date. February twenty-eighth, 1905.

16. Fourteen Months

[CUE: singing-bowl chapter chime] [MUSIC: scottish3 from 20s, ducked] [AMBIENCE: wind.mp3] [MOTIF: Jane, Goldberg Aria phrase]

Narrator (Kore): Fourteen Months. Narrator (Kore): [SILENCE] Count forward from the date the workman cut into the marble. Fourteen months later, on the morning of April eighteenth, 1906, the ground is going to move. The spire she built, and the laurel tie from the railroad's last spike, are in the path of it. Narrator (Kore): Five days after that, Cixi orders one hundred thousand taels for the sufferers in San Francisco, as a mark of friendship, and a separate forty thousand for the Chinese residents, sent to the Chinese minister in Washington. The American minister in Peking is handed the hundred-thousand-tael note. Roosevelt has decided to refuse the gifts of foreign governments. That note comes back, with word that this is not a slight. The forty thousand is the sum that stays on its errand. Narrator (Kore): Six months after that, the San Francisco school board will order the city's Japanese schoolchildren into the Oriental School in Chinatown. Chinese children are already there. They have been sent to separate schools since 1885, after Mamie Tape won the right to a public education and the legislature answered by allowing separate schools for children of Mongolian or Chinese descent. Roosevelt will call the 1906 order a wicked absurdity, because Japan has a navy and a treaty. The Chinese children stay segregated. That fight is the next hour. The dead hand says the land cannot be sold. It does not say the sandstone cannot crack.

The Engineer's Century — full audio script (voice/cue markup)

The Engineer's Century — full audio script (voice/cue markup)

Episode 5 of 8. The Engineer's Century (1905–1945). Narrator = Kore. Bland documentary/official quotes = Charon. Other voices are listed per line. [HARD] = music gated out. [SILENCE] = 2 s of silence before a reveal. Reuse only cue, music, and ambience names from the Spike in the Museum pipeline. A note on sources, the pronunciation sheet, and the fact-check live in the review file and are not narrated.

Pronunciation, not narrated: Amadeo Pietro Giannini, ah-mah-DEH-o pee-EH-tro jah-NEE-nee. Sons of Gwalia, gWAH-lee-ah. issei, EES-say. nisei, NEE-say. klystron, KLY-

stron. bionomics, by-oh-NOM-iks. Comité, ko-mee-TAY, not used on air.

01. The Spire and the Plank

[CUE: singing-bowl chapter chime] [MUSIC: hebrides from 30s, ducked] [AMBIENCE: fs_636552.mp3, church.mp3]

Narrator (Kore): The Engineer's Century. Narrator (Kore): April 18, 1906. Five twelve in the morning. Narrator (Kore): This hour covers forty years, and we are not going to march them. We will land on six years and say so when we jump. Nineteen six. Nineteen sixteen. Nineteen nineteen. Nineteen thirty-two. Nineteen thirty-nine. Nineteen forty-two. Hold those six. Everything else is the road between them. Narrator (Kore): The Geological Survey marks the foreshock at five twelve. The break came twenty to twenty-five seconds later and tore nearly three hundred miles of the San Andreas. William James, the Harvard philosopher, was in a campus bedroom. He timed his room at five thirteen. LETTER (Puck: William James, sixty-three, writing home the week of the earthquake, shaken and precise): "My room was shaken like a rat by a terrier." Narrator (Kore): The wooden houses held. The chimneys came down. The higher university buildings, he wrote, were largely piles of ruins. Narrator (Kore): At the center of the Quadrangle stood Memorial Church, opened by Jane Stanford in nineteen three. It had an eighty-foot spire, twelve-sided, on a crossing that was not tied hard enough to the roofs around it. The crossing moved by itself. The spire fell through the chancel roof. The mosaic on the north face blew out into the Quad. The apostles commissioned for the altar were smashed. Hundreds of feet of arcade lay flat. The spire was not put back. You can stand in that Quad today and see a church whose tower stops where the earthquake cut it. Narrator (Kore): [HARD] The same morning, in San Francisco, a different kind of monument burned. Narrator (Kore): The golden spike from last hour sits in the art museum at Stanford. Gordon Chang records the other half. The polished laurel tie it was driven into burned in the San Francisco fire that followed this quake. Narrator (Kore): [SILENCE] [HARD] The object that marked the ceremony survived. The object the workers built toward did not. Narrator (Kore): On Nob Hill the Stanford mansion went the next day. Ah Wing, the family's old cook and the caretaker on the hill, locked the door, walked to Jones Street, looked back, and saw the house in flame. The university's earthquake timeline puts that loss on April nineteenth. The fire finished what the fault had started. Narrator (Kore): The other man that morning was Amadeo Pietro Giannini, A. P., thirty-six, born in San Jose in eighteen seventy. The Giannini Foundation says he left school at thirteen. He did not go to Stanford, and he did not go to Berkeley. Hold that. The building with his name is not where the legend puts it. Narrator (Kore): On October seventeenth, nineteen four, a remodeled saloon in North Beach opened as the Bank of Italy. Italian-speaking tellers. Small deposits. Loans to tradesmen. By the end of nineteen five, assets above a million dollars. Then the ground moved. Narrator (Kore): The story goes like this, and you have probably heard it. Giannini was thrown from his bed in San Mateo. He borrowed a produce wagon, drove into the city, pulled about two million dollars in gold, coins, and securities out of the bank, hid the money under vegetables, and rolled past the looters. The next days, while other bankers waited for their superheated vaults to cool, he set a wooden plank across two barrels on a North

Beach wharf, Washington Street, and lent money on a face and a signature. Narrator (Kore): Time has told that version. So has Berkeley's own page for Giannini Hall. Two million. Vegetables. A plank. A face and a signature. Narrator (Kore): The record is messier, and the mess is the lesson. The Office of the Comptroller of the Currency, the federal office that charters national banks, tells the same morning with a different number. Eighty thousand dollars in coins, under crates of oranges, and then loans on a handshake from a wharf. The scholarly biography later historians lean on says two of his stepfather's produce wagons, loaded with all the cash, gold, and records they would hold, and a plank on two barrels on the Washington Street wharf within two days. Narrator (Kore): The plank, the wagon, and the loans to people with nothing left to pledge are in every serious account. The two million is the legend. The eighty thousand is the comptroller. Narrator (Kore): The dollars in this series all use one method. The Minneapolis Fed's consumer price index, nineteen sixty-seven equals one hundred, puts nineteen six at twenty-seven and its twenty twenty-six estimate at one thousand four point eight. Before nineteen thirteen those figures are estimates, so the early ones are rough. Two million in nineteen six is roughly seventy-four million in grocery money now. Eighty thousand is roughly three million. Narrator (Kore): A railroad is a chokepoint. One line, and you pay to pass. A bank is a chokepoint too, if it only lends to people who already have collateral. Giannini did the opposite, and he gave it a shape. Narrator (Kore): Character lending is a loan on the person, not the pledge. A face, a signature, a handshake. Branch banking is one bank with many offices, instead of a separate little bank in every town. By nineteen eighteen the Bank of Italy was the first with branches across a whole state. In November nineteen thirty it became Bank of America. In nineteen thirty-seven that bank lent Walt Disney the money to finish Snow White, and it stayed with the studio through Fantasia. At Giannini's death in nineteen forty-nine the Comptroller could call it the largest bank in the world. More than five hundred branches, about six billion in assets, roughly eighty-four billion in today's grocery dollars. Narrator (Kore): A saloon to the largest bank on earth, and the university that will spend this hour ranking human beings never taught him. His name goes on a hall at Berkeley in nineteen twenty-eight because he wrote a check. First we go and get the engineer. Narrator (Kore): This is Cardinal. Episode five. The Engineer's Century.

02. The Orphan and the Mine

[CUE: singing-bowl chapter chime] [MUSIC: borodin from 0s, ducked] [AMBIENCE: wind.mp3, horses.mp3]

Narrator (Kore): The Orphan and the Mine. Narrator (Kore): Jump back fifteen years, to a campus that is not finished. Narrator (Kore): Herbert Hoover, the Iowa orphan last hour let into Stanford because he passed mathematics and failed the rest, arrived in the summer of eighteen ninety-one, before opening day, living on the farm while the sandstone was still going up. He ran a laundry agency and a lecture bureau. He wrote a friend that he was working awful hard and had schemes for making more. The hunger was not a joke. Narrator (Kore): At the start of his second semester a geologist named John Casper Branner arrived to chair the department. Hoover could type. Branner hired

him as office assistant at twenty-five dollars a month, and Hoover kept the job until he graduated. Twenty-five dollars in the eighteen nineties, roughly, is on the order of a thousand dollars a month in our grocery money. It was not a living. It was a door. Hoover's own rule, told to a classmate, was simple. Do your work so they notice it, and always be on the job. Narrator (Kore): He graduated in the pioneer class of eighteen ninety-five, in geology, with a degree and almost no money. His first job was pushing ore carts in a Nevada mine. The degree had not delivered a career. It had delivered a person a London banker could be talked into trusting. Narrator (Kore): Call him the engineer as a fixer. In the eighteen nineties, investors in the City of London, the financial district, not the city of churches, were putting money into holes in Western Australia, South Africa, and China. They could not see the holes. Somebody had to go and look, price the risk, buy the machinery, hire the men, keep books an auditor in London could read, and talk to whatever government claimed the ground. The scarce thing was not dynamite. It was a report a stranger would believe. A degree from a new university on a horse farm was, for a few years, a strange but usable proof. Narrator (Kore): The recommendation that moved him was not an application he filled out. Louis Janin, a San Francisco mining engineer, passed him to the London firm of Bewick, Moreing and Company. In eighteen ninety-seven, at twenty-three, they sent him to Western Australia. Narrator (Kore): Coolgardie and Kalgoorlie were inland, hot, short of water, and full of money. He told the firm which holes to buy. He recommended the Sons of Gwalia. The name is Welsh. The mine made his reputation. Narrator (Kore): Jump to nineteen five, the year Jane Stanford dies in Honolulu, and eight years after Hoover first went down that mine. The parliament of Western Australia appointed a Royal Commission on the Ventilation and Sanitation of Mines. A royal commission is not a royal visit. It is a government inquiry with the power to compel testimony and write a judgment. This one was hard on most of the operators in the goldfields. Workers, the record of the inquiry said, were unduly exposed to dust and fumes and falling rock. At the Sons of Gwalia, the commission found the air notably bad. Narrator (Kore): Kenneth Whyte, in his life of Hoover, notes that the same parliament was launching six or eight royal commissions a year, some of them on questions as small as stone or stucco for the new law courts. A government that inquires into everything will inquire into you. That takes some of the sting out. It does not take the finding out. The question of how a young engineer got his results followed him. Narrator (Kore): He was twenty-three at the mine. The commission spoke when he was thirty-one. By then the firm had sent him to China. In February eighteen ninety-nine he married Lou Henry in Monterey, and they sailed the next day. Narrator (Kore): Lou Henry is not a wife in the margin. A guest lecture by Branner in Monterey captured her. She talked her parents into geology, the only woman in the program, degree in eighteen ninety-eight. Whyte sets their first meeting in Branner's lab. She expected someone brawny. What walked in was speechless, hair uncombed, tanned from summer work. She did not look away. Narrator (Kore): Two people who can read the same rock. When the war comes, the skill that matters is the mining engineer's list. Survey, account, price the risk, move food, talk to governments that hate each other. He had practiced it underground. Narrator (Kore): By the time the guns started he was a London partner on London Wall, two doors from the rooms where he was about to try to feed a country. The orphan from Iowa had become a man the City used to look at the underside of the earth. Narrator (Kore): Jump forward.

Not to the war yet. To the boardroom. The mine boss is about to sit down at the table that owns the farm.

03. The Trustee

[CUE: singing-bowl chapter chime] [MUSIC: aase from 0s, ducked] [AMBIENCE: fs_42139.mp3] [ERA: phono]

Narrator (Kore): The Trustee. Narrator (Kore): Autumn, nineteen twelve. Herbert Hoover joined the board of trustees of Leland Stanford Junior University. Narrator (Kore): A trustee is a person who legally holds an institution for somebody else. At Stanford that somebody was written down in eighteen eighty-five as the children of California. The board hires the president, watches the endowment, and decides what the founder's forever clauses still mean. If you want to know who a university belongs to, read the board. Narrator (Kore): Whyte's sentence on Hoover's arrival is dry, and I am going to give it to you whole. QUOTE (Charon, flat documentary): He joined the Stanford board of trustees in the autumn of 1912 in circumstances as cloudy as the awarding of his geology degree seventeen years earlier. Narrator (Kore): The cloud was a residency rule. Trustees were supposed to live close enough to do the work. Hoover escaped it by claiming he had immediate plans to move to San Francisco. He knew he was at least a year away from leaving London. He still called the university, in his own phrase, the best place in the world. He wanted his hand on it. Once he was on the board he said the place was deficient in a very great proportion of the departments, and that too many of the professors were not good enough. The orphan had come home as a critic with a vote. Narrator (Kore): Set him next to another man who came for a different reason, six years earlier, and you can see the two ways a famous outsider sits down at Stanford. Narrator (Kore): Jane Stanford had wanted William James, the most famous philosopher in America, and chair of the American branch of the Society for Psychical Research. He went to séances looking for one honest medium, what he called a white crow. The San Jose Mercury announced him in February nineteen five. Less than a week later she was dead in Honolulu. Narrator (Kore): James came anyway, in January nineteen six, nearly sixty-four and tired of lecturing. He told his wife Alice why. LETTER (Puck: William James, writing to his wife Alice, January 1906, wry and mercenary): "If I can only get through these next 4 months and pocket the $5,000 I shall be the happiest man alive." Narrator (Kore): Five thousand dollars for one term. In our grocery money, using the nineteen six index, that is roughly one hundred eighty-six thousand dollars. James was being paid a young engineer's youth for four months of talk. Narrator (Kore): His only class was a history of philosophy. Three hundred students enrolled, and another hundred and fifty visitors sat in. He wrote that they did not know what hypothesis and concept meant, and that he found it hard to aim too low. One of his few mentions of the woman who had wanted him there was a thank-you. The holiday for her death let him cancel a class. Narrator (Kore): Then the earthquake let him go. He walked into San Francisco with a colleague looking for the colleague's sister. A whole population stood in the streets with whatever they had pulled from houses about to burn. He called the fires most beautiful in the effulgent sunshine. By early May he was back in Cambridge. Before he left he wrote Horace Davis that the faculty did not trust the

people governing the university, and that Mrs. Stanford's death had been a chance for a new start which nobody had taken. Jordan answered that James was not speaking for the responsible members of the faculty. Narrator (Kore): Two outsiders. One came for five thousand dollars and told the truth on the way out. One came onto the board while still living in London and stayed for the rest of his life. Governance is who is allowed to remain in the room after the visiting professor's train has gone. Narrator (Kore): Jump four years, to nineteen sixteen. The university is about to put a number on a child.

04. Bionomics

[CUE: singing-bowl chapter chime] [MUSIC: mozart40_2 from 0s, ducked] [AMBIENCE: fs_42139.mp3]

Narrator (Kore): Bionomics. Narrator (Kore): Nineteen sixteen. A Stanford psychologist named Lewis Madison Terman published a book called The Measurement of Intelligence. Inside it was a test. He called the test the Stanford-Binet. He called the number it produced an intelligence quotient. The world shortened that to IQ. Narrator (Kore): Say the word bionomics once, because David Starr Jordan loved it. He was a great ichthyologist, the first president of this university, a pacifist at real cost, and a eugenicist. Bionomics, in his mouth, was the claim that one science could cover heredity, history, economics, and ethics, and then tell a society what to do. Narrator (Kore): Eugenics is the idea that a nation should breed human beings the way a farmer breeds livestock, encouraging the births it calls fit and discouraging or preventing the births it calls unfit. In those decades this was mainstream progressive science. University presidents held it. So did public health officers. The biology was wrong. The structure was useful. If poverty and what they called feeblemindedness are mostly inherited, then the hierarchy you see is nature, and a university endowed by a railroad monopoly is not defending a class. It is defending the species. Narrator (Kore): California wrote it into law. In nineteen nine the state authorized the medical superintendents of state institutions to operate on inmates when, in the superintendent's judgment, it would improve the person's physical, mental, or moral condition. No consent from the patient. No court order. Over the following decades California sterilized more people under that authority than any other state. Later counts put the state's share at roughly a third of the recorded sterilizations in the United States. The counts do not agree with each other, because the people this was done to were not carefully counted. The disagreement is part of the harm. Narrator (Kore): Lewis Terman arrived at Stanford in nineteen ten, with a wife, Anna, and two small children, Fred and Helen. Malcolm Harris notes why they were in California at all. Terman's young adulthood was marred by a long and debilitating course of tuberculosis, and the illness required the move to a mild climate. Fred, the son who will build the industry, came west because his father could not stay east and breathe. Then the father set out to measure minds. Narrator (Kore): The test he adapted was not his invention. In nineteen five, in France, Alfred Binet had built a test to find children who were falling behind in school, so they could be helped. Binet warned against using it as a measure of inborn worth. Terman and his Stanford team did the thing Binet had warned against. They translated the items, reset the age norms for American schools, and forced the scores

into a curve whose average would always be one hundred. Mental age divided by actual age, times one hundred. A single number for a whole child. Narrator (Kore): Stanford Magazine, looking back, is blunt. Terman believed general intelligence was an original endowment. Genetics alone dictated it. School and hard work did not change it. In nineteen sixteen he put the test and the manifesto in the same book. A child could finish in about fifty minutes. The next year he helped design the Army's tests. More than one point seven million draftees took them. A campus instrument became a government instrument in a year. Narrator (Kore): One point seven million soldiers, set against a classroom. His long study of gifted children began with six hundred forty-three California children at or above one hundred thirty-five. He called it Genetic Studies of Genius. He pledged not to publish their names. He enrolled his own son and daughter. Narrator (Kore): And he missed one. Malcolm Harris records that the study passed over the only child of a Stanford engineering professor. William Shockley. Score, one hundred twenty-nine, just under the line Terman was using for genius. Harris's phrase is that the IQ ranked slightly subgenius. Narrator (Kore): Remember the number. One hundred twenty-nine against a line at about one hundred thirty-five. Six points. The boy who missed the cut would share the Nobel Prize in Physics in nineteen fifty-six for the transistor, and then argue a eugenics of his own. Fred, already inside his father's study, would build the industry. We are not there yet. The measuring stick is about to walk out of the psychology department and into the world. Narrator (Kore): A rank is a moral decision pretending to be a ruler. Once you have a single number, you can sort a school, an army, an immigration desk, a sterilization ward, and say you are only following the measurement. Binet wanted a tool for help. Terman built a tool for order. Narrator (Kore): Jordan would leave the presidency in nineteen thirteen, but the science he had sheltered did not leave with him. Lewis Terman taught until nineteen forty-three and died in nineteen fifty-six, the year of Shockley's Nobel. The father ranks children. The son builds an industry. That sentence is the hinge of this hour. Fred Terman is the son. We will meet him in a garage, and then at a laboratory at Harvard, and then on a train home. Narrator (Kore): Jump back two years from the book, to the summer the war began. The engineer is about to discover that logistics is a kind of power, and that paper is a kind of weapon.

05. Feeding Belgium

[CUE: singing-bowl chapter chime] [MUSIC: brahms3_3 from 0s, ducked] [AMBIENCE: telegraph.mp3] [ERA: radio]

Narrator (Kore): Feeding Belgium. Narrator (Kore): June twenty-eighth, nineteen fourteen. A Serbian nationalist killed the Archduke Franz Ferdinand in Sarajevo. Hoover, in London, told himself it was just another Balkan lapse into barbarism, and went back to work. He was wrong, and he was in the right city to find out how wrong. Narrator (Kore): On the afternoon of August fourth, a few hours after German troops crossed into neutral Belgium, Hoover walked a little more than a city block to the American consulate. That walk is the scene. One block. A mining engineer with an office on London Wall becomes the man who will try to feed seven and a half million people. Narrator (Kore): Learn Belgium in three facts. About seven and a half million

people. It grew only about twenty percent of the food it ate. The German army took the harvest, and the British blockade, meant to starve Germany, also closed the occupied country. Between them, a nation that imported four pounds of every five it ate began to beg. Narrator (Kore): His first errand was small. An American engineer named Millard Shaler had fifteen hundred tons of cereals stuck on a dock. Hoover saw that fifteen hundred tons would not touch a country. He built the Commission for the Relief of Belgium, the CRB. A private, neutral machine in London, buying food around the world, landing it at Rotterdam, moving it by canal, handing it to Belgians to distribute. He wanted, in his own words, the strongest centralization and an effective monopoly. Absolute command of his half. No knitting bee. Narrator (Kore): He also ran a campaign he named himself. QUOTE (Charon, flat documentary): enormous propaganda. Narrator (Kore): That is Hoover's phrase, recorded by Whyte. He browbeat American newspapers and governors. He hinted to one state that another state had already given more. He told reporters to go see the famine and to play up what he called the detailed personal horror stuff. He sent a motion-picture crew to film the food lines in Brussels. He hired famous writers. George Bernard Shaw sent a sentence. QUOTE (Charon, flat documentary): one Belgian goes without a full meal whilst thousands of lapdogs are living in luxury from Canterbury to San Francisco. Narrator (Kore): A cable went out. I will read it the way a cable sounds. LETTER (Algieba: Herbert Hoover, a clipped cable from London, autumn 1914, no adjectives): "Over one million people on bread line in Belgium at present moment with supplies estimated to last from one to three weeks. We shall require upwards of twenty thousand tons foodstuffs monthly." Narrator (Kore): Twenty thousand tons a month. The first gift ship, thirty-five hundred tons of beans, bacon, flour, and rice, reached Holland on November twenty-first. One ship, a sixth of a month. He got two great British lines to carry the food free, and the piers and the telegraph too. He treated charity as a logistics firm. Narrator (Kore): You cannot move twenty thousand tons a month on pity alone. The ships sail on a schedule, so the story has to sail on a schedule. A true story, repeated with discipline, is a kind of track. The food runs on it. He named the craft himself. When he later hates the stories told about him, remember that he had already called the method propaganda. Narrator (Kore): When the United States entered the war, Wilson made him Food Administrator. His name became a verb. To Hooverize meant to clean your plate, to skip the meat, to waste nothing. Reporters expected a giant and found an unobtrusive man at a desk. The verb was larger than the body. A successful campaign replaces the person with an instruction. Narrator (Kore): Now jump to nineteen nineteen. The war is over. Hoover is in Paris, advising Wilson at the peace conference, and he does something that will outlast the verb. Narrator (Kore): He had conceived, while he was feeding Belgium, of gathering the paper of the war. In June nineteen nineteen he sent a telegram to Stanford's president, Ray Lyman Wilbur. The Hoover Institution still calls that telegram its founding document. Hoover offered fifty thousand dollars and told Wilbur to send Professor Ephraim D. Adams to Paris to start the collecting. Fifty thousand dollars in nineteen nineteen is roughly nine hundred sixty thousand dollars in our grocery money. For that, he bought the beginning of an archive. Narrator (Kore): Adams went. So did Ralph Lutz and Frank Golder. They gathered pamphlets, newspapers, posters, and government documents across Europe. The founding collections include the files of the Commission for the Relief of Belgium, the United States Food Administration, and the American Relief Administration. In

nineteen twenty-two the collection was renamed the Hoover War Library. That year it took in forty thousand documents, housed in the Stanford library. Narrator (Kore): Forty thousand documents in one year. A library is a ledger. Whoever keeps the record owns the ground. He put the record of a war on his own campus, under his own name. After he lost the presidency he would watch the anti-Hoover literature pile up on shelves he had paid for. The archive does not love you. It keeps you. Narrator (Kore): While he was feeding a country and filing its papers, another industry, down the coast, was learning to sell the West back to the people who lived in it. Jump to a tavern on a corner in Hollywood. Late October, nineteen eleven. We are between our anchor years on purpose. The myth machine does not wait for a round number.

06. The Myth Machine

[CUE: singing-bowl chapter chime] [MUSIC: scottish3 from 0s, ducked] [AMBIENCE: static.wav] [ERA: phono]

Narrator (Kore): The Myth Machine. Narrator (Kore): On October twenty-seventh, nineteen eleven, the Nestor Film Company opened a studio at the northwest corner of Sunset Boulevard and Gower Street. The building had been the Blondeau Tavern. A local liquor law had hurt the trade. David Horsley, who had been making pictures in Bayonne, New Jersey, leased it. Behind the tavern, a wooden stage about twenty by forty feet, muslin overhead to soften the sun. The actors brought their own clothes. The carpenters brought their own tools. Narrator (Kore): Not the first movie set in Los Angeles. The first permanent studio in Hollywood. Its first California picture was a western, The Law of the Range, out December thirteenth, nineteen eleven. One reel, about nine hundred fifty feet. The ad called it a rousing Western foto-drama, taken in the West. The review said it was full of struggles, a chase, a posse, a hold-up, and that the photographs were not very good. NEWS (Sadachbia: a 1911 trade-paper reviewer, brisk, slightly disappointed): "The story is interesting, but the photographs are not very good." Narrator (Kore): Start there. Not with a masterpiece. With a cheap chase, shot in the sun, by men who had left New Jersey because Thomas Edison's lawyers were winning. Narrator (Kore): The Motion Picture Patents Company was the Trust. Edison and a handful of licensed companies held the patents on cameras and projectors. Pay them, or fight them. A trust, here, is a combination that uses patent law as a gate. We have seen this gate. Huntington's railroad. The Associated Press franchise. A rule that looks like property and behaves like a toll. The independents went west in part because California was far from the process servers. Narrator (Kore): The sun was a product before it was a backdrop. On January thirty-first, eighteen seventy-four, Thomas Balch Elliott and his Indiana neighbors formed the colony that became Pasadena. Many of them had come for a milder climate, sick with asthma or tuberculosis. Lyra Kilston, following John Baur's history of those health seekers, records the house that joked about the cure. In eighteen eighty-two Emma Bangs kept a Pasadena boarding house the town called the busted lung brigade. A parrot there had learned to cough exactly like a lunger. Hollywood inherited that migration. A cure, then a chase. The wage is the part with a number. Narrator (Kore): Los Angeles was an open-shop town. Open shop means you can work without joining a union, and the courts helped employers keep it that way. When the

studios needed carpenters and electricians to build a street that looked like Wyoming, the wage was the point. On The Squaw Man, Cecil B. DeMille noted that carpenters and other hands cost twenty-five to fifty percent less than on the East Coast. Public television's Lost LA records the note. Narrator (Kore): Twenty-five to fifty percent. That is the number. A carpenter in Los Angeles, nineteen thirteen, at three-quarters of a New York wage, or at half. Myth-making locates itself where the payroll is cheaper and the lawyers are farther away. Narrator (Kore): The Squaw Man is the feature people mean when they say Hollywood grew up. December nineteen thirteen. DeMille got on a train. The plan was Flagstaff, Arizona. The story he later told is the one you have heard. It was colder than when he left New York. Everybody back on the train, all the way to California. Narrator (Kore): The story goes. Public television says that version is widely taken to be apocryphal. A different memory, kept by the American Film Institute's historians, has him saying he got to Flagstaff, saw high mountains, and wanted plains with mountains in the distance. The star, Dustin Farnum, said they ought to go where the other companies were. They reached Los Angeles on December twentieth. On Christmas Eve he signed a lease on a barn at Selma and Vine. Shooting started December twenty-ninth. To save three dollars, he played a faro dealer himself. Narrator (Kore): The cold-train story is the legend, and the people who love it call it shaky. The Christmas Eve lease is a document. The wage gap is a document. Hollywood's origin is a budget and a patent fight, wearing a cowboy hat. Narrator (Kore): And the prehistory of the screen was shot on this farm. In eighteen seventy-eight, on Leland Stanford's stock farm, Eadweard Muybridge set a row of cameras to see whether a horse ever had all four hooves off the ground. It did. Those frames are the ancestor of the motion picture, paid for by the railroad president whose laurel tie would burn in nineteen six. Cinema begins here as a tool for proving something about a rich man's horse, and becomes a factory that sells his West back to the country at a discount on the carpenters. Narrator (Kore): The banker and the myth machine meet next, in a name on a building and a column in a square. The story of our streets.

07. The Column in the Square

[CUE: singing-bowl chapter chime] [MUSIC: brahms4_2 from 0s, ducked] [AMBIENCE: crowd.mp3]

Narrator (Kore): The Column in the Square. Narrator (Kore): If you have stood in Union Square, you have stood under a granite shaft with a bronze Victory on top, wreath in one hand, trident in the other. A lot of people think that column is for General MacArthur. It is not. Narrator (Kore): On May first, eighteen ninety-eight, Commodore George Dewey's squadron destroyed the Spanish fleet in Manila Bay in a morning. The war was short. The empire was not. On May thirteenth, nineteen three, Theodore Roosevelt dedicated the monument. Ground had been broken on May twelfth, nineteen one, by President McKinley. James D. Phelan gave the shaft to the city. Mayor Eugene Schmitz accepted it. Roosevelt said Manila had opened the Pacific to American commerce. The shots that count, he said, are the shots that hit. Narrator (Kore): Five years from the battle to the column. A morning in Manila becomes a stone in a shopping square before the uniforms are out of fashion. Union Square was already named, for pro-Union rallies

on the eve of the Civil War. The shaft is not about that war. It is about an ocean. Narrator (Kore): While he was in the city the Union League Club gave him a farewell banquet. Covers for eight hundred. The Morning Oregonian printed the toast list, and the list is the correction. The Army, General MacArthur. The Navy, Secretary Moody. California, Governor Pardee. The Presidential Party, Senator Perkins. The Citizens' Committee, M. H. de Young. NEWS (Sadachbia: a 1903 newspaper, reading a toast list, formal): "The Army, General MacArthur. The Navy, Secretary Moody. California, Governor Pardee. The Citizens' Committee, M. H. de Young." Narrator (Kore): General Arthur MacArthur. Douglas's father. Not Douglas. Arthur had commanded in the Philippines. He toasted the Army at the banquet for the column, a naval victory in the islands he had helped to hold. M. H. de Young, of the Chronicle, toasted the citizens. The column outside was for Dewey. If someone points at Union Square and says MacArthur, say Dewey, May first, eighteen ninety-eight, and Arthur was inside at dinner. Douglas, the son, gets a freeway, a tunnel, and a later hour. Narrator (Kore): One East Coast name, then it leaves. After the military government, the civilian governor-general of the Philippines from nineteen nine to nineteen thirteen was W. Cameron Forbes, of the Boston Forbes family, the family that had fought western railroad men over rates in the eighteen seventies. Boston, Manila, and a column in Union Square are one circuit. Narrator (Kore): Now the other name, and it belongs to the banker on the wharf. Narrator (Kore): In nineteen twenty-eight Giannini gave one and a half million dollars that founded the Giannini Foundation of Agricultural Economics at Berkeley and built Giannini Hall. Roughly twenty-nine million in our grocery money. He had left school at thirteen. He did not study in that hall. He bought it. The man who lent on a face put his name on the public university, not on Stanford. Narrator (Kore): The new fortunes named a hall for a banker who had skipped school, and a square for an admiral who had won a harbor in a morning. A fortune wants to become geography. Narrator (Kore): Jump to nineteen thirty-two. The engineer who fed Belgium is president, and the country has put his name on the opposite of a monument.

08. Hooverville

[CUE: singing-bowl chapter chime] [MUSIC: tragic from 0s, ducked] [AMBIENCE: wind.mp3]

Narrator (Kore): Hooverville. Narrator (Kore): In November nineteen twenty-eight Herbert Hoover was elected president of the United States. He had been secretary of commerce. He had been the man who Hooverized. He believed, with a mining engineer's faith, that a problem of production and distribution could be administered. Then the floor gave way. Narrator (Kore): The stock market broke in the autumn of nineteen twenty-nine. What followed was a collapse of work. The Bureau of Labor Statistics did not yet run the monthly survey we use now. That begins in nineteen forty. For the earlier years the Bureau later built estimates, and it says they are estimates. The estimate for nineteen thirty-three is an unemployment rate of twenty-four point nine percent. Twelve million eight hundred thirty thousand people. About one worker in four. The Department of Labor's history calls March the worst month. Farmers losing their land were often not even in the count. Narrator (Kore): One in four. Picture a street of

four households and dark windows in one of them, every street, at once. Irving Bernstein, writing for the Labor Department, notes that United States Steel had about two hundred twenty-five thousand full-time employees in nineteen twenty-nine. By April first, nineteen thirty-three, the full-time number was zero. Whoever was left was part time. Narrator (Kore): The people who lost their rooms built camps of cardboard, tar paper, tin, and packing crates. The camps were named for the president. Hoovervilles. HistoryLink, documenting Seattle's, reports that the word first showed up in print in nineteen thirty, credited to Charles Michelson of the Democratic National Committee. The largest Seattle camp stood on the tidal flats south of Pioneer Square, where a shipyard had been. It began in October nineteen thirty-one, when an unemployed logger named Jesse Jackson and about twenty men put up shacks. It grew to as many as twelve hundred people on about nine acres, and it lasted until nineteen forty-one, when the war wanted the land back for ships. The city tried to burn it. The men rebuilt it. Narrator (Kore): Twelve hundred people on nine acres, a few city blocks, named for a man who had moved twenty thousand tons of food a month. Belgium could be treated as logistics, because the food existed somewhere else. A depression in the country that grew the food was a question of who owned the work. You cannot employ a quarter of a nation by being competent. The engineer-state had met its limit. Narrator (Kore): On July twenty-eighth, nineteen thirty-two, the limit had a photograph. World War veterans, the Bonus Army, had camped in Washington asking for early payment of a bonus the law had promised them later. Troops under General Douglas MacArthur drove them out. Cavalry, tanks, the camps burned. How much of the order to cross the Anacostia Hoover had given, and how much MacArthur decided once he was over the bridge, is still argued. The record shows the eviction, the general, and the president who took the blow. The family that toasted the Army in Union Square in nineteen three now had a son clearing veterans off the capital's grass. Narrator (Kore): Hoover lost the nineteen thirty-two election to Franklin Delano Roosevelt. One East Coast fact, then we leave it. Roosevelt was the grandson of Warren Delano Junior, a partner in Russell and Company, the American firm that had made a great part of its China fortune in opium. The man who beat the Stanford engineer came from a family that already knew empire as a trading house. That is a family tree, not a conspiracy, and the tree is not blank. Narrator (Kore): After the defeat he went back to Palo Alto. Whyte has him in the study, as many as fifteen hundred letters a day, walking over to the Hoover War Library to watch the anti-Hoover literature accumulate. He spoke of the relief of no longer making ten vital decisions a day. The library held the verb and the insult in one building. He had built a machine for keeping the record. The record kept him. Narrator (Kore): A witness on that campus refused the credential. John Steinbeck came in nineteen nineteen, the year the war collection was founded, and left in nineteen twenty-five without a degree. He used the place as a library and went to ranch work, and later wrote the books that make the migrants hard to look away from. The pipeline produces presidents. It also, by accident, produces the people who write the presidents down. Only one of those outputs gets a tower. Narrator (Kore): The tower comes in nineteen forty-one. We are not there. Jump to nineteen thirty-nine, to a garage on a flat street in Professorville, where the engineer's century changes trades. From mines and food to a box that makes a tone.

09. The Garage

[CUE: singing-bowl chapter chime] [MUSIC: schubert2 from 0s, ducked] [AMBIENCE: hum.mp3] [ERA: hum]

Narrator (Kore): The Garage. Narrator (Kore): Three six seven Addison Avenue, Palo Alto. A house built about nineteen five for John Spencer, a physician who would be mayor. By the late nineteen thirties the house was two flats. Three six seven downstairs. Three six nine up. In the back, a garage, twelve by eighteen feet, the kind of outbuilding a Sanborn fire-insurance map bothers to draw. In September nineteen thirty-eight David Packard and his wife Lucile moved into the downstairs flat. William Hewlett moved into a shed behind the house. The landlady, Ione Spencer, lived upstairs. They chose the house because it had a garage. Narrator (Kore): Hewlett and Packard had finished electrical engineering at Stanford in nineteen thirty-four. Packard went to General Electric in Schenectady, New York. Hewlett went on to graduate work at Stanford and at MIT, and came back. Their professor, Frederick Terman, told them to make a run for it in California instead of becoming permanent employees of a firm back east. In nineteen thirty-eight Terman arranged a Stanford fellowship so Packard could work on vacuum tubes with Russell Varian. Packard took a leave from GE. He and Lucile drove west with a used Sears drill press in the car. Lucile went back to work in the Stanford registrar's office, and that paycheck was the steady money. In the evenings she kept the books. Narrator (Kore): The number is five hundred thirty-eight dollars. That is the working capital Hewlett-Packard's own history and the National Register of Historic Places agree on. Cash, plus the value of the drill press. They began part-time work in the garage with that sum. On January first, nineteen thirty-nine, they formalized the partnership. A coin toss decided the order of the names. Hewlett before Packard. Their first real product was an audio oscillator, the Model 200A, a box that makes a clean tone. In nineteen thirty-eight a sound engineer from Walt Disney's studio heard it and asked for changes. Hewlett-Packard's own history says the changes produced the Model 200B, and Disney ordered eight of them, at seventy-one dollars and fifty cents each, to test the sound system for Fantasia. Narrator (Kore): Five hundred thirty-eight dollars in nineteen thirty-eight. The index that year is forty-two point three. Roughly twelve thousand eight hundred dollars in twenty twenty-six grocery money. The rent on the flat, in the company's own telling, was forty-five dollars a month, roughly eleven hundred dollars a month now. The firm that will anchor the industrial park of the next episode started in a garage behind a mayor's old house, because a professor told two students not to leave. Narrator (Kore): Terman was already building companies, before that job had a name. Terman did not write them a check for a million dollars. Venture capital, as a profession, does not exist yet. What he wrote was permission, a fellowship, a problem to work on, and a sentence: stay here. A university that can do that is no longer only a school. It is a place that chooses which of its students get to become companies, and where those companies sit. The Founding Grant still said the land could not be sold. Nobody in the garage was thinking about leases. Terman was starting to. Narrator (Kore): Packard was six foot five, a high school class president, an athlete, the kind of man David Starr Jordan had liked to imagine as the Stanford type. Hewlett had impressed the admissions office less. Terman took both. The eugenic dream of the right stock and the engineering dream of the right circuit are not the same dream, but they passed through the same family. Lewis measured the boy. Fred deployed him. By nineteen thirty-nine Fred was a

professor with a laboratory habit and two former students in a garage on Addison. The war was about to take the professor east and turn the habit into a system. Narrator (Kore): Jump to nineteen forty-two. The last anchor year. The garage is already too small. The century of the engineer is about to be paid for by the government, in secret, at a university on the other ocean.

10. Total War

[CUE: singing-bowl chapter chime] [MUSIC: eroica2 from 240s, ducked] [AMBIENCE: static.wav] [ERA: radio]

Narrator (Kore): Total War. Narrator (Kore): December seventh, nineteen forty-one. Pearl Harbor. Malcolm Harris, following the record of Frederick Terman's war, writes that three weeks later the country called him. Not to Manila. Not to a trench. To Cambridge, Massachusetts, to run a laboratory. Narrator (Kore): The Radio Research Laboratory at Harvard was the other half of radar. Not how to see the enemy. How to keep the enemy from seeing you. Strips of metal dropped from airplanes so a radar screen fills with ghosts. The British called them window. The Americans called them chaff. Harris credits that work, done with British scientists, as the laboratory's biggest success, and he has Terman in England coordinating the teams. The test was Normandy. Planes dropped enough chaff, and jammed enough signals, to paint ghost fleets on German screens. Terman did not storm a beach. He helped decide what the other side's instruments would believe. Narrator (Kore): A laboratory wearing a university's name takes federal money to do work the public cannot read. Call that the contractor state. The money is public. The result is classified. Vannevar Bush wrote the plea to keep the arrangement after the peace. Science, the Endless Frontier, to the president in July nineteen forty-five. Basic research, he argued, is where practical power comes from, and private firms will underfund it because they cannot capture all the benefit. So the government should fund it permanently, through universities, with scientists deciding where the money goes. Terman read that as a business plan. In this hour you only need to see it being born while he is still in Cambridge. Narrator (Kore): Behind him, the klystron was already a weapon. A klystron is a vacuum tube that makes high-power microwaves. Russell and Sigurd Varian built it with the Stanford physicist William Hansen in the late nineteen thirties. Radar eats microwaves. A tube invented next to a university became part of how the Allies saw in the dark, and the seeing was not for publication. Narrator (Kore): Now the other Stanford man the IQ test had missed. Narrator (Kore): William Shockley had grown up in Palo Alto, gone through Caltech and MIT, and joined Bell Laboratories. In nineteen forty-two Philip Morse called him into anti-submarine operations research. Research director. A roving commission. Blow up submarines with mathematics. Narrator (Kore): Operations research means using numbers to decide how to fight, not only what to build. Shockley, Harris reports, solved his first problem in days. Ship-dropped depth charges were set to explode seventy-five feet down, right if a submarine has had time to dive. The same charges, dropped from airplanes onto boats still near the surface, exploded too deep to hurt them. Reset the airplane charges to thirty-five feet. Within two months, Harris writes, successful anti-submarine attacks rose by a factor of five. Narrator (Kore): Five times. A six-point miss

on a childhood IQ test, and a fivefold increase in submarines hit, because a man moved a fuse forty feet. The regret of hiring him comes later. Tonight the regret is on Bryant Street. Narrator (Kore): February nineteenth, nineteen forty-two. Seventy-four days after Pearl Harbor. Franklin Roosevelt signed Executive Order ninety sixty-six. An executive order is a president's directive. It is not a statute, but it moves armies until Congress or a court stops it. This one let the secretary of war mark military areas and exclude any or all persons from them. It did not say Japanese. It did not have to. Lieutenant General John DeWitt drew the zones and ordered the removal. Congress then made disobedience a crime. Up to a year in prison, and a five thousand dollar fine. Five thousand dollars then is roughly one hundred thousand now, plus the year. Narrator (Kore): About one hundred twenty-two thousand people of Japanese ancestry on the West Coast, most of them citizens, were forced into guarded camps the government called relocation centers. No person of Japanese ancestry in the United States was convicted of a serious act of espionage or sabotage during the war. That sentence is the National Archives, not a speech. The order was ancestry, not evidence. Narrator (Kore): Palo Alto's own historians count about one hundred forty-four Japanese Americans in that city. In March they registered their property at the police station on Bryant Street. In April, an eight o'clock curfew, only for them. On May nineteenth DeWitt ordered them out. Days, not months, to sell a nursery or a shop. On May twenty-sixth, one hundred forty-four people boarded buses for the Santa Anita racetrack, renamed a reception center. What you could carry. No pets. Friends passed candy and oranges. Then the buses left, and by the middle of June not one Japanese American remained in Palo Alto. Narrator (Kore): One hundred forty-four, against about one hundred twenty-two thousand on the coast. Most of the Palo Altans went on from Santa Anita to Heart Mountain, in Wyoming. Fifty-nine students from the Palo Alto schools were incarcerated. Cherry Ishimatsu, of Palo Alto High School, later described Scouts parading inside the wire. SPOKEN (Achird: Cherry Ishimatsu, remembering Heart Mountain, quiet, exact): "Here we are, all prisoners behind barbed wire fences and a military guard surrounding us, and here we have young people doing a parade with their American flags flying." Narrator (Kore): Before the buses, Arthur Okado, president of the Palo Alto Japanese American Association, said who they were in the Palo Alto Times. SPOKEN (Zubenelgenubi: Arthur Okado, a formal public statement, controlled, no plea in the voice): "Without reservation, we are loyal to this, our country, the United States of America." Narrator (Kore): The association had bought about twenty-seven hundred dollars in defense bonds, roughly fifty-five thousand now, and seven of their men were already in the Army. Ten weeks later the FBI arrested Okado. He came home in February nineteen forty-five, cleared. The bonds were not a shield. Narrator (Kore): One widely repeated estimate puts the property loss at about four hundred million dollars then, roughly eight point two billion in grocery dollars now. Orchards, nurseries, shops. The Alien Land Law of nineteen thirteen had already told these families they could not own farmland. First you forbid ownership. Then you force a sale. Then you call it military necessity. Narrator (Kore): Two words. Issei. First generation, born in Japan. Nisei. Second generation, born here, citizens. The buses did not sort them carefully. In Cambridge that same season, a Stanford professor helped drop metal foil so a radar set would lie, and the lie saved lives on a beach in France. In Palo Alto, a city registered its neighbors at a police desk and put them on a bus to a racetrack. Both are the contractor state. One contract is with a

laboratory. The other is with a general. The argument about what the laboratories are for waits until nineteen sixty-nine. This hour ends on a train.

11. The Plan

[CUE: singing-bowl chapter chime] [MUSIC: brahms4_2 from 0s, ducked] [AMBIENCE: hum.mp3, wind.mp3]

Narrator (Kore): The Plan. Narrator (Kore): Nineteen forty-five. The war in Europe ends in May. The war in the Pacific ends in August, after two atomic bombs and a Soviet invasion and a surrender signed in September. Frederick Terman comes home. Narrator (Kore): The histories put him back in Palo Alto as the war ends, nineteen forty-five into nineteen forty-six, to a small department and a university that had spent the war lending its best people to other people's laboratories. MIT had the radar lab. Harvard had him. Berkeley had a hand in the bomb. Stanford had individuals. His diagnosis, repeated for twenty years, was that Stanford had been the wrong shape. Federal money goes to places already visibly excellent in the fields the government wants, and to men the decision-makers already know. Adequate at many things means invisible when the contracts are written. Narrator (Kore): He had learned how the money moves, what a proposal has to look like, and how to run rooms where most of the staff cannot discuss most of the work. He had also learned that the money would not stop. A government that has found science as a strategic asset does not give it back. Bush wrote the theory in July. Terman brought the practice west in his head. Narrator (Kore): The land was waiting, and the land had a lock on it. Narrator (Kore): In eighteen eighty-five the Founding Grant told the trustees they had no power to sell or convey the land. A grieving father, protecting a dead boy's memorial from trustees who might liquidate it. For sixty years the clause froze a fortune in acres. Horses, then orchards, then a war that barely touched the ground, because the war work had been done in other men's buildings. Narrator (Kore): Terman looked at the same sentence and saw a door. Narrator (Kore): [SILENCE] [HARD] The grant says the land cannot be sold. It does not say the land cannot be leased. Narrator (Kore): A lease is not a sale. The university keeps the title. The tenant pays rent, builds a laboratory, hires the graduates, and the ground is still Stanford's when the term ends. The appreciation stays with the landlord, and the landlord chooses the neighbors. On a few thousand acres of orchard, that choice is about to become an industrial policy. Narrator (Kore): Who said anything about selling. Narrator (Kore): Next hour, the first signature on that idea is a company called Varian, in nineteen fifty-three, and the professor from the garage is on the company's board. The boy the IQ test called slightly subgenius opens a laboratory because his mother lives nearby, wins the Nobel, and drives eight of his best people out the door. Across the creek, Black families cannot buy the houses near the jobs. In April nineteen sixty-nine, students occupy a lab and call the arrangement the Stanford Empire. Narrator (Kore): Not yet. Tonight the spire is still down, the laurel tie is ash, the plank has become the largest bank in the world, the verb Hooverize has curdled into a shack on a tidal flat, and a man is riding west with a plan that fits in a sentence. Narrator (Kore): [SILENCE] [HARD] Do not sell the farm. Rent it to the future. Narrator (Kore): This is Cardinal.

The Park — full audio script (voice/cue markup)

The Park — full audio script (voice/cue markup)

Episode 6 of 8. The Park (1945–1969). Narrator = Kore. Bland documentary/official quotes = Charon. Other voices are listed per line. [HARD] = music gated out. [SILENCE] = 2 s of silence before a reveal. Reuse only cue, music, and ambience names from the Spike in the Museum pipeline. A note on sources, the pronunciation sheet, and the fact-check live in the review file and are not narrated.

Pronunciation, not narrated: Noyce, NOYSS. Hoerni, her-NEE. Kleiner, KLY-ner. Engelbart, ENG-el-bart. Dinkelspiel, DING-kul-shpeel. Zoetrope, ZOH-ee-trope. Pitzer, PIT-zer. Tanforan, TAN-for-an. Korematsu, kor-eh-MAT-su. Buri Buri, BOO-ree BOO-ree. Kesey, KEE-zee. Goebbels, GUR-bulz. Bechtel, BEK-tul. Quezon, KAY-zon.

01. The Lease

[CUE: singing-bowl chapter chime] [MUSIC: brahms4_2 from 0s, ducked] [AMBIENCE: hum.mp3]

Narrator (Kore): The Park. Narrator (Kore): Nineteen fifty-three. A company signs a piece of paper with a university, and the paper is not a deed. Narrator (Kore): Last hour Frederick Terman came home from Harvard with one sentence. The grant would not let him sell the farm. It did not forbid a lease. This hour is that lease, then the laboratory, the creek, the demonstration, and the occupation. Narrator (Kore): The company is Varian Associates. The brothers are Russell and Sigurd Varian, the men who built the klystron before the war. Malcolm Harris records the first lease at the Stanford Industrial Park. It went to Varian's research department in nineteen fifty-three. No surprise, he writes, because Terman was an investor and a member of the board. Narrator (Kore): The lease is the harvest. The planting was earlier, and it was small. Narrator (Kore): The money in this hour uses the same method as last hour. The Minneapolis Fed's consumer price index, nineteen sixty-seven equals one hundred, and the twenty twenty-six index at one thousand four point eight. When a check appears, that is the only converter. Narrator (Kore): In January nineteen forty-five, while the war in Europe was still on and Terman was still the man Harvard had borrowed, he talked Stanford's president, Donald Tresidder, out of twenty-five thousand dollars. Harris is particular about the name. Not a microwave laboratory. Not the Stanford microwave laboratory. The Microwave Laboratory. Twenty-five thousand dollars in nineteen forty-five is about four hundred sixty-five thousand in our grocery money. It was a drop, Harris says, and Terman knew where the spigot was. The Office of Naval Research. A university that had spent the war

lending its people to other people's buildings was about to have a building of its own, aimed at the tubes the Varian brothers had already taught the military to want. Narrator (Kore): Then a fight over what kind of institute a university is allowed to be. Terman wanted the Stanford Research Institute, founded in nineteen forty-six, to take the big military contracts. Sally Denton's history of Bechtel puts Steve Bechtel senior, the builder, among the founding directors. The years when a Bechtel president sits down with a future cabinet are not this hour. The signature on the institute is. Tresidder, a Hoover man, feared a university that lived on government money. Harris records that Tresidder fired the institute's director rather than let one Navy contract turn SRI into a public charge, and that he died before he could name a replacement. The next president, J. E. Wallace Sterling, promoted Terman to provost in nineteen fifty-four. Rebecca Lowen, the historian of the Cold War university, is quoted by Harris for the fit. Sterling was sufficiently anticommunist, and sufficiently anti-Roosevelt, to please Hoover. The man who had feared the government's check was gone. The man who knew how to write the proposal was in charge of the campus. Narrator (Kore): Hold the two hats on the Varian lease. Professor, and director. The man who tells the university what its land is for is also a man who owns a piece of the tenant. That is not a scandal hidden in a footnote. It is the design. The university's best advertising is that its own provost has already bet. Narrator (Kore): Learn the instrument. A ground lease. The university keeps the title. The tenant pays rent, builds a laboratory, hires the graduates, and when the term ends the buildings and the ground are still the landlord's to lease again. Selling would have broken the grant of eighteen eighty-five, the clause that told the trustees they had no power to sell or convey the farm. Leasing uses the clause as a business. The appreciation stays with Stanford. The neighbors are chosen by Stanford. A landlord who cannot sell is not a landlord who is stuck. He is a landlord who collects forever. Narrator (Kore): A sale would have been a goodbye. A lease brings the tenant back, and the deed never changes hands. Horses, then orchards, then microwave tubes. Narrator (Kore): This is Cardinal. Episode six. The Park.

02. Why Here

[CUE: singing-bowl chapter chime] [MUSIC: aase from 0s, ducked] [AMBIENCE: fs_42139.mp3] [ERA: hum]

Narrator (Kore): Why Here. Narrator (Kore): After Varian, Harris writes, the park filled with companies Terman already knew. Hewlett-Packard put its headquarters on the land. General Electric came. Kodak came. The garage on Addison Avenue, five hundred thirty-eight dollars and a drill press, had grown up and walked onto the farm. Narrator (Kore): In nineteen fifty-four Terman added a door for the people inside those buildings. The Honors Cooperative Program. A company in the park could send its employees to Stanford for advanced classes, ahead of the ordinary admissions line, if it paid double tuition. Harris records that they paid it. The same firm now pays rent to the university and tuition to the university, and the university trains the next person the firm will hire. One circuit. Three invoices. Narrator (Kore): Walk it, because the mechanism is a distance. A student finishes a class in a building Terman's school of engineering controls. She crosses Page Mill, which we will name properly in a few minutes, and

badging into a company that already paid her tuition at double the ordinary rate. Her professor may be on that company's board, or in its laboratory on a weekday the university has agreed to lend him. The government, which bought the tubes during the war, is still the customer. Harris draws the loop in one breath. The government sponsors the microwave research at Stanford. Stanford professors consult for Varian. Varian hires Stanford students trained to make microwave tubes. Varian sells the tubes to the government. Four steps. One parking lot. Narrator (Kore): Call the pattern a cluster. Firms sit next to the laboratory that trained their people, and next to each other, because the next hire, the next part, and the next idea are a short drive, not a letter to Boston. Economists have a longer name for it, agglomeration, the simple fact that specialists get more done when they can see each other's doors. You only need the picture. A park is a neighborhood of specialists. The university is the landlord of the neighborhood and the school of its children. A landlord who also writes the syllabus is not a neutral host. He is choosing which problems count as an education, because those are the problems the tenant will pay to have taught. Narrator (Kore): An investor who watched it happen said the quiet part in two sentences. Arthur Rock, in a two thousand seven conversation at the Computer History Museum, was asked where Stanford fit. SPOKEN (Alnilam: Arthur Rock, a 2007 interview, dry, no romance in the voice): "They allowed their professors to work part-time, and this was unknown in the East. And also, they had all this land, and they allowed these companies to rent or lease space from them." Narrator (Kore): Part-time, he said, was unknown in the East. MIT and Princeton, in his account, would not let their professors start companies on the side. Stanford would. And Stanford had the acres. One East Coast fact, then it leaves. The rule on the other ocean was that a professor stayed a professor. The rule here was that a professor could also be a founder, and the founder could rent the university's dirt. Narrator (Kore): Rock thought there would not be much of a Silicon Valley without Terman and that combination. You can disagree with the size of the claim. You should not miss the mechanism. Talent was allowed to leave the classroom on Tuesdays, and the land was allowed to leave the orchard without leaving the deed. Narrator (Kore): Years later, in a two thousand fourteen panel at the same museum, he put the geography in a joke that is also a map. All the money was on the East Coast. All the brains, by then, were on the West Coast. It did not take much, he said, to put two and two together. Most of the entrepreneurs he was meeting out here had started as easterners. The gumption was the move. Stanford's contribution was that, once they arrived, the university did not punish them for starting a company, and it had dirt to rent them. Narrator (Kore): The park could recruit a professor the way it recruited a tenant. Harris's example is Carl Djerassi, a co-inventor of the birth control pill, who came to teach chemistry and brought his company, Syntex, along behind him. A pill is not a missile and not a transistor. It is the same real estate logic. A man with a product sits down in a department, and the company follows the man onto university land. Teachers and students move between the park and the school until profit and nonprofit are hard to tell apart at the parking lot. Narrator (Kore): Varian first, in nineteen fifty-three. Hewlett-Packard on its heels. Then the companies that wanted to sit next to both. Narrator (Kore): Jump three years, to a laboratory that is not on the leased farm, and to a man the IQ test had already misfiled.

03. The Laureate

[CUE: singing-bowl chapter chime] [MUSIC: mozart40_2 from 0s, ducked] [AMBIENCE: hum.mp3]

Narrator (Kore): The Laureate. Narrator (Kore): February nineteen fifty-six. William Shockley is back in the valley where he grew up. Narrator (Kore): Last hour he was the boy Lewis Terman's genius study passed over. Score one hundred twenty-nine. Slightly subgenius. Then he was the man at Bell Laboratories who helped move a depth-charge fuse from seventy-five feet to thirty-five. Now he is the most famous physicist in the most promising trade in electronics, and he has left Bell to start a company. Narrator (Kore): The account that has held is personal before it is strategic. His widowed mother was still in Palo Alto. He had grown up in that town, gone away to Caltech and MIT and Bell Laboratories, and when he left Bell in nineteen fifty-five to start a company, he put it where she was. Not on the leased farm. In Mountain View, the next town down the peninsula. Beckman Instruments, a Los Angeles company that made laboratory instruments, financed it. Shockley Semiconductor Laboratory opened in nineteen fifty-six. Narrator (Kore): Look at the town he chose, because the legend will later pretend the valley was waiting for him. In nineteen fifty-six Mountain View is orchards, war housing, and the edge of Moffett Field. There is no chip industry to join. There is no supply chain, because there is nothing yet to supply. The supply chain will be the people in the building. He recruits them one by one, from laboratories that already work, and they come because the man who wrote the textbook asked. A cluster, before anyone is using the word, assembled by a son who wanted to live near his mother. The world's chip business sits on this peninsula in substantial part because of that want. Let the contingency sit before the legend of destiny arrives to tidy it. Narrator (Kore): In November of that same year the Nobel Prize in Physics went to Shockley, John Bardeen, and Walter Brattain, for the transistor. Learn the object, because the rest of the hour is people fighting over who gets to make it. A transistor is a small solid switch. A vacuum tube, the thing in the radios and the radar sets of the last war, is glass, heat, and a filament that burns out. A transistor does the switching inside a crystal. Less heat. No glass to break. A size that can shrink again and again until a building's worth of calculating fits in a pocket. The Nobel was for the effect. The industry was for the shrinking. Narrator (Kore): Shockley's book on electrons and holes was the field's textbook. That matters more than the medal for the young people who got on trains. You move to a town with no semiconductor industry because the man who wrote the book asked you, and because the book is what every serious laboratory is teaching. Bardeen and Brattain stayed identified with the first, point-contact device. Shockley's name sat on the junction transistor that could be manufactured. The prize made them a trio. The company made him a boss. Young physicists moved because he asked them to. They did not yet know they were moving into a management problem disguised as a laboratory. Narrator (Kore): A field can be right and a boss can still empty the building. Shockley had the standing to recruit the best young people in the trade. He did not have the temperament to keep them. Jay Last, who was one of them, said later that Shockley had no business sense at all, which was not surprising, until the moment the work required it. Relationships with Beckman deteriorated. Then relationships inside the lab deteriorated. Individually, Last remembers, most of them decided they were going to leave. Narrator (Kore): They did not leave because the science was boring. They left because they

wanted to make a transistor, a device the market could use, and the man who had recruited them would not run a laboratory they could stand. The IQ test had missed him by six points. The management test, which nobody had written down, he failed in public. Narrator (Kore): Remember who ranked him. Lewis Terman, the father, had come west because tuberculosis required a mild climate. His son Fred leased the farm. The boy the father scored at one hundred twenty-nine came home to his own mother and built the laboratory the son of the tester would have wanted in the neighborhood. Three Termans and one Shockley, and only one of them can keep a staff. Narrator (Kore): By the summer of nineteen fifty-seven a group of the young men had decided the way out was not eight separate jobs. It was one exit, together.

04. The Eight

[CUE: singing-bowl chapter chime] [MUSIC: scottish3 from 0s, ducked] [AMBIENCE: static.wav] [ERA: radio]

Narrator (Kore): The Eight. Narrator (Kore): September nineteen fifty-seven. Eight men leave Shockley Semiconductor. Their names, as Harris lists them: Julius Blank, Victor Grinich, Jean Hoerni, Eugene Kleiner, Jay Last, Gordon Moore, Robert Noyce, and Sheldon Roberts. Late twenties, early thirties. Last was twenty-seven. Kleiner, he thought, was about thirty-two. Narrator (Kore): Harris gives two of them a past you can see. Noyce, a preacher's son. Kleiner, a Jewish refugee. Moore was the only one from Northern California. The rest had been ambitious enough to leave the East, where big science still felt like the comfortable job, and take a risk in a valley that did not yet make chips. Shockley had offered them the role of rebellious younger brothers, inside the prestige of his Nobel and outside the mass of Bell Laboratories. What they met, Harris writes, was an awful boss. Unbearable. He compared them to one another and to himself. With the prize secured, his ego was at its height, and he treated the group as lackeys. Narrator (Kore): Then he changed the work. He became obsessed, in Harris's account, with a four-layer diode of his own design, a device he believed could do more logic than a transistor, and he forbade the staff from growing the silicon devices they had been hired to make. That is the scene. Not a vague clash of personalities. A door closed on the product. The people who had crossed the country to build transistors were told to stop building them. Narrator (Kore): The timing makes the loss sharper. Sputnik, the Soviet satellite, went up in October nineteen fifty-seven, a few weeks after the eight had gone. Suddenly the country wanted tough, reliable transistors for a space race. Shockley's business plan, Harris notes, had just become better than he first thought. It was too late. The men who could have built them for him were down the road, looking for a check. Narrator (Kore): They did not begin by founding a company. Last is plain about it. There were seven of them first. They hoped some employer would hire the group, so they could keep making the transistor instead of scattering. Eugene Kleiner's father had a relationship with a Wall Street firm. Kleiner wrote. The letter reached a junior man named Arthur Rock at Hayden Stone, who had the wit, Last says, not to throw it in the wastebasket. Rock and his boss, Bud Coyle, came out. Narrator (Kore): Rock's idea was the one that felt foreign. Do not join a company. Start one. SPOKEN (Orus: Jay Last, a 2007 interview, young in the memory, precise): "There wasn't any

venture capital really at that time." Narrator (Kore): Last is the one who should teach this. Venture capital, the profession of funding a company that does not exist yet in exchange for a piece of it, is not an industry in nineteen fifty-seven. There are investment bankers. There are rich companies. There is no row of firms on Sand Hill Road whose only job is the bet. Rock is inventing the motion by doing it once. Narrator (Kore): The two men do not remember the number the same way. Narrator (Kore): Rock, in that two thousand seven evening, says they went to thirty-five companies, and all thirty-five turned them down. The firms could not see how to invest in a subsidiary without upsetting their own employees. Last remembers a list of about twenty or thirty, and that all thirty said no, and that the ones who got close were stopped by lawyers. Same shape. A long list of no. Different counts. The record of a refusal is solid. The exact headcount is two memories. Narrator (Kore): They were about to give up, Rock says, when someone suggested Sherman Fairchild. Fairchild was an heir, an inventor, a pilot, and the largest individual shareholder in IBM. He was intrigued. He got one of his companies, Fairchild Camera and Instrument, to invest one and a half million dollars. That is Rock's figure for the check that started Fairchild Semiconductor. Narrator (Kore): One and a half million dollars in nineteen fifty-seven. The index that year is eighty-four point four. In our grocery money, about seventeen point nine million dollars. Thirty-five companies, or thirty, looked at eight young men and a transistor and said no. One heir said yes. The industry that follows is not proof that the check was obvious. It is proof that obvious and available are different words. Narrator (Kore): Rock remembers the split. The eight took about eighty percent. Hayden Stone took twenty. He told a room of later venture capitalists that they could thank him for the twenty. That is his memory of where a custom began. Narrator (Kore): Fairchild Semiconductor was set up in Mountain View, the same town as the laboratory they had left. Noyce joined the seven and made eight. Moore stood beside him. Within a year of opening, Harris writes, Fairchild was shipping circuits to IBM. The people Shockley had recruited were now a few miles from his door, making the device he had not let them make. Narrator (Kore): Now the name, because the name is a fight, and both sides go on the air. Narrator (Kore): Malcolm Harris writes that Shockley later tarred the group as the traitorous eight. QUOTE (Charon, flat documentary): the traitorous eight. Narrator (Kore): Jay Last was asked, in the same season of interviews, where the phrase came from. The interviewer added that Shockley's wife, Emmy, said her husband never used it. Last's answer is the correction. SPOKEN (Orus: Jay Last, correcting a legend, unhurried): "This was some newspaper guy way after the fact." Narrator (Kore): He had not heard the term for at least ten or fifteen years. Then it caught on, because people liked it. Rock, years later, slips and says treacherous. A widow denied it. A historian credits Shockley. A banker cannot keep the adjective still. They left, and the leaving worked. Narrator (Kore): Why the leaving was legal is the California part, and it is worth a slow sentence. Business and Professions Code section sixteen thousand six hundred voids a contract that restrains anyone from engaging in a lawful profession. The idea is older than the park. It comes down from the Civil Code of eighteen seventy-two, section sixteen seventy-three, and it was recodified in nineteen forty-one. In plain speech: an employer in this state generally cannot stop you from going across the street and competing. A noncompete is the clause, common elsewhere, that says if you leave you may not work in the same trade, in the same region, for a year or two. On the other ocean that clause

could be a lawsuit. Here the lawsuit was mostly unavailable. Narrator (Kore): Picture the Friday. Eight men have a check, a statute, and a town small enough that nobody has to sell a house. Mountain View to Mountain View. The children stay in the same schools. The cost of betrayal, if you want to call it that, is a different badge on Monday. Route 128 around Boston had universities, federal money, and talent in these same years. It did not throw off companies the way this peninsula did. The missing lawsuit is part of why. A law written in eighteen seventy-two, for a state of shops and farms, turns out to be a permission slip for an industry that did not exist when the legislature voted. Narrator (Kore): Eight men. One check after a corridor of noes. A statute from eighteen seventy-two that lets them do it again.

05. Defense Money

[CUE: singing-bowl chapter chime] [MUSIC: eroica2 from 240s, ducked] [AMBIENCE: wind.mp3]

Narrator (Kore): Defense Money. Narrator (Kore): The park was not only a nursery for transistors. It was a place to put the missile work where the engineers already lived. Narrator (Kore): Harris dates the decision to nineteen fifty-five. Lockheed Aircraft planned to move its expanding missiles division from Los Angeles to Sunnyvale, to sit near the Ames laboratory at Moffett Field. Ames was not new money. The field had been built with one of the region's rare large defense contracts before the war, in nineteen thirty-one, while Hoover was president. In nineteen fifty-six, with some prodding from Terman, Lockheed put a research complex in the industrial park, and then a manufacturing complex nearby that became the county's defining employer. Narrator (Kore): A research office in the park is a tenant of the university. A factory in Sunnyvale, and a test range in the Santa Cruz mountains, are not slices of the founding grant. Narrator (Kore): What the valley felt was the payroll, and the payroll had a number. Accounts of the missiles division, Steve Blank's history of it among them, put Lockheed's local employment at zero in nineteen fifty-six and at more than twenty-eight thousand people by nineteen sixty-five. Twenty-eight thousand. Set that against Hewlett-Packard and Fairchild, which were famous and still small beside a missile factory. Set it against the other number Harris records. Thirty thousand California engineers were laid off in nineteen sixty-three and sixty-four when a defense cycle turned. One company's hiring, at its peak, was on the order of the statewide layoff. A cluster that lives on contracts can fill a county and then empty a profession in the same decade. People bought houses and drove to work on missiles. Some of those houses, within a few years, belonged to people who no longer had the job. The park is not a promise of permanent work. It is a promise that the work, while it lasts, will be near the campus. Narrator (Kore): Learn the product the payroll was for, without pretending to a tour of the factory. A missile division is not a clever gadget in a garage. It is a prime contractor, a company the Navy or the Air Force pays to assemble a weapon out of everyone else's parts. Guidance, airframe, fuel, the warhead that belongs to someone else. The contractor's job is the system. Stanford's job, if Terman has done it right, is to be where the system's engineers were trained, and where a research office can sit on a ground lease while the factory itself stands on private land in Sunnyvale. The university gets the prestige and

the rent. The county gets the traffic. The contract gets a return address that sounds like a campus. Narrator (Kore): On January seventeenth, nineteen sixty-one, leaving office, Dwight Eisenhower warned the country about the arrangement in a phrase that stuck. NEWS (Sadachbia: a farewell address, January 1961, grave, unhurried): "In the councils of government, we must guard against the acquisition of unwarranted influence, whether sought or unsought, by the military-industrial complex." Narrator (Kore): He was describing a machine, not a conspiracy. A military that needs machines, an industry that needs the orders, and a university that needs the research will learn to need one another. Sought or unsought. The Defense Department's budget went from thirteen billion dollars to fifty-eight billion in nineteen fifty-one. In grocery money, using that year's index, about one hundred sixty-seven billion to about seven hundred forty-seven billion. Harris writes that while bombers were leveling cities in North Korea, military support for electronics research at Stanford tripled. By nineteen sixty-seven, he adds, citing the historian Rebecca Lowen, Stanford was passing every American university but MIT and Caltech in defense contracts. Terman did not have to corrupt anyone. He had to build a park where the three could share a parking lot. Narrator (Kore): One more beat, so the map is not a single campus. The Advanced Research Projects Agency was spreading this kind of money to a handful of universities. MIT. Carnegie Mellon. Utah. Berkeley. Stanford. Stanford was on the map. It was not the map. The East Coast laboratories were real. What they did not have was Terman's dirt, and a state that let the professors walk. Narrator (Kore): The bay had already learned how to build for a war budget. The yards were up the water, before the park had a name.

06. The Yards and the Stalls

[CUE: singing-bowl chapter chime] [MUSIC: hebrides from 30s, ducked] [AMBIENCE: wind.mp3, hum.mp3]

Narrator (Kore): The Yards and the Stalls. Narrator (Kore): The war this hour inherits did not begin at Pearl Harbor. In Asia it began on July seventh, nineteen thirty-seven, at the Marco Polo Bridge outside Beijing. That is the Second Sino-Japanese War. The first, in the eighteen nineties, is a different war. Say which one. Rana Mitter, a historian of China's war, puts the dead in a range of fourteen to twenty million, and the fighting had already run four years before the United States was in it. Narrator (Kore): When the United States did enter, the Bay built the ships. Henry Kaiser's four yards at Richmond turned out seven hundred forty-seven of them, five hundred nineteen of those Liberty ships. The National Park Service keeps the figures. Richmond went from about twenty thousand people to more than one hundred thousand. One hull, the Robert E. Peary, went from keel to launch in four days, fifteen hours, and twenty-nine minutes, from November eighth to November twelfth, nineteen forty-two. Four days for one hull. A town of twenty thousand became a town of more than a hundred thousand. Bechtel's partner yards and Bechtel's Pacific bases, from Alameda out across the islands, are a different company on the same water. Narrator (Kore): Earlier that year, on El Camino Real in San Bruno, a racetrack became a prison. Narrator (Kore): Tanforan. In use from April twenty-eighth to October thirteenth, nineteen forty-two. Densho, the Japanese American history project, counts eight thousand thirty-three people held there. Sixty-

four percent of them were citizens of the United States. About thirty-seven hundred lived in horse stalls, three to six people in a stall built for one horse. The artist Miné Okubo was held there. SPOKEN (Zubenelgenubi: Miné Okubo, remembering Tanforan, flat, no plea): "I was given a number and placed in a horse stall." Narrator (Kore): Fred Korematsu was held there too. Stanford sent books to the camp library. Food, Densho records, cost thirty-seven cents a day for each person. The Army was spending fifty cents a day on a soldier. The track was converted for about one million one hundred eighty-eight thousand dollars, about twenty-four million in grocery money. Narrator (Kore): The name on the gate is a rancho. Toribio Tanforan was the grandson-in-law of José Antonio Sánchez, the grantee of Rancho Buri Buri. The stalls stand on land from that grant, beside the old royal road. Last hour the buses left Palo Alto for a different track, Santa Anita. This is the other assembly center, the one on the Sánchez grant. Eight thousand people, and a horse's floor. Narrator (Kore): Set that spring against a check signed in January, on an island. On January third, nineteen forty-two, on Corregidor, President Quezon of the Philippines awarded Douglas MacArthur five hundred thousand dollars, about ten million in grocery money, with smaller sums for his aides. Carol Petillo found the order in nineteen seventy-nine. The same year a general is paid a fortune on a fortress, Bay Area families are numbered into stalls at thirty-seven cents a day. Both are the war. They are not the same experience of it. Narrator (Kore): Jump back to the peninsula the leases will cover. The jobs arrive. The houses do not, not for everyone.

07. Across the Creek

[CUE: singing-bowl chapter chime] [MUSIC: tragic from 0s, ducked] [AMBIENCE: fs_636552.mp3]

Narrator (Kore): Across the Creek. Narrator (Kore): San Francisquito Creek is a ditch with a county line in it. Palo Alto, in Santa Clara County, on the Stanford side. East Palo Alto, in San Mateo County, on the other. The name says east. The map says north. A later newspaper would have to remind readers that the poorer city is not east of the rich one. It is across the water, and across Highway 101. Narrator (Kore): The name itself was a compromise, and it has a date. The city's own general plan records an informal election on December twelfth, nineteen twenty-five. The area had been called Ravenswood, and Runnymede. East Palo Alto won the vote. There was talk, even then, of annexing to Palo Alto. The enabling law never arrived. The place kept a name that sounds like a neighborhood of the rich city, and a government that was not that city's. Narrator (Kore): Richard Rothstein, in The Color of Law, uses Palo Alto as a case, and Harris follows him onto the ground. After the war the federal government, through the Federal Housing Administration, became a superbank for private mortgages. The loans were designed to cost the Treasury nothing, which meant the government only backed borrowers and neighborhoods it called solid. The old Home Owners' Loan Corporation maps colored districts by investability. Red was the color for black neighborhoods. The practice took the name redlining. A red line on a map is a refusal of a loan, drawn in advance of the applicant. Narrator (Kore): Restrictive covenants did the same work in the deed. A covenant is a promise written into the title. These promises said a buyer

could sell later only to white buyers. In nineteen forty-eight the Supreme Court said courts would not enforce racial covenants. The FHA did not therefore start insuring black buyers. Harris is blunt. The agency did not lend to black buyers under any circumstances. It brought the red line along. Narrator (Kore): Here is the scene where liberal intentions met the form. In the late nineteen forties the Peninsula Housing Association of Palo Alto, middle-class and working-class residents, bought an undeveloped ranch near Stanford and planned a cooperative subdivision. Hundreds of houses. Common spaces. Of the first one hundred fifty families, Harris writes, two percent were black. That was enough. The federal insurers would not back the construction loans. Banks, including Bank of America, the bank from the North Beach plank, did not make the loans the FHA refused. The association gave up and sold the land to a developer who accepted the condition. Whites only. The subdivision is called Ladera. It is still there. The co-op that tried to include black families is not. Narrator (Kore): Two percent of one hundred fifty is three families. Three families, and a two hundred sixty acre ranch next to Stanford in nineteen forty-eight loses its insurance. That is the number. A policy can be triggered by a handful of people and still cover a landscape. Narrator (Kore): Blockbusting was the other tool, and it used a sale rather than a refusal. Into the nineteen fifties a white resident on the eastern side of Palo Alto, across Highway 101, sold his house to a black family. Speculators went door to door and told the white neighbors the block was changing, so prices would fall, so they should sell now. The speculators bought at fire-sale prices and sold at a premium to black workers who were starved for houses they were allowed to buy. Within six years, Rothstein writes, and Harris repeats, East Palo Alto went from exclusively white to eighty-two percent black. Narrator (Kore): Eighty-two percent, against a starting point of nearly none, in six years. Not because a law ordered black families into one town. Because the loan, the deed, and the rumor did it, and because Palo Alto zoned less than ten percent of its housing land for multifamily buildings and set a minimum size for a single-family lot. Density, which is how working people share a mortgage, was planned out of the city next to the jobs. Harris adds one name from the payroll. Art Fong, a Chinese-American recruit at Hewlett-Packard, came home from the war and could not find a rental in Palo Alto. The park would hire people the city would not house. Narrator (Kore): Housing policy is industrial policy. You can lease an orchard to a missile company and still decide, with a zoning map and a federal form, who sleeps near the gate. The university does not have to pass the covenant. It only has to sit inside a region that did, and to collect rent from the employers on the white side of the freeway. Narrator (Kore): The eighty-two percent is the number. The creek is the border you can walk. Narrator (Kore): The story of our streets, while we are already outside.

08. Three Landings

[CUE: singing-bowl chapter chime] [MUSIC: brahms1_1 from 0s, ducked] [AMBIENCE: crowd.mp3]

Narrator (Kore): Three Landings. Narrator (Kore): April seventeenth, nineteen fifty-one. San Francisco airport. Douglas MacArthur comes down the stairs into the floodlights. Six days ago Harry Truman fired him from command in Korea. He has not stood in the

continental United States in fourteen years. Governor Earl Warren is there to take his arm. Signs in the crowd say MacArthur for president. The next morning he stands on the steps of City Hall and renounces any run for office. Narrator (Kore): The years he is coming home from are the occupation of Japan. As supreme commander he broke up the tenant farms. Tenant-farmed land fell from forty-five point nine percent in November nineteen forty-five to nine point nine percent by August nineteen fifty. About four point seven six million acres changed hands, on the order of six hundred Stanford farms. His father, Arthur MacArthur Junior, had been military governor of the Philippines from May fifth, nineteen hundred, to July fourth, nineteen oh one, clashed with Taft, and left the landlords where they were. The son was a military adviser there in the nineteen thirties, not a civil reformer. The breaking-up happened in Japan. Narrator (Kore): Oakland had already put the son's name on a boulevard. In March and April of nineteen forty-two the council stitched old streets into one, and the signs went up on April eleventh, while he was still in the Pacific. The freeway you drive now, Interstate 580, got the name later, a nineteen sixty-eight resolution on a road that was already open. Narrator (Kore): In July nineteen fifty-two he gave the keynote at the Republican convention and let his name be put in nomination. Eisenhower won on the first ballot. Narrator (Kore): The peace was signed without him, a short walk from City Hall. On September eighth, nineteen fifty-one, at the War Memorial Opera House, Japan and forty-eight other nations signed the Treaty of San Francisco. The Soviet Union, Poland, and Czechoslovakia refused. The same day, at the Presidio, the United States and Japan signed the security treaty that kept American troops in Japan. MacArthur's name is not on either one. Narrator (Kore): March seventeenth, nineteen forty-six. LaGuardia Field, five years before MacArthur's landing. Herbert Hoover is seventy-one. He boards a military C-54. Truman has asked him to go and count a famine. He tells the reporters he has been called back like an old family doctor. Five weeks. About thirty-five thousand miles. Twenty-five countries. In Germany he walks the ruins of the offices and the bunker. In Vienna he rides in Hitler's Mercedes. That same year the Hoover Institution was offered the only original portions of the Goebbels diaries. Handwritten pages from nineteen twenty-five and twenty-six. Typed pages from nineteen forty-two and forty-three. The larger Nazi party archive in the library is a microfilm copy. The paper originals of that archive went to the German federal archive. Narrator (Kore): The tower those pages sit under was already up. Arthur Brown Junior designed it, two hundred eighty-five feet, and in nineteen thirty-eight Hoover had settled on six hundred thousand dollars to build it, about fourteen million now. The bells had come from the Belgian pavilion at the nineteen thirty-nine World's Fair. The great bell says, for peace alone do I ring. They dedicated the tower on June twentieth, nineteen forty-one, Hoover in the room, less than six months before Pearl Harbor. During the war it was a listening post, recording broadcasts from the Far East. In nineteen fifty-nine the trustees set the Institution apart, inside the university, and Hoover wrote that its job was to show private enterprise could best serve the people. A bell for peace. A diary of the man who ran the opposite. Same shaft. Narrator (Kore): Page Mill Road is the lumber road under the park. William Page, a timber man from New York, cut it in the early eighteen sixties out to a mill near today's California Avenue, in the settlement then called Mayfield. On December twelfth, nineteen sixty-nine, Francis Ford Coppola and George Lucas open American Zoetrope in a warehouse on Folsom Street. Narrator (Kore): Nineteen

seventy-three. A small private plane, and two men hopping California for a ballot measure. Milton Friedman, the Chicago economist, spent the day with Ronald Reagan barnstorming for Proposition One, a cap on state spending written into the state constitution. By the last stop Friedman could answer the question a reporter put to him. SPOKEN (Puck: Milton Friedman, remembering the day, plain, decided): "I was able to give an enthusiastic yes to a reporter's question as to whether I would support Reagan for president." Narrator (Kore): The proposition lost, narrowly. The day in the plane did not. Back to nineteen sixty-eight, and a hall in San Francisco, where an institute that keeps the war's contracts is about to show the public a mouse.

09. The Demo and the Contract

[CUE: singing-bowl chapter chime] [MUSIC: schubert2 from 0s, ducked] [AMBIENCE: hum.mp3] [ERA: hum]

Narrator (Kore): The Demo and the Contract. Narrator (Kore): December ninth, nineteen sixty-eight. San Francisco. The Fall Joint Computer Conference, in Brooks Hall. Douglas Engelbart, of the Stanford Research Institute in Menlo Park, sits at a console and shows a room what a computer can be if it is a tool for a mind instead of a batch of cards. Narrator (Kore): The name Mother of All Demos comes later. On the day, the Smithsonian's account of it, he had asked for the slot in March and been given about ninety minutes in a hall that could hold two thousand. He said later he was nervous as hell. The stage lights were bright enough that he could not see the audience. His terminal was tied, over telephone lines, to a computer thirty miles away at SRI, and to colleagues in Menlo Park who joined the picture on the screen. A mouse under his hand. The prototype, the museum notes, he and Bill English had built at SRI in nineteen sixty-four. Text that could be edited, pointed at, rearranged, copied. He typed a word, then the same word again, and showed that a mistake could be backed up without a card punch. The Defense Department's research agency and the Air Force had paid for the work. A contract cover from the following spring lists an ARPA order and a sum of about one point five million dollars for the study of computer-augmented management. The index for nineteen sixty-eight is one hundred four point five. That check is roughly fourteen and a half million in grocery money. A demonstration of a mouse, funded like a weapon, because in nineteen sixty-eight a great deal of advanced computing is a military purchase. Narrator (Kore): The man on the stage had been a Navy radio man. He shipped for the Philippines on V-J Day and, on the way, read Vannevar Bush, the Atlantic piece reprinted in Life. In nineteen fifty he was an engineer at Ames, at Moffett Field, and he decided the rest of his life was for a screen that made hard problems solvable. Pretty much everyone, Harris notes, thought it was nuts. SRI hired him. A small Air Force grant let him write it down. The demo is that idea, in public, twenty-three years later. SPOKEN (Rasalgethi: Douglas Engelbart, on a stage, December 1968, curious, not sales): "If in your office, you, as an intellectual worker, were supplied with a computer display backed up by a computer that was alive for you all day, and was instantly responsive to every action you have, how much value could you derive from that?" Narrator (Kore): That is his question, as the Smithsonian records it. Value, for an intellectual worker. Not a weapon. Not a market. A desk. Narrator (Kore): The day had a

second audience, and it was not the Pentagon's. Stewart Brand, who had been around the acid tests, stood with the camera. Bill Paxton, who worked the demo, remembered him behind the lens, and remembered the feedback as psychedelic. SPOKEN (Alnilam: Stewart Brand, remembering the demo, still impressed): "I've never seen anything that was so dangerous, and undertaken with such bravado." Narrator (Kore): Harris puts the cost of that single day at one hundred seventy-five thousand dollars, about one point seven million in grocery money, inside a contract year that was already about one and a half million. A truck had hauled the gear over Skyline Boulevard. A projector borrowed from NASA put the picture on the wall. The tool for the mind was staged like a show. Narrator (Kore): The show and the drug are neighbors on this peninsula, and they are not the same event. Around the turn of the sixties, Ken Kesey, a Stanford writing fellow, took part in a government drug study at the Menlo Park veterans hospital on Willow Road. The published accounts split the year between nineteen fifty-nine and nineteen sixty-one. He was paid about seventy-five dollars a session, on the order of eight hundred dollars now. Historians tie that study to CIA money. They do not file those trials under the name MKUltra. The Palo Alto veterans hospital is a different place. In nineteen fifty-five the university ceded eighty-seven and a half acres on Miranda Avenue for that one. Willow Road is the other door. Narrator (Kore): On May fifth, nineteen sixty-five, a band called the Warlocks played Magoo's Pizza in Menlo Park. On December fourth, at an acid test in San Jose, the same musicians went on as the Grateful Dead, the first time under that name. Stewart Brand did not debut them. He co-produced the Trips Festival in San Francisco the next January. That is a later night. Al Hubbard, a veteran of the wartime OSS, turned up in the same years at an SRI project called Alternative Futures, in khakis, with a badge and a pistol, running trips and gathering opposition data on the New Left. Harris's account, drawing on Lee and Shlain, says he was not a CIA operative as such. He was in the room. Narrator (Kore): There is a theory, written mostly about Laurel Canyon in Los Angeles, that intelligence agencies manufactured the counterculture. It is a theory. What the Palo Alto record shows is smaller. A government drug study. A man from the old intelligence service inside an SRI project. And a mouse, paid for by the same research agency that paid for the war. Nobody has shown that the Grateful Dead were a government project. The overlap is the fact. The conspiracy is the theory. The free-love surface and the Pentagon money are both on the map. They are not the same plot. Narrator (Kore): Dual-use means one letterhead and two products. SRI's history says the campus learned in nineteen sixty-five of two chemical-warfare contracts, and in nineteen sixty-seven of contracts tied to the war in Southeast Asia. The institute says the chemical work was almost exclusively defensive, and that many of the demands were unfounded. The students said defensive did not excuse it, and that the trustees owned the institute. Engelbart's contract is the mouse. The chemical contracts are other rooms. Narrator (Kore): The Advanced Research Projects Agency pays for the demo and, through the same war economy, pays for the work the students want stopped. Harris writes that SRI helped develop the CIA's strategic-hamlet plan, and that by nineteen sixty-eight the institute's counterinsurgency work was running to millions of dollars. The landlord of the idea and the landlord of the weapon are not always the same man. In Menlo Park, in nineteen sixty-eight, they share a name. Narrator (Kore): Four months later the students stop asking from the sidewalk.

10. April Third

[CUE: singing-bowl chapter chime] [MUSIC: borodin from 0s, ducked] [AMBIENCE: crowd.mp3]

Narrator (Kore): April Third. Narrator (Kore): April third, nineteen sixty-nine. Students meet and take the date as a name. The university's news service, writing the next day, records what they opposed. All classified research. Research in chemical and biological warfare, and in counterinsurgency. Studies used against the people of Vietnam, Laos, and Thailand. They voted to ask the trustees, who were to meet on April eighth, to end that research at the Stanford Research Institute. Narrator (Kore): They are asking the trustees to govern an institute the trustees already control. SRI's history says the mood had shifted. By the end of nineteen sixty-eight many protesters wanted the institute cut loose, and then understood that a separate SRI could keep the contracts with nobody on campus left to answer. Closer ties, not a sale. SRI's president, Charles Anderson, and much of the staff wanted independence, and no campus board deciding which contracts were moral. The fight is over who is responsible. Narrator (Kore): On April tenth the news service reports a second meeting. Students who packed Dinkelspiel Auditorium voted the trustees' responses unacceptable, and decided to sit in. The same release says that on the evening of April eighth, students filled the Applied Electronics Laboratory. They intended to stop classified research there. Narrator (Kore): The Applied Electronics Laboratory is Terman's building. Harris calls it a postwar version of the Radio Research Laboratory he had run at Harvard. Built in the early nineteen fifties with the Office of Naval Research, to house the offspring of the wartime electronics. Its director, Bill Rambo, was one of Terman's men from Cambridge. When the complaints started, Rambo tried to lower the temperature. The building had a sinister reputation not fully deserved, he said. Most of it was freely accessible. The students had decided that accessibility was not the question. The question was the contracts. Narrator (Kore): They held the building for more than a week. Harris records that they took the laboratory's printers and used hundreds of thousands of its blank sheets to print and distribute their own paper. An early flyer, he notes, misnamed the place the Atomic Energy Lab, which is almost funny, because atomic energy was one of the few military sciences Stanford did not lead. The mistake did not last. The occupiers turned the building into their own research office, reading the institute they had entered. Narrator (Kore): The newsletter has a name, and the movement's own archive keeps it. The first issue, April tenth, was called Unclassified. The ones after were called Declassified. They were printed, the archive says, on the presses inside the laboratory, by the authority of the people in the building. Students in a room they had occupied, using the machine that belonged to the classified work, to publish their own account of a laboratory they could finally see. Narrator (Kore): On April fourteenth, when faculty and staff came through to identify people for a discipline process, the demonstrators gave only one name. The April Third Movement. President Kenneth Pitzer, in his first year, sent written notice that they were violating university policy by stopping the ordinary work of the lab. He did not, in those first days, clear the building with police. The next day the news service records a debate inside the occupation about classified files, and a decision that is easy to miss. They reaffirmed that they would not tamper with the files. They had come to

stop the work, not to steal it. A later sit-in, at Encina Hall, ended at the start of May when the sheriff was called. The laboratory occupation was allowed to become a meeting. Narrator (Kore): On April seventeenth the news service records the line that ties the occupation back to the lease. If they succeeded here, the next step would be the industrial park. SPOKEN (Achird: an occupier, April 1969, level, not a chant): "If we are successful here the next step will be to move to the Stanford Industrial Park." Narrator (Kore): The park. Varian's lease. Hewlett-Packard's headquarters. Lockheed's research complex. The students had followed the circuit Terman built: from the classified bench, to the institute in Menlo Park, to the tenants on the orchard. A ground lease does not only collect rent. It points. Anyone who can read a map of the contracts can walk it. Narrator (Kore): The month after the laboratory, the map left the building. On Friday, May sixteenth, the day after the fight over People's Park in Berkeley, a flyer went out under the words smash war research. It drew a map. The Stanford Research Institute. The Hoover Institution. The business school. Varian. Hewlett-Packard. Lockheed. IBM. The caption, in the flyer as the archive keeps it, reads: since World War Two, how to run an empire from Palo Alto. About three hundred people blocked the entrance to the industrial park, at Oregon Expressway and Hanover. They used a school bus and let the air out of its tires. About one hundred fifty police. Tear gas around eleven in the morning. Fifteen arrests. Leslie Berlin and Malcolm Harris both keep the day. Berlin counts nine days inside the laboratory. Harris counts more than a week. Narrator (Kore): Inside the argument over the word defensive, the physicist H. Bruce Franklin answered a provost in the words Harris records. SPOKEN (Achird: H. Bruce Franklin, 1969, sharp, not shouting): "Those are electronic countermeasures, used by our bombers to jam enemy radar. If you think that's defensive, you're out of your mind." Narrator (Kore): The same season, the newsletter of the occupation printed a song in the voice of a researcher at the institute. SPOKEN (Zubenelgenubi: a song from the occupation newsletter, quiet, almost to himself): "I have never killed a man. I have never seen him die. These formulas: they could be for jelly. I don't think I know." Narrator (Kore): That is the sentence the hour has been walking toward. A person can do the work and still not have to watch it land. Narrator (Kore): A poster from those weeks says strike against the Stanford Empire. Empire, here, is the students' word for one board over a university, a research institute, a park, and a war.

11. Off the Books

[CUE: singing-bowl chapter chime] [MUSIC: brahms3_3 from 0s, ducked] [AMBIENCE: wind.mp3]

Narrator (Kore): Off the Books. Narrator (Kore): The trustees' answer was not the answer the occupiers had asked for when they asked for control. Narrator (Kore): SRI's institutional history dates the independence to March thirty-first, nineteen seventy. The institute was sold, in effect, to itself. The formula that stuck was a payment to the university of about one percent of gross receipts, dropping to half a percent after twenty-five million dollars had been paid. By nineteen eighty-nine, SRI records, more than twenty-five million had gone back to Stanford. The institute could keep the Stanford name for a term of years. It could not be told, by a campus meeting, which contracts to

refuse. Narrator (Kore): One percent. A small number that behaves like a royalty. The university stops being the owner of the work and becomes a payee of the work. Responsibility moves off the campus. The invoice does not. Narrator (Kore): The institute's own people took to calling the percent alimony. Some years it was larger than the net profit. The staff had not wanted a campus meeting to define moral work. They got their independence. The students who had wanted the work stopped got a letterhead that no longer said Stanford, and a check that still did. Narrator (Kore): Picture the door on the morning after March thirty-first. The same scientists. The same contracts. A new line on the letterhead, or the old line kept for a few more years by agreement. A student who had sat in the Applied Electronics Laboratory can no longer point at the institute and say you own this. The trustees can say we sold it, and here is the percent. Down on Page Mill the leases do not notice. Varian is still a tenant. Hewlett-Packard is still a tenant. Lockheed's research complex is still a tenant. The Honors Cooperative still charges double tuition to the companies that send their engineers to class. The statute from eighteen seventy-two still lets a researcher walk out and start again. The creek is still the county line. A new letterhead did not move any of it. Narrator (Kore): [SILENCE] [HARD] The rebellion did not stop the machine. It made the machine look less like a weapons lab. Narrator (Kore): Terman's sentence held. The grant still said the land could not be sold. Nobody had to sell it. They leased it, they staffed it, they classified it, and when the classification became a scandal they moved the signature a few miles down the road and kept the percent. Narrator (Kore): Next hour the gate is a road on university land, and the people waiting at it are investors. Not yet. Tonight the orchard has tenants, the laboratory has been sat in, and the institute has been given its own letterhead. Narrator (Kore): [SILENCE] [HARD] The farm was never for sale. That is why they could keep it. Narrator (Kore): This is Cardinal.

The Road — full audio script (voice/cue markup)

The Road — full audio script (voice/cue markup)

Episode 7 of 8. Cardinal. Narrator = Kore. Documentary quotes = Charon. Newspapers = Sadachbia. Other voices are listed per line. [HARD] = music gated out. [SILENCE] = about 2 seconds before a reveal. Cue and ambience names are reused from the series pipeline. Sources are not narrated; they live in the review file.

01. Sand Hill

[CUE: singing-bowl chapter chime] [MUSIC: schubert2 from 0s, ducked] [AMBIENCE: wind.mp3, hum.mp3] [MOTIF: railroad stinger]

Narrator (Kore): The Road. Narrator (Kore): Cardinal. Episode seven. The Road. Nineteen sixty-nine to two thousand two. Narrator (Kore): Nineteen seventy-two. Menlo Park, California. A suburban road runs west from the freeway, past the edge of a university, and up toward the coastal hills. The buildings are low. They sit back among oak trees, with more parking than a visitor can explain. There is nothing on the road that looks like power. That is the point of a gate that wants to be mistaken for geography. Narrator (Kore): At number three thousand, a partnership is opening its doors. The partners are Eugene Kleiner and Tom Perkins. Kleiner had a private office in the building before the firm even had a name. The historian Leslie Berlin, who wrote the history of these years, says the partnership began in nineteen seventy-two with an eight-million-dollar fund, one of the largest in a business that was still small enough to fit in a handful of rooms. Narrator (Kore): Hold the number. Eight million dollars in nineteen seventy-two is about sixty-four million dollars in today's money. Half of it came from one man, Henry Hillman, a Pittsburgh industrialist. The other half came from individuals, trust funds, and institutions, among them an insurance company and Rockefeller University. Narrator (Kore): A gatekeeper is the whole hour, and a gatekeeper is not the person who does the work. A gatekeeper is the person who decides who is allowed to pass. A century earlier, the railroad through this same valley decided which towns could ship their wheat. The charge was called a freight rate, and the men who set it said it was whatever the traffic would bear. Sand Hill Road is about to do a quieter version of the same job. The thing being moved is no longer wheat. It is a company that does not exist yet. Narrator (Kore): Kleiner was not a banker. He was one of the eight young researchers who had walked out of William Shockley's laboratory in nineteen fifty-seven and started Fairchild Semiconductor. As a boy in Vienna he had watched Nazis force his father to open the family safe. A police officer who knew the family pulled the father out of a line bound for the camps. Kleiner never finished high school. He volunteered for the American army, he said, because he had a personal thing to settle, and the G.I. Bill put him through engineering school. Perkins was different. He had made Hewlett-Packard's computer division the biggest in that company, and he had the polish of a man who had studied how a businessman was supposed to look. At Harvard Business School he had sat in the class of Georges Doriot, the French professor who taught a generation of Americans that venture investing was a craft. Narrator (Kore): The same year, nineteen seventy-two, Don Valentine founded a firm called Sequoia Capital. Valentine had run marketing at Fairchild, and then at National Semiconductor. Berlin notes that Sequoia was moving onto Sand Hill Road while other people's money was still learning the address. The operators had become the financiers. They already knew the people worth funding, because they had been those people's bosses. Narrator (Kore): Then Perkins did something that complicates the postcard. Shortly before a young associate named Bob Swanson joined the firm, Kleiner and Perkins left three thousand Sand Hill Road. They moved to the twenty-ninth floor of Two Embarcadero Center, in San Francisco, looking out at the bay. Perkins did not want the neighbors to see whom he took to lunch. Quote (Charon, flat documentary): We didn't want to be considered part of the flock. Eagles don't flock, was our joke. Narrator (Kore): The joke lost. The other firms stayed on the road. Within a decade the low buildings among the oaks were the largest concentration of venture capitalists in the valley, and, Berlin writes, perhaps in the world. You can move your own office to a

skyscraper. You cannot move the gate, once everyone else has agreed that the gate is a road. Narrator (Kore): One more fact, because the money did not come only from rich families. In nineteen seventy-four Congress passed a pension law with a stern name, the Employee Retirement Income Security Act, and a sterner idea inside it: a prudent person does not gamble with workers' retirement. Fund managers read that as a ban on anything speculative. Venture capital is speculative by design. Most of the companies fail. In nineteen seventy-eight Congress cut the tax on capital gains and eased those rules, and the next year the Labor Department clarified the point that mattered. Prudence would be judged on the whole portfolio, not on each wild bet inside it. A pension fund could put a small slice into companies that might go to zero. Narrator (Kore): Berlin's number: within a year, venture capitalists had more than half a billion new dollars to invest. In twenty twenty-six grocery money, that is about two point three billion dollars. The retirement savings of teachers, truck drivers, and clerks had become the raw material of a road in Menlo Park. The beneficiaries were not asked. That is the public-money leg of a private gate, and it is the part the origin stories usually leave on the cutting-room floor.

02. The Ticket Booth

[CUE: singing-bowl chapter chime] [MUSIC: brahms3_3 from 0s, ducked] [AMBIENCE: hum.mp3] [ERA: hum]

Narrator (Kore): The Ticket Booth. Narrator (Kore): January eleventh, nineteen seventy-one. A weekly trade paper called Electronic News lands on desks from Boston to Sunnyvale. The headline, over a series by a columnist named Don C. Hoefler, reads "Silicon Valley U.S.A." It is the first time those two words appear in print as the name of a place. Hoefler was writing about semiconductors, the chips cut from silicon, the second most common element in the earth's crust. He was not writing about social networks or search engines. Those did not exist. He was naming a factory district. The name outlived the factories. Narrator (Kore): By the time Hoefler published the name, Stanford had already made a small office permanent. The story of the office begins with a man named Niels Reimers. In the late nineteen sixties he ran a pilot: take what a professor invents, file a patent, and license the patent to a company, in exchange for a royalty. By the summer of nineteen seventy-four, Berlin writes, the pilot was a four-year-old official operation called the Office of Technology Licensing. Reimers and an assistant worked out of a trailer behind Encina Hall. On the wall, in capital letters, a sign said MAKE IT HAPPEN. On Friday afternoons they filed the week's paper over popcorn and a beer called Henry Weinhard's. They called the ritual Management by Beer. Narrator (Kore): Teach the words, because the rest of the hour is built out of them. A patent is a temporary monopoly. The government gives you the exclusive right to an invention for a term of years, and in exchange you publish how it works, so that when the term ends anyone can use it. A license is the permission slip. The university keeps the patent and rents the right to use it. A royalty is the rent, usually a percentage of sales. Equity is different. Equity is a slice of the company itself. Reimers would spend the nineteen eighties arguing that Stanford should take stock, not only cash, because a fast company might design the licensed invention out of its product before a royalty check ever arrived. He estimated that by refusing equity the office had left more than fifty

million dollars on the table. The university let him start taking stock in nineteen eighty-nine. Narrator (Kore): The scale, when it arrived, was not modest. Berlin, writing about the office as of twenty sixteen, says cumulative income from technology licensing was nearly two billion dollars. An Office of Technology Licensing overview a year later puts the cumulative figure at about one point nine billion, through fiscal year twenty seventeen. Three inventions, out of roughly twelve thousand the office had evaluated, accounted for the fortune: recombinant DNA, a set of antibody patents, and Google. The rest was a long tail. That is the shape of this business. You run a ticket booth for decades. A few tickets pay for the building. Narrator (Kore): Compare it with the year the booth first beat everyone else. In nineteen eighty-three Stanford earned more from technology licensing than any other university in the country: two point five million dollars, about eight million in twenty twenty-six grocery money. Almost sixty percent of that year's income came from a single patent, the one Reimers had filed on a way of moving genes from one organism into another. We are going to that patent now, because it is where the university claimed ownership of a piece of life, and where a man who had just been fired decided to start a company anyway.

03. Patenting Life

[CUE: singing-bowl chapter chime] [MUSIC: mozart40_2 from 0s, ducked] [AMBIENCE: hum.mp3]

Narrator (Kore): Patenting Life. Narrator (Kore): Summer, nineteen seventy-four. Niels Reimers is at his desk in the trailer, going through the mail. Two scientists have done something that will split the century. Stanley Cohen, at Stanford, and Herbert Boyer, at the University of California, San Francisco, have found a way to cut a gene out of one organism and stitch it into the DNA of another. DNA is the molecule that carries heredity, the long instruction sheet inside a cell. Their method let a person rewrite a line of that sheet. Reimers decided the method should be patented, and he began the fight to file. Narrator (Kore): The fight started immediately. Berlin writes that the application was criticized for its potential to undermine the university's basic mission, which was to increase the world's stock of knowledge, not to fence it. Faculty said the claim reached too far. The press ran nightmares. The mayor of Cambridge, Massachusetts, asked whether a strange orange-eyed creature, supposedly sighted nearby, could be connected to the experiments, and said he would be damned if his city was going to be diseased over academic freedom. In nineteen seventy-five scientists met at Asilomar, on the Monterey coast, and agreed to regulate themselves. Berlin's dry observation is that the gesture advertised the fear instead of calming it. By the end of nineteen seventy-seven Reimers's clipping file was three inches thick. He kept going. His own line, later, was that he kept thinking about the potential for good: drugs, hormones, vaccines that did not depend on a weakened virus. Narrator (Kore): Now the fired kid. Robert Swanson was twenty-eight. He had talked his way through an MIT chemistry degree and a management degree in five years, opened a Citicorp venture office in San Francisco, and been hired by Eugene Kleiner, who saw a version of his own nerve. Swanson lasted less than two years. The break came over a company called Cetus. At a long lunch, a Nobel laureate mentioned recombinant DNA as something Cetus might explore. Nobody else

at the table wanted it, including Tom Perkins, who had come to the lunch to show how worried the firm was. Swanson still thought he had a job. Perkins told him the partners would like to have just the two of them working there. Swanson could stay on the payroll until the end of nineteen seventy-five. SPOKEN (Puck: Bob Swanson, twenty-eight, just fired, quietly to himself): "Maybe I'm not very good." Narrator (Kore): He had time, a Ping-Pong table that doubled as a desk, and a stack of cheap meals. He cold-called the scientists who had been at Asilomar. He had never heard of Cohen, Boyer, or Reimers. Boyer, unusually for a biologist in nineteen seventy-six, was willing to talk about a company. His cats were named Watson and Crick. Most of his colleagues would not have taken the call on a university phone. Arthur Levinson, who would later run Genentech and sit on the boards of Apple and Google, remembered that as a young biochemist he walked down the street to a pay phone if he needed to speak to a company. A venture capitalist named Brook Byers compared a scientist thinking about business to Bob Dylan going electric. People wondered what was up, and whether bad would come of it. Narrator (Kore): Swanson asked Kleiner and Perkins to pay him a salary while he incubated the company, the way they had done for the founders of Tandem Computers. They said there was no room. The fund was barely making money. So Swanson and Boyer started it themselves. Swanson's first proposed name was Herbob, a mash of Herbert and Bob. Boyer countered with a word built from genetic engineering technology. Genentech. Narrator (Kore): The record of what happened next is a set of dates and a stock table. Genentech was founded in nineteen seventy-six. On October fourteenth, nineteen eighty, it began trading on the NASDAQ exchange under the symbol GENE. The opening price was thirty-five dollars a share. In the first twenty minutes it hit eighty-nine, the fastest first-day gain Wall Street had seen. At the close the company was valued at five hundred thirty-two million dollars. In twenty twenty-six grocery money, that is about two point two billion. Swanson and Boyer had each put in five hundred dollars at the start. Berlin says each of those stakes was worth about sixty-five million dollars by the closing bell. The slice Kleiner and Perkins had eventually bought, for a hundred thousand dollars and then another hundred thousand, was worth the same. The firm that had fired Swanson was now rich because of him. Narrator (Kore): Sebastian Mallaby, in The Power Law, uses that fund as the picture of the math. As of nineteen eighty-four, the fourteen investments in Kleiner Perkins's first fund showed a combined profit of two hundred eight million dollars, about six hundred seventy million today. Ninety-five percent of the profit came from two companies, Tandem and Genentech. Without those two, the fund would still have beaten the stock market. With them, it became the story the industry tells itself. Narrator (Kore): That story has a name. A power law is a distribution in which the winners do not win a little more than the losers. They win so much, and so increasingly, that the middle barely matters. Most venture bets go to zero. A handful return the fund. The rational response, if you believe the math, is to swing only at companies that could be enormous, and to ignore the company that might decently double. A decent double is what an ordinary investor calls a success. On Sand Hill Road it is a waste of a seat. Narrator (Kore): And the patent? Stanford and the University of California licensed it widely, on purpose, because an exclusive deal would have been impossible. Cohen was consulting for a company that would compete with Genentech. Over the life of the patents, which expired in December nineteen ninety-seven, four hundred sixty-eight companies took a

license. Maryann Feldman and her colleagues added up the money: about two hundred fifty-five million dollars in licensing revenue to Stanford and the University of California by the end of two thousand one. In twenty twenty-six grocery money, that is about four hundred eighty million dollars. Ninety percent of it was royalties on actual products. The other ten percent was the cover charge, the fees companies paid just to be allowed in the room. A license, for a young biotech firm, was also a permission slip. It said a famous university had looked at your science and not thrown you out. Narrator (Kore): In two thousand nine the Swiss drug company Roche bought the rest of Genentech for forty-six point eight billion dollars. Every share that had cost thirty-five dollars at the opening bell in nineteen eighty was worth four thousand five hundred sixty dollars. Swanson did not live to see that sale. He stepped down as chief executive in nineteen ninety, after Roche bought a majority, and he died of brain cancer in nineteen ninety-nine, at fifty-two. Levinson's line is the one worth keeping. The man who spent his life turning this technology into drugs that saved other people did not get a drug that saved him.

04. Pong to Apple

[CUE: singing-bowl chapter chime] [MUSIC: borodin from 0s, ducked] [AMBIENCE: hum.mp3, crowd.mp3]

Narrator (Kore): Pong to Apple. Narrator (Kore): Nineteen eighty. A meeting at Atari, the video-game company that had taught a generation to lose quarters in a bar. Al Alcorn, the engineer whose work had helped launch the company in nineteen seventy-two, wants to build a new kind of cartridge system, one that would throw a hologram behind the game. The chief executive, Ray Kassar, tells him to talk to marketing. The marketing manager tells him to write a business plan. SPOKEN (Zubenelgenubi: Al Alcorn, an Atari engineer, dry and annoyed): "We didn't need a business plan for Pong or the VCS." Narrator (Kore): He needed one now. The line is Berlin's, from Alcorn's mouth, and it is a eulogy for a decade. Pong was the table-tennis game that made Atari. The VCS was the box that brought the games into the living room. Both had been built by people who shipped first and wrote the memo second. By nineteen eighty the memo had won. Kassar nodded at prototypes and released almost none of them. Alcorn's hologram machine, called Cosmos, drew thousands of orders at a trade show and was never sold. He had four words for what had happened to the company. All creativity had ceased. Narrator (Kore): Across the valley, a different company was turning the memo into a religion. Apple had been started by Steve Jobs and Steve Wozniak, and saved as a business by Mike Markkula, a former Intel manager who wrote the checks, hired the grown-ups, and cared about how the thing looked. In December nineteen seventy-nine he wrote a one-page creed called "The Apple Marketing Philosophy." It had three words at the top: Empathy, Focus, Impute. Under the third, he wrote this. Quote (Charon, flat documentary): Our image is the combined result of everything the customer sees, hears, or feels from Apple. Narrator (Kore): Read it again, slowly, because it is also a description of a university. The sandstone, the red tile, the church at the center of the Quad, the tour guide walking backward: those are an image, and the image is doing work. Markkula called the work "impute." If you present a product in a slipshod way, he

wrote, people will perceive it as slipshod. If you present it in a creative, professional way, you impute the qualities you want them to feel. You do not argue them into it. You surround them. Narrator (Kore): This is the cult energy of the Protestant fortune, and it is worth naming for a listener who has never stood on this peninsula. Catholic California put saints on the land. San Francisco, Santa Clara, the missions, the royal road. The fortunes that arrived after the gold rush were largely Protestant, and then, in the late twentieth century, frankly secular. They did not build missions. They built companies that asked for the feelings a mission used to ask for: belief, pilgrimage, a story about the future in which you are already late. Apple is the clearest example, and it did not happen by accident. Markkula wrote the instructions down. Narrator (Kore): The money that made the instructions matter came, in part, from the East, and this is the one eastern beat of the hour. In the fall of nineteen seventy-seven Markkula and Jobs flew overnight to New York. They changed into new blue suits in a men's room at Thirty Rockefeller Plaza and rode the elevator to the fifty-sixth floor, to the offices of Venrock. Venrock is the venture firm of the Rockefeller family. The partner they had to convince was Peter Crisp. Jobs talked about a computer in every living room. The partners, by their own later account, barely heard the specifics. The whole territory was too speculative to take literally. They stepped into the hall. SPOKEN (Algieba: Peter Crisp, a Rockefeller venture partner, remembering a shrug): "We went out in the hall, four or five of us, we looked at each other, and we shrugged our shoulders and said, What the hell?" Narrator (Kore): They put in three hundred thousand dollars for ten percent of Apple. The deal valued the company at three million dollars. In twenty twenty-six grocery money, the check is about one point seven million dollars, and the company, on that day, is about sixteen million. Crisp later said people gave Venrock too much credit for being smart. Arthur Rock, the financier who had backed Intel, took a fifty-thousand-dollar slice of Venrock's own allocation, because Crisp owed him a favor from nineteen sixty-eight. Old eastern money, a shrug in a hallway, and a company that would spend the next forty years imputing a feeling to anyone who held its objects. Narrator (Kore): The feeling is the product. That is Markkula's sentence, and it is why a machine for playing Pong and a machine for sending email can both be sold as a way of life. The university had been practicing the same sale since the day it opened, with a chapel in the middle and a railroad fortune underneath. Apple just had the honesty to put it in a memo.

05. Myth Machines

[CUE: singing-bowl chapter chime] [MUSIC: hebrides from 30s, ducked] [AMBIENCE: crowd.mp3, wind.mp3]

Narrator (Kore): Myth Machines. Narrator (Kore): December twelfth, nineteen sixty-nine. San Francisco. A warehouse at eight twenty-seven Folsom Street, south of Market. Francis Ford Coppola and George Lucas open a film company and call it American Zoetrope. The word is Greek before it is English. Zoe, life. Tropos, a turn. A zoetrope is a nineteenth-century toy, a drum with slits, spun so that a strip of still drawings looks alive. They were naming a machine that turns still pictures into a world. Narrator (Kore): They had driven up from Los Angeles with a crew of film-school friends, among them the editor and sound designer Walter Murch. None of them, Murch said, had

family in the industry. They wanted American movies made under the influence of the films they loved from everywhere else. In nineteen seventy-two the company moved into the Sentinel Building, at nine sixteen Kearny Street, in North Beach, the old Italian neighborhood where A. P. Giannini had opened a bank in a remodeled saloon in nineteen oh four. Hold that address in your ear. A bank, a film company, and later a tunnel will all turn out to be stories about who gets to tell California what California is. Narrator (Kore): Lucasfilm, the company Lucas built after he left the Folsom Street experiment, started a computer division in nineteen seventy-nine to drag high technology into movies. By the early nineteen eighties Lucas needed cash. He put the division up for sale. Steve Jobs, pushed out of Apple and looking for a next act, bought it. The popular sentence is "Jobs bought Pixar." The record is more precise, and Alvy Ray Smith, who was there, has kept the paperwork. On January thirtieth, nineteen eighty-six, Jobs wrote a certified check for five million dollars. Smith and Ed Catmull signed it over to Lucasfilm. Jobs then put another five million into the new company to capitalize it. Ten million dollars in all. About thirty million in today's money. Narrator (Kore): The first press release, dated February tenth, nineteen eighty-six, was coy. It said Jobs had paid an undisclosed sum in the millions. It also said the new firm would sell a graphics computer for about a hundred twenty-five thousand dollars, a machine faster at pictures than a six-million-dollar supercomputer. About forty employees came along, John Lasseter among them. They owned thirty percent. Jobs owned seventy, and he was chairman, not the founder in the way the brochures later implied. Catmull was president. Smith was vice president. Smith is blunt about the title: Jobs was the investor. Narrator (Kore): The computer was not the fortune. The fortune was the toy, scaled up. In November nineteen ninety-five Pixar released Toy Story. A week later the company went public, in what Variety called the largest initial public offering of the year. In January two thousand six the Walt Disney Company bought Pixar. The announced price was about seven point four billion dollars, about twelve billion in twenty twenty-six grocery money. Smith's rueful arithmetic, looking back, was that Disney could have had the group for five million from Lucasfilm, and ended up paying billions. A drum that makes drawings look alive. Hollywood had not moved to the Bay Area. The Bay Area had learned to manufacture myth in software, and then sold the factory back to Hollywood. Narrator (Kore): There is a third Coppola, and he is the one with a theater named after him. August Coppola, Francis's brother and Nicolas Cage's father, was dean of the College of Creative Arts at San Francisco State from nineteen eighty-four to nineteen ninety-two. He persuaded the state legislature to put up fifteen million dollars for a Fine Arts Building, in the middle of a budget crunch. It opened in nineteen ninety-three: labs, a sound stage, and a hundred-and-fifty-seat theater with his name on it. In nineteen ninety-one, instead of a groundbreaking, he held what he called a skybreaking. Alumni tossed beans into a hole. He set a glass model of the building in the hole, poured wine over it, and said they would see how the dreams grew. Then he explained the ceremony. Quote (Charon, flat documentary): The idea is to look up, rather than down. Look up to the sky, the clouds, teach young people to dream. Narrator (Kore): Same impulse as Markkula's memo and the zoetrope: the product is a direction for the eyes. Fifteen million public dollars is six times what Stanford earned from all its licenses in nineteen eighty-three. A state college's new arts building cost more to house than the country's most successful technology office was then earning from owning ideas. The

myth machine was never only a startup. It was also a dean, a legislature, and a room with a hundred and fifty seats.

06. The Story of Our Streets

[CUE: singing-bowl chapter chime] [MUSIC: scottish3 from 0s, ducked] [AMBIENCE: wind.mp3, horses.mp3]

Narrator (Kore): The Story of Our Streets. Narrator (Kore): Drive north out of the valley, across the city, toward the Golden Gate, and the road ducks under a hill in the Presidio. For years the official name of that tunnel was the Presidio Tunnel. Before that, locals called it the Funston Avenue tunnel, after the approach. On November fifth, nineteen eighty-six, the state gave it another name. Senate Concurrent Resolution eighty-six, introduced by San Francisco's state senator Milton Marks, added the words General Douglas MacArthur Tunnel. Caltrans, the state transportation department, still carries both names on its list of structures. Narrator (Kore): Say the first name carefully. Douglas. Not Arthur. Arthur MacArthur was the father, a general who commanded the Department of the Pacific, and who, on the night in nineteen oh three when Theodore Roosevelt dedicated the column in Union Square, stood up and toasted the Army. That column is not his. It belongs to Admiral George Dewey and the battle of Manila Bay. The son is the one on the tunnel: Douglas MacArthur, who commanded in the Philippines, who left Corregidor in nineteen forty-two and said he would return, who ran the occupation of Japan, and whom President Truman fired in nineteen fifty-one for refusing to keep a war inside the lines a president had drawn. The Peninsula and the city have a habit of welding a famous last name to the wrong man. The tunnel is the son's. The monument in the square is an admiral's. The father is a toast at a dinner, and almost nobody orders the right one. Narrator (Kore): Sand Hill Road is the other sign, and it does not commemorate a general. It commemorates a function. The name is plain English. A hill, some sand, a road. What it means, if you can read a gate, is that a company which cannot get a meeting in one of those low buildings does not merely lack money. It lacks the introductions to customers, the executives the partners know by their first names, and the next round of capital, which tends to follow the first round's letterhead. Proximity is the product. Judgment about people travels badly. You have to be in the room, and the room is four minutes from the campus that supplies the people being judged. Narrator (Kore): And one departure, so the map stays honest. In nineteen ninety-eight NationsBank, of Charlotte, North Carolina, bought the San Francisco company that still carried A. P. Giannini's name. The merged bank kept the name Bank of America and put the headquarters in Charlotte. The bank that had lent on a face and a signature from a North Beach wharf, after the nineteen oh six fire, is run from North Carolina. Giannini Hall is still at Berkeley. His bank is not in his city. Narrator (Kore): The jobs stayed. The bedrooms did not keep up. A gate that lets companies through, and does not let roofs through, bids up the ground. Henry George, watching the railroad do this to wheat towns, had a name for who captured the gain: the person who held the land. Stanford, by its founding grant, could not sell the farm. It could lease it. The road at the edge of the farm became the permission slip, and the rent on a bedroom followed.

07. Bayh-Dole

[CUE: singing-bowl chapter chime] [MUSIC: aase from 0s, ducked] [AMBIENCE: fs_42139.mp3]

Narrator (Kore): Bayh-Dole. Narrator (Kore): December second, nineteen eighty. The United States Patent and Trademark Office issues patent number four two three seven two two four, "Process for producing biologically functional molecular chimeras." The application that issued had been filed on January fourth, nineteen seventy-nine, after years of argument. A chimera, in the old stories, is an animal sewn together from different animals. In the patent, it is Cohen and Boyer's method. Stanford and the University of California own it. Ten days later, on December twelfth, President Jimmy Carter signs a law with a dull name, the Patent and Trademark Law Amendments Act. Everyone calls it Bayh-Dole, after the two senators, Birch Bayh of Indiana and Bob Dole of Kansas. Narrator (Kore): Here is what it did, in plain words. If a university, or a small business, or a nonprofit made an invention using federal research money, the institution could elect to own the patent. Before this, the federal agency that had paid for the work generally kept the rights, and a great many inventions sat on a shelf because a company would not spend a decade turning an idea into a product it was not allowed to own. Bayh-Dole flipped the default. The university takes title. The university has to file for the patent, has to try to license it, has to share the royalties with the human inventor, and has to give the government a license to use the invention for government purposes. The taxpayer funded the science. The university owns the paper. A company rents the paper and builds the product. Narrator (Kore): Stanford did not wait for the law. Reimers had been running the office for a decade, and he had already fought the Department of Health, Education, and Welfare for the right to hold title on particular inventions. What Bayh-Dole did was make the Stanford arrangement the national rule, for every campus that took federal money. A local habit became a statute. That is how a model spreads when it is too useful to leave as a local habit. You do not argue the other universities into it. You write it into the condition of their funding. Narrator (Kore): The number to put next to the statute is the one from nineteen eighty-three. Two point five million dollars in licensing income, first in the country, sixty percent of it from the chimera patent. The law did not create that patent. The patent was already filed, and the company built on it had already gone public. The law nationalized the booth. Within a generation, an office like Reimers's was normal equipment at a research university, the way a library was. Berlin's count of the change is the right one. When Reimers started, there were a handful of such offices in the country. After the recombinant DNA patent and the statute that followed it, they were simply how a university did business. Narrator (Kore): Hold the sequence, because it is the sequence. A trailer behind Encina Hall. A patent on a method for rewriting a cell. A fired associate and a scientist with two cats. A public offering that made a five-hundred-dollar check worth sixty-five million dollars in an afternoon. Then a federal law that told every other campus to build the trailer. The gate was no longer a road in Menlo Park alone. It was a clause in the United States Code.

08. The Bechtel Cabinet

[CUE: singing-bowl chapter chime] [MUSIC: brahms4_2 from 300s, ducked] [AMBIENCE: telegraph.mp3]

Narrator (Kore): The Bechtel Cabinet. Narrator (Kore): George Shultz did not take the route the shorthand usually gives him, so the record goes first. He was an economist. He taught at M.I.T. and then at the University of Chicago, where he became dean of the business school. In nineteen sixty-eight and sixty-nine he was a fellow at the Center for Advanced Study in the Behavioral Sciences, on the Stanford campus, in the hills above the Quad. Then Richard Nixon put him in the cabinet three times: secretary of labor, the first director of the Office of Management and Budget, and secretary of the Treasury. He left the Treasury in nineteen seventy-four. Narrator (Kore): He went to Bechtel. Bechtel is a San Francisco engineering and construction company, private, vast, and in the habit of building the things governments decide to want: pipelines, power plants, cities in the desert. Shultz became an executive, and then president. While he was there, he joined the Stanford faculty part time and taught management and public policy. Laton McCartney, who wrote the history of the firm, says Shultz and his wife lived in a modest house on the campus, and that he drove an Oldsmobile when his rank entitled him to a Cadillac. The professorial manner was not a costume he put on later. His colleagues told McCartney that in a company where engineers shouted, George spoke beautifully, logical, calm, the collected professor. It worked on them. Narrator (Kore): In June nineteen eighty-two Reagan asked him to be secretary of state. He was sworn in on July sixteenth. He served until January twentieth, nineteen eighty-nine, six and a half years, the longest stretch in the job since Dean Rusk. Senators asked, at his confirmation, whether a man who had just run Bechtel could conduct the foreign policy of a country in which Bechtel had business. He was confirmed anyway. Sally Denton, writing later about the company's nuclear work, quotes a line from those hearings. SPEECH (Schedar: George Shultz, at his 1982 confirmation, controlled and formal): "Private interests have no business with nuclear weapons." Narrator (Kore): Denton sets the sentence down in order to watch him walk away from it, years afterward, when the laboratories that design those weapons were being handed to private managers and Bechtel was in line. The sentence is his, in nineteen eighty-two. A revolving door is simpler than the scandal sheets make it. A revolving door is a career that passes through a public office and a private firm that does business with that office, in either order, more than once. The person does not have to fix a contract on the way through. The knowledge, the phone numbers, and the benefit of the doubt travel with the body. Shultz's own path is the clean diagram. Campus fellowship. Three cabinet posts. President of the company. Secretary of state. Then, after nineteen eighty-nine, back to Stanford as a professor of international economics, and a distinguished fellow at the Hoover Institution, the think tank in the tower named for the orphan from the first class. Narrator (Kore): He was not the only one. Caspar Weinberger, who had served in Nixon's cabinet and then as a Bechtel executive, became secretary of defense in nineteen eighty-one and stayed through most of the Reagan years. Two of the offices that buy the largest things a government buys, diplomacy and weapons, were held by men who had just been signing Bechtel's paper. The Grove, as a detail and only a detail: the Bohemian Grove, a private encampment in the redwoods north of the city, where McCartney places both the Bechtels and Shultz. The point of mentioning it is scale. The network fit in a

camp, a campus, and a company. It did not need a conspiracy. It needed a small enough number of dining rooms. Narrator (Kore): When this story reaches a blood-testing company valued at nine billion dollars, Shultz will be on its board, a Hoover elder whose presence is itself a kind of license. That is later. For now, keep the mechanism. A university that licenses ideas, a company that builds what governments order, and a man who can walk from one to the other without changing his voice. The door revolves because the rooms were never far apart.

09. The Grievance Paper

[CUE: singing-bowl chapter chime] [MUSIC: tragic from 0s, ducked] [AMBIENCE: fs_42139.mp3] [ERA: radio]

Narrator (Kore): The Grievance Paper. Narrator (Kore): June ninth, nineteen eighty-seven. A monthly tabloid appears on the Stanford campus, aimed at conservative readers, edited by a philosophy undergraduate named Peter Thiel. He founded it that spring with Norman Book, a friend from high school who was about to enroll. The masthead of the first issue lists twelve names. Thiel is editor in chief. All twelve are men. Max Chafkin, who wrote the biography, notes that the pattern held for the rest of Thiel's career. Narrator (Kore): The Stanford Review was a student paper, which is to say it was a place where young people tried on sentences they might later be paid to mean. Chafkin's account of it is not a story about a campus club. It is a story about a network's first node. The names on those early mastheads, Keith Rabois, David Sacks, Norman Book, and a half dozen younger staffers, turn up again inside PayPal, the payments company this hour is walking toward. Sacks, who had put together one of the Review's most inflammatory issues, became PayPal's chief operating officer. The paper was the audition. The company was the production. Narrator (Kore): The concept is older than the internet, and this series has already seen it. Collis Huntington put an Associated Press reporter on a railroad payroll. A memo listed which New York papers would do as they were told. A student paper is not that. Nobody was being bribed to print the Review. What a paper can be, even a small one, is a seedbed. It sorts people. It teaches them who the enemy is supposed to be, and it gives them the experience of publishing the sort of sentence that makes a room go quiet. Thiel, in Chafkin's telling, got better at that sentence than anyone else in the room, and then he hired the room. Narrator (Kore): He was not yet rich. He was a student with a grievance and a masthead. The grievance was against the university's culture, against what he and his writers treated as a closed, moralizing elite. The method of answering it was to build a counter-elite in the same zip code, using the same credential. You do not leave Stanford in order to fight Stanford. You take the diploma, you keep the friends, and you build an institution that answers to you. By the time he had an office of his own, it was on Sand Hill Road. The road had a new tenant, and he had learned the trade in a newsroom that fit in a dorm.

10. BackRub

[CUE: singing-bowl chapter chime] [MUSIC: schubert2 from 150s, ducked] [AMBIENCE: hum.mp3] [ERA: hum]

Narrator (Kore): BackRub. Narrator (Kore): The middle of the nineteen nineties. A dormitory room at Stanford is an operations center. Larry Page, a graduate student, has stuffed a mismatched collection of computers into a cabinet he built out of Lego bricks. The project running on them is called BackRub. It is a search engine. The name is a small joke about the way the system crawls the web by following links backward, looking at who points at whom. Page also named the ranking math after himself. PageRank. People assumed it meant a web page. Steven Levy, who later wrote the history from inside the company, calls it a sly vanity. Narrator (Kore): The problem was mechanical before it was philosophical. The campus network into the Gates building could carry about ten megabits a second. The university's main line, a T3, could carry about forty-five. Scott Hassan, an early collaborator, remembered what happened when the team found a switch in the basement that had been set wrong. SPOKEN (Zubenelgenubi: Scott Hassan, an engineer, amused, remembering a dorm room): "We were using all the bandwidth of the network. And this was from a single machine." Narrator (Kore): At times, Levy writes, the crawler consumed half of Stanford's entire internet capacity. The university let the project run. Page's professors were using it to answer questions. So were their friends. A piece of student work was eating the institution's pipe, and the institution's response was to become a customer. Narrator (Kore): In January nineteen ninety-seven Page wrote the plan down, in emails to a venture capitalist named Vinod Khosla, who had money in a search company called Excite. Excite would buy BackRub. Page, and only Page, would go work there for seven months, long enough to install it, and then he would leave in time for the fall term and his doctorate. He claimed the technology would lift Excite's traffic by ten percent, which he translated into a hundred thirty thousand dollars a day in extra advertising, forty-seven million dollars a year. The price for all of this was one point six million dollars. Three hundred thousand of it was for Stanford, because the university owned the patent. Page and Brin had done the work as Stanford graduate students. Two hundred thousand was salary. Four hundred thousand was a bonus if he installed it in three months. Seven hundred thousand was Excite stock. Narrator (Kore): One point six million dollars in nineteen ninety-seven is about three point three million in twenty twenty-six grocery money. The university's piece, three hundred thousand, is about six hundred twenty-five thousand. Khosla came back at about seven hundred fifty thousand, total. The deal died in a conference room. Excite's chief executive, George Bell, a magazine executive the investors had installed as adult supervision, watched a bake-off. The query was the single word "internet." Excite's first results, Hassan remembered, were Chinese pages where the English word stuck out of a block of characters. BackRub's first results told you how to use a browser. Bell was upset. If people found the answer immediately, they would leave the site, and his advertising depended on them staying. Stickiness, in the dialect of that year, was the virtue. SPOKEN (Zubenelgenubi: Scott Hassan, still amazed years later): "He told us he wanted Excite's search engine to be eighty percent as good as the other search engines." Narrator (Kore): Hassan and the others looked at each other. The buyers did not want a better product. They wanted a product worse than the one they were being offered, because a worse product would keep the visitor confused, and a

confused visitor is a longer visit. Page and Brin left with the patent still in the university's name. Narrator (Kore): On September fourth, nineteen ninety-eight, they filed the papers and moved off campus. Sergey Brin's girlfriend knew a manager at Intel, Susan Wojcicki, who had just bought a house on Santa Margarita Street in Menlo Park with her husband, for six hundred fifteen thousand dollars. To help pay the mortgage, the couple rented Google the garage and several rooms for seventeen hundred dollars a month. In today's money the house is about one point three million dollars, and the rent is about thirty-five hundred a month. They hired a fellow student, Craig Silverstein, who had offered to show them how to compress the links so the system would fit in memory. He said later that the offer was a way to get his foot in the door. They also kept a hand on campus that fall, coteaching a class. Narrator (Kore): Stanford licensed PageRank to the company for stock. One point eight million shares. Not a royalty on each search. A piece of the firm. That is the equity decision Reimers had fought for, applied to a pair of graduate students who had been eating half the campus bandwidth. The Office of Technology Licensing's own later note is almost shy. Many search companies already existed. Most of the companies the office called were unimpressed. The licensing associate who sat with Page and Brin thought a startup was the right licensee. The office's job, it said, was to find the best one, and Google was it. Narrator (Kore): The sale of those shares is its own scene, and it has a moral the university wrote down. Google went public in August two thousand four. Stanford sold a hundred eighty-four thousand two hundred seven shares in the offering, about a tenth of the stake, for fifteen point seven million dollars. The policy, printed in the office's annual report, was to sell public stock as soon as the university was allowed to, for reasons of conflict of interest and insider trading, rather than to hold out for the peak. The rest went in early two thousand five. Katherine Ku, who directed the office, confirmed the total to the San Jose Mercury News: three hundred thirty-six million dollars. Set it next to the founding gift. Leland and Jane Stanford's twenty million dollars in eighteen eighty-five is about seven hundred forty-four million in the same grocery dollars. The Google shares, sold, were on the order of three-quarters of the fortune that built the university, measured by the cost of living. Narrator (Kore): The Mercury News did the cruel arithmetic. By the first of December, two thousand five, the stock had closed at four hundred four dollars and ninety-one cents. Held to that week, the same one point eight million shares would have brought in more than seven hundred million dollars. The university had chosen the rule over the windfall. You can call that prudence. You can also call it the price of a policy written to keep a licensor from becoming a speculator. Either way, three hundred thirty-six million dollars moved from a company started in a dorm room into the treasury of the landlord. Narrator (Kore): And Page already knew what the company was. At a party, Kevin Kelly, the writer and editor who had helped start Wired, asked him why the world needed another search engine. Page's answer, as Kelly later told it: SPOKEN (Umbriel: Larry Page, a young founder, flat, at a party): "We are not really interested in search. We are making an AI." Narrator (Kore): [SILENCE] [HARD] AI is artificial intelligence. A machine that does tasks people had assumed required a mind. Page was saying, before the phrase became a product category, that search was the training ground, not the destination. The university had licensed the training ground. It had taken the shares. It had sold the shares. The machine Page was describing was already, by his own account, the point.

11. PayPal

[CUE: singing-bowl chapter chime] [MUSIC: brahms1_1 from 0s, ducked] [AMBIENCE: hum.mp3, crowd.mp3]

Narrator (Kore): PayPal. Narrator (Kore): Thanksgiving, nineteen ninety-nine. A group of young men leaves an office in Palo Alto and walks to an automated teller machine, an ATM, the box in the wall that pays out cash. One of them puts in a debit card from a company he has named X.com. He types a number. The machine thinks, and then it produces bills. Jimmy Soni, who wrote the history of the company from the inside, records the reaction of a colleague. SPOKEN (Zubenelgenubi: an X.com employee, delighted, years later): "Elon was very, very happy with that." Narrator (Kore): Elon Musk was not yet thirty. He had sold a first company, Zip2, and put the money into a financial website that wanted to be a bank without the marble. The card working at a machine near the office was the product, made physical. Money had come out of a wall because a database said it could. A few years later, in a talk at Stanford in two thousand three, Musk compressed the idea. SPOKEN (Rasalgethi: Elon Musk, in a 2003 talk at Stanford, matter of fact): "Money is low bandwidth. You don't need some sort of big infrastructure improvement to do things with it. It's really just an entry in a database." Narrator (Kore): Low bandwidth means the message is small. A payment is not a barrel of wheat and it is not a reel of film. It is a line in a ledger that says this account is lighter and that account is heavier. Whoever runs the ledger runs the payment. That is an old insight in a new machine. The custom house at Monterey, in the hide trade, had the same job. So did the railroad's rate desk. The difference is speed, and the difference is that the ledger can now sit in a company that did not exist when the card was printed. Narrator (Kore): X.com collided with another Palo Alto company, Confinity, where Peter Thiel and Max Levchin were trying to beam money between handheld computers. Levchin had come from Kiev. As a boy he wrote programs longhand, on paper, because he did not yet have a machine that could run them. The merged firm took the name customers actually used. PayPal. The internal war was bitter and, in places, operatic. Musk was pushed out of the chief executive's chair. Thiel's circle, several of them alumni of the Stanford Review, resented the rebranding and the software rewrites. Fraud leaked through the system faster than they could patch it. The fight is not the lesson. A network effect is simple enough to say in a minute. Narrator (Kore): A network effect is what happens when a service becomes more useful to each person because other people are already using it. A telephone is worthless alone. A telephone is valuable in a city where everyone has one. A payment system works the same way. If the person you are buying from does not accept it, the system is a hobby. If the marketplace where both of you already are accepts it, the system is a road. PayPal found that road on eBay, the auction site, where strangers were already trading with strangers and needed a way to pay a person they would never meet. By the summer of two thousand two, eBay's own press release said, about sixty percent of PayPal's business was happening on eBay. eBay had forty-six million users. PayPal had the ledger those users were choosing, even though eBay was pushing a payment service of its own, called Billpoint. Narrator (Kore): The story about Musk and Stanford belongs here, and it has to be labeled. The story goes that he enrolled in a Stanford doctorate program in physics and materials science, and

dropped out after two days because the internet was more interesting. Ashlee Vance records it that way, with the phrase "as the story goes." Vance also records what happened when a lawyer later asked the university for the file. The graduate admissions office wrote back that, based on the information given, it was unable to locate a record for Elon Musk. Professors who were teaching physics there in nineteen ninety-five did not remember him. Some of his early enemies think the admission itself was a story told to decorate a young entrepreneur. The record, on the enrollment, is thinner than the legend. What the record does show is the work: Zip2, then X.com, then the card in the machine, then the companies after. Narrator (Kore): July eighth, two thousand two. eBay announces it will buy PayPal. The document is a merger agreement dated the day before. Each share of PayPal becomes zero point three nine shares of eBay. On eBay's stock price of July fifth, the release says, the deal is worth one point five billion dollars. About two point eight billion in today's money. The release is careful about the roundness of the number. The final price would move with eBay's stock, and with how long the deal took to close. Narrator (Kore): It closed on October third, two thousand two. PayPal's stockholders approved it. The same day, Peter Thiel resigned as chief executive. An eBay executive took the payments job. Billpoint would be phased out. PayPal would keep its name. The young men who had been trying to throw one another out of the building walked out of it, in the same week, holding stock in the buyer.

12. The Network

[CUE: singing-bowl chapter chime] [MUSIC: brahms4_2 from 0s, ducked] [AMBIENCE: wind.mp3, hum.mp3]

Narrator (Kore): The Network. Narrator (Kore): October third, two thousand two. The sale closes, and Peter Thiel writes the company a note. Jimmy Soni prints it. Effective at the close of the markets, he says, this is his last day. Then the sentence that matters. In founding PayPal, he and Max started by hiring their friends, and then their friends' friends, moving out in concentric circles. He calls those friendships the lasting testimony. Narrator (Kore): Stay with them for a minute, because they are not a chart yet. Levchin was still the engineer who had built the system that fought the fraud. He was not the face anyone would put on a magazine. Reid Hoffman, whom Thiel had made chief operating officer against the advice of his own board, had already spent a weekend at his grandparents' house in Gualala trying to imagine what came after a collapse. On the third day he floated a network of professional names. The world would later call it LinkedIn. David Sacks had been left in charge while Thiel traveled, and then learned he would not run the company he had been running. Musk had been pushed out of that chair two years earlier. He was already putting what he had into a car, and into rockets. Jack Selby, who went with Thiel, said they were trading again before the month was out. Ken Howery went too. Quote (Charon, flat documentary): I have always known that people are the most valuable part of any business, and today I'm more certain than ever that this is the case. Narrator (Kore): They kept in touch, which is what the note had promised, and the keeping in touch became the next gate. Hoffman's network. Levchin's later companies. Musk's car and his rockets. Thiel's fund, and the firm he would name for a stone that sees. Five years later Fortune photographed a group of them dressed as a

crime family and gave the joke a name, the PayPal Mafia. Soni, who wrote the history from the inside, says the picture left people out. It made the group look more alike than the company had been. The engineers who stayed, and argued about passwords, and hated the khaki, do not get the caption. What the caption did was tell the men in it that they were a class. Narrator (Kore): A class is a gate with a memory. Sand Hill had been the gate for the companies. This was the gate for the people. You did not apply. You had been in the room when the card worked at the machine in the wall, or you had been hired by someone who had. A century earlier the railroad had put its clerks' sons into the next offices. This road did it with stock, and with dinner. The product was sold. The address book was not. Narrator (Kore): [SILENCE] [HARD] The machine does not stop when the company is bought. It changes desks.

The Open Era — full audio script (voice/cue markup)

The Open Era — full audio script (voice/cue markup)

Episode 8 of 8. Cardinal. Narrator = Kore. Documentary quotes = Charon. Other voices are listed per line. [HARD] = music gated out. [SILENCE] = about 2 seconds before a reveal. Cue and ambience names are reused from the series pipeline. Sources are not narrated; they live in the review file.

The coda is a separate post-roll. It is not part of the sixty-minute count.

01. The Birthday

[CUE: singing-bowl chapter chime] [MUSIC: tragic from 0s, ducked] [AMBIENCE: hum.mp3, static.wav]

Narrator (Kore): The Open Era. Narrator (Kore): Cardinal. Episode eight. The Open Era. Two thousand two to twenty twenty-five. Narrator (Kore): May sixteenth, twenty twenty-two. Just after midnight. A young man in Uvalde, Texas, turns eighteen. The families who will sue say that twenty-three minutes later he bought a rifle, a Daniel Defense DDM4V7. Eight days after that, at Robb Elementary, nineteen children and two teachers were killed. His name was Salvador Ramos. The suits come two years later, on the anniversary, and they are allegations. They are not findings. Hold that distinction, because the hour will depend on it. Narrator (Kore): The lawyer is Josh Koskoff, of Connecticut, the same lawyer who sued the maker of the rifle used at Sandy Hook. On May twenty-fourth, twenty twenty-four, he files two cases. One, in Los Angeles, for about forty-five family members, against Meta, the company that owns Instagram and

Facebook, and against Activision Blizzard, the company that makes the video game Call of Duty. The other, in Uvalde County, for thirty-one family members, against Daniel Defense. The Los Angeles complaint says the companies rake in billions from the comfort of Menlo Park and Santa Monica. Menlo Park is the town next to the university this series has been walking. Santa Monica is the beach. SPEECH (Algieba: Josh Koskoff, the families' lawyer, to reporters, controlled): Just twenty-three minutes after midnight on his eighteenth birthday, the Uvalde shooter bought an AR-fifteen made by a company with a market share of less than one percent. This three-headed monster knowingly exposed him to the weapon, conditioned him to see it as a tool to solve his problems, and trained him to use it. Narrator (Kore): That is the theory. The complaint says that in November twenty twenty-one he downloaded Call of Duty: Modern Warfare, and that the rifle appeared on the game's opening title page. It says Daniel Defense's Instagram called a rifle with a holographic sight "totally murdered out," and captioned a photo of soldiers "Hunters Hunt." It says that within weeks he was back on the company's website, and that Instagram's own rules prohibit firearm advertising, and that Meta enforces other rules and not this one. Activision's lawyers answered that a war game is speech, and that the company owed no duty to stop a third person's violence. None of this has been found by a jury. Narrator (Kore): What has been found, on paper, is older, and it is not this case. Go back to two thousand nine. Remington, through its parent, Freedom Group, made a confidential product-placement deal with Activision. The gun was Remington's Adaptive Combat Rifle. The game was Call of Duty: Modern Warfare Two. No money changed hands. Both sides agreed to keep it secret. The documents came out years later, in the Sandy Hook litigation, and the Wall Street Journal reported them. One memo said this. Quote (Charon, flat documentary): A primary means for young potential shooters to come into contact with firearms and ammunition is through virtual gaming scenarios. Narrator (Kore): Another said that people seek out the brands of the guns they have seen, and that a lack of direct branding helps shield the company from looking like an endorsement. Executives wrote about winning their fair share of these young consumers. A former vice president, John Trull, told the Journal that the executives did not understand multiplayer team deathmatches when they made the deal. The memo is real. The understanding may have been thin. Both of those sentences can be true. Narrator (Kore): Sandy Hook was December fourteenth, twenty twelve. Twenty first graders and six educators. Remington argued that its ads did not target young people at risk, and that nothing linked the ads to the shooting. One of the ads, in Maxim, read "Consider Your Man Card Reissued." The families could sue at all because of a door in a shield. In two thousand five Congress passed the Protection of Lawful Commerce in Arms Act. Teach the name by the job. The law makes gunmakers nearly impossible to sue for what a criminal does with a gun. It has an exception. If the maker knowingly violates a statute about how guns are sold or marketed, the shield drops. On March nineteenth, twenty nineteen, the Connecticut Supreme Court held that the state's unfair-trade law can be that kind of statute. The United States Supreme Court declined to review it. On February fifteenth, twenty twenty-two, nine Sandy Hook families settled for seventy-three million dollars, paid by Remington's insurers, the maximum the policies covered, after the company's second bankruptcy. Narrator (Kore): The same key has not opened Menlo Park. On September fifth, twenty twenty-five, Judge William F. Highberger of Los Angeles Superior Court

dismissed Meta from the Uvalde case. He wrote that liability would impose limitless liability on a creator or a disseminator for the violent acts of others anywhere in the world. The defense firm later summarized the ruling this way: the shooter was the proximate cause, and Meta's conduct was so far attenuated from the harm that it was only negligible or theoretical. He reaffirmed it that December. The families appealed. The same September, he denied Activision's motion to throw the case out as an attack on free speech. Activision then argued it owed no duty, on the same ground as Meta. Narrator (Kore): Why a gunmaker can be made to settle, and a platform can be let out, is a statute from nineteen ninety-six. In nineteen ninety-five a New York court held the online service Prodigy liable as a publisher, because Prodigy moderated its boards. Moderate, and you own the words. Leave them alone, and you might not. Representatives Chris Cox, a California Republican, and Ron Wyden, an Oregon Democrat, wrote the answer into the Telecommunications Act. Twenty-six words. Quote (Charon, flat documentary): No provider or user of an interactive computer service shall be treated as the publisher or speaker of any information provided by another information content provider. Narrator (Kore): That is Section two thirty. A railroad act, a century earlier, gave a company the right of way. This sentence gave a platform the right not to be treated as the speaker of what its users post. The families say Menlo Park built the hallway that led a teenager to a rifle. The court, so far, says a hallway is not a cause. The allegation stands. The dismissal stands. The appeal is the part that is not over.

02. The Ledger

[CUE: singing-bowl chapter chime] [MUSIC: mozart40_2 from 0s, ducked] [AMBIENCE: hum.mp3]

Narrator (Kore): The Ledger. Narrator (Kore): The company in that dismissal was born on the other coast, and this is the one eastern beat of the origin, not of the bloodline. February two thousand four. A dormitory at Harvard. Mark Zuckerberg and his roommates put a site online and call it Thefacebook. By summer the company is in Palo Alto. It does not stay a college directory. The product becomes the habit, and the habit becomes the crop. Narrator (Kore): Teach the crop, because the rest of the complaints are about it. The attention economy is a market in which the thing being sold is not the song, or the post, or the game. It is the time you spend, and the prediction of what you will do next. Shoshana Zuboff gave the harsher name, surveillance capitalism. In plain words: the service looks free because you are not the customer. The customer is whoever pays to know what you are likely to do, and then to nudge you toward doing it. A recommendation is not a suggestion in the old sense. It is a bid. Narrator (Kore): By August twenty fourteen, Christopher Wylie writes, two months after they launched the harvesting app, Cambridge Analytica had collected, through that app, built by a Cambridge researcher named Aleksandr Kogan, Facebook data on more than eighty-seven million people. Facebook later confirmed the figure. Eighty-seven million is more than twice the population of California. The word for what the firm said it was selling is psychographics. Not your age and your zip code. Your personality, inferred from what you liked, and then a message aimed at that inference. Christopher Wylie, who helped build it and then described it, wrote the sentence that belongs to this university.

SPOKEN (Alnilam: Christopher Wylie, uneasy, remembering): And if their colleagues at universities like Harvard or Stanford were also getting interested in our work, I thought that surely we must be onto something. Narrator (Kore): That is legitimation. A famous campus does not have to audit the work. It only has to be nearby, and the nearby is read as a permission slip. Wylie also says that staff from Palantir, Peter Thiel's data company, asked about access to the set and pointed at a possible legal loophole, at a meeting in Palantir's London office in the summer of twenty fourteen. That is his account. It is not a court's. Narrator (Kore): In July twenty nineteen the Federal Trade Commission fined Facebook five billion dollars, a record, and imposed a new privacy order. Five billion dollars is about fifteen times what Stanford received when it sold the Google shares. The fine did not move the ledger. It priced a scandal. The company kept the users, the prediction, and the headquarters in Menlo Park.

03. Wonderland

[CUE: singing-bowl chapter chime] [MUSIC: brahms3_3 from 0s, ducked] [AMBIENCE: hum.mp3]

Narrator (Kore): Wonderland. Narrator (Kore): April sixteenth, twenty twenty. The New York Times releases the first episode of a podcast, not a documentary, called Rabbit Hole. The reporter is Kevin Roose. The person at the center of the reporting is Caleb Cain, then twenty-six, in West Virginia. He had dropped out of college. In twenty fourteen, when he thought of himself as liberal, he went looking for self-help videos. He found a host named Stefan Molyneux. SPOKEN (Puck: Caleb Cain, a young man in West Virginia, plain): He was willing to address young men's issues directly, in a way I'd never heard before. Narrator (Kore): Recommendations pulled him toward the far right. He later said the videos made him feel as if they were giving him power and respect and authority. He handed Roose the whole history. More than twelve thousand videos. Twenty-five hundred searches. The machine's own number, from those years, is the one to teach. Recommendations drove more than seventy percent of watch time on YouTube. From twenty twelve the company optimized for watch time. Think of a television network that is paid only when you do not change the channel. The rational program is the one you cannot leave. Narrator (Kore): The same machine carried him out. A change in the recommendations brought him creators on the left who used the same style, Natalie Wynn among them, and Cain became a creator himself. On January twenty-fifth, twenty nineteen, YouTube began reducing recommendations of what it called borderline content, at first in the United States, and it said the change touched under one percent of videos. That December it claimed a seventy percent average drop, in the United States, in watch time of such content coming from recommendations to people who had not subscribed. YouTube has denied that the system radicalizes people. Narrator (Kore): The research that followed is the honest complication, and it should be said at the same volume as the story. A twenty twenty-one paper in the Proceedings of the National Academy of Sciences found far-right consumption on YouTube small, stable, and matched to what people already did elsewhere, with no clear recommendation-driven rabbit hole. A twenty twenty-three paper in Science Advances found extremist channels concentrated among a small group of already resentful users, who arrived by

subscriptions and outside links. People who had not subscribed rarely got those recommendations. A twenty twenty-four study, using bots, found that recommendations had a mild moderating effect compared with what users chose for themselves. Most of that data comes from after the twenty nineteen changes. Cain's years were twenty fourteen to twenty eighteen, before them. The rabbit hole can be real for a few and a myth for most. A documentary that picks one of those sentences and hides the other is doing the machine's job. Narrator (Kore): The chief executive Roose interviewed, in a later episode, was Susan Wojcicki. The last hour had her as the Intel manager who rented a Menlo Park garage to the founders of Google for seventeen hundred dollars a month. By twenty twenty she was running the largest video service on earth. The garage and the recommendation are the same peninsula, twenty-two years apart.

04. The Studio

[CUE: singing-bowl chapter chime] [MUSIC: borodin from 120s, ducked] [AMBIENCE: hum.mp3]

Narrator (Kore): The Studio. Narrator (Kore): August twenty-ninth, nineteen ninety-seven, in Scotts Valley. Reed Hastings and Marc Randolph start Netflix. The company later moved to Los Gatos. Hastings has a Stanford master's in computer science, nineteen eighty-eight, in artificial intelligence. In December twenty twenty-five Netflix agreed to buy Warner Bros., HBO, and HBO Max, an enterprise value of eighty-two point seven billion dollars. It lost. Narrator (Kore): The fortune that won began as a database that did not work. At Ampex, Larry Ellison worked on a failed CIA contract called Project Oracle. Three employees left and started a company. On the third name, Oracle. The first customer was the CIA. Founded in nineteen seventy-seven, with Bob Miner and Ed Oates. In the Reagan years, Malcolm Harris writes, the agency mailed checks in plain white envelopes and did not mind that the first versions did not work. An engineer recalls being told this. Quote (Charon, flat documentary): We bought this thing knowing damn well it wasn't going to work. We're buying an idea. Narrator (Kore): Ellison has no Stanford degree. When Musk was buying Twitter, Faiz Siddiqui records the exchange. Any interest in participating in the Twitter deal? Yes, of course. Ellison put in about a billion dollars. Narrator (Kore): His son David's company is Skydance. In January twenty eighteen Tencent bought just under ten percent, at about a one point five billion dollar valuation, with co-financing and the right to distribute in China. In the twenty nineteen trailer for Top Gun: Maverick, the Taiwan and Japan flags on the flight jacket were swapped out while Tencent was a backer. Tencent withdrew. The twenty twenty-two film restored the flags, and did not open in mainland China. Skydance later told the FCC that Tencent would hold a non-voting, passive stake under five percent. Narrator (Kore): In two thousand eight, Sheehan records, a businessman sat with China's deputy minister of culture. The official opened with a question. SPOKEN (Schedar: a deputy minister of culture, formal, puzzled): Who is the CEO of Hollywood? Narrator (Kore): There wasn't one. Wang Jianlin, of Dalian Wanda, bid four point nine billion dollars for forty-nine percent of Paramount, and was turned down. In twenty seventeen Beijing's capital controls, and a ban on what it called irrational investment in entertainment, stopped the rest. Narrator (Kore): Paramount and Skydance signed on

July seventh, twenty twenty-four. On July second, twenty twenty-five, Paramount paid sixteen million dollars to settle the president's Sixty Minutes lawsuit. On July seventeenth, days after Stephen Colbert called the settlement, in his words, a big fat bribe, CBS canceled The Late Show. The last season would end in May twenty twenty-six. On July twenty-fourth the FCC approved the purchase, about eight billion dollars. It closed August seventh as Paramount Skydance. Reuters put it at eight point four billion. On October sixth, twenty twenty-five, Paramount bought The Free Press, reported at about one hundred fifty million dollars, and named Bari Weiss editor in chief of CBS News, reporting directly to David Ellison. Narrator (Kore): Tie it to the start. Leland Stanford paid for the horse photographs. In January eighteen eighty Eadweard Muybridge showed the horses moving, on a machine he called a zoopraxiscope, at the Stanford mansion in San Francisco. Malcolm Harris writes the line. The age of movies began in California, not in Hollywood but in the Bay. Paramount began as a western. In nineteen fourteen Cecil B. DeMille shot The Squaw Man, the title of that year, in a barn at Selma and Vine. It was the first feature made in Hollywood. The Jesse L. Lasky Feature Play Company became Paramount.

05. Four Companies

[CUE: singing-bowl chapter chime] [MUSIC: brahms1_1 from 0s, ducked] [AMBIENCE: static.wav, hum.mp3]

Narrator (Kore): Four Companies. Narrator (Kore): August sixth, twenty eighteen. In about a day, Apple removes most of the InfoWars podcasts, YouTube terminates Alex Jones's channel, and Facebook removes four of his pages. Spotify has already pulled the show. Twitter gives him a seven-day suspension, then bans him for good on September sixth. Four companies, most of them on this coast, switch one man off. The lie they were switching off had been on the air for years. Jones said the Sandy Hook murders were a hoax, staged with actors, to tighten the gun laws. At trial he conceded the shooting was one hundred percent real. Parents testified that his followers sent death threats and rape threats, and harassed them in person. Narrator (Kore): The judgments, updated to the date of this episode, are not the numbers people memorized. In Texas, a jury's award of about fifty million dollars to the parents of six-year-old Jesse Lewis was cut on August twenty-first, twenty twenty-six, by the state's Third Court of Appeals, to about six million. Compensatory damages of four point one million, and punitive damages capped at about one point five million. Jones said he would appeal to the Texas Supreme Court. The families' lawyer, Mark Bankston, called the cut irrelevant. In Connecticut, a judge had already found Jones liable by default, for discovery violations, and a jury awarded nine hundred sixty-four million, with the judge adding four hundred seventy-three million more. Appeals brought the Connecticut judgment to about one point two five billion. The United States Supreme Court refused to hear it on October fourteenth, twenty twenty-five. As of August twenty twenty-six, Jones had paid none of it. He had been in personal bankruptcy since late twenty twenty-two. Narrator (Kore): The microphone, meanwhile, found another plug. On October twenty-seventh, twenty twenty, months into a Spotify deal reported at about a hundred million dollars, Jones sat as a guest on Joe Rogan's show, with Tim Dillon. Spotify employees protested on the

company's internal chat. The general counsel's guidance, as reported, was that Spotify was not going to ban specific individuals from being guests on other people's shows. YouTube's position was that it bans channels, not guests. Rogan said he fact-checked every crazy thing Jones said. He is not InfoWars. He pushed back on some claims, and Spotify has said it does not edit him. The economic fact is plainer than the guilt. The companies that had switched Jones off were not the only companies that could carry a voice, and the one that paid the most for a single voice decided a guest was not a channel. In February twenty twenty-four Rogan renewed with Spotify for about two hundred fifty million dollars, the Wall Street Journal's figure, and the new deal was no longer exclusive. That October he recorded a three-hour interview with Donald Trump in Austin. On November fourth, after posting an interview with Elon Musk, he endorsed Trump, and said Musk had made the most compelling case. Narrator (Kore): And the last attempt to buy the microphone, as of this recording, is frozen. In November twenty twenty-four The Onion, backed by Sandy Hook families, won a bankruptcy auction for the company that owned InfoWars. A federal bankruptcy judge threw the result out over process and sent the families to state court. A Texas state court later ordered the assets to a receiver. In April twenty twenty-six The Onion announced a deal to license and then buy the brand. The chief executive said the families had not received a penny. A Texas appeals court stayed the deal while Jones appealed. On September ninth, twenty twenty-six, the Texas Supreme Court refused to lift the stay. Jones had already given up the InfoWars name and moved the show. The satire and the judgment are both real. The handover is not finished.

06. Specific Deterrence

[CUE: singing-bowl chapter chime] [MUSIC: schubert2 from 0s, ducked] [AMBIENCE: fs_42139.mp3]

Narrator (Kore): Specific Deterrence. Narrator (Kore): December two thousand seven. A blog called Valleywag, part of Gawker, publishes a post by Owen Thomas. The headline is a taunt. Peter Thiel is totally gay, people. Nick Denton, Gawker's founder, had suggested the story. Thiel is a Stanford graduate, twice, bachelor's and law, and the founder of the Stanford Review, the student paper from the last hour. He did not sue over the headline. He funded someone else's suit, in secret. Terry Bollea, known as Hulk Hogan, had a sex tape, and Gawker published it. In March twenty sixteen a Florida jury awarded one hundred forty million dollars. One hundred fifteen million, then twenty-five million in punitive damages. Thiel later told the New York Times he had spent about ten million dollars on the case. Quote (Charon, flat documentary): It's less about revenge and more about specific deterrence. Narrator (Kore): Specific deterrence is a lawyer's phrase. You punish one person so that person, and the people watching, do not do the thing again. To his friends, Max Chafkin records, Thiel called it the most philanthropic thing he had ever done. Gawker went bankrupt. Univision bought assets for one hundred thirty-five million dollars. The site shut down. The company later settled the Bollea case for thirty-one million. The jury's number and the check that cleared are different numbers, and both belong in the record. A man who had learned to publish in a dormitory newsroom had funded a wrestler's lawsuit until a newsroom died.

The railroad, in an earlier hour, put reporters on a payroll. This is the later version. You do not buy the copy. You buy the plaintiff.

07. The Commencement

[CUE: singing-bowl chapter chime] [MUSIC: aria from 0s, ducked] [AMBIENCE: crowd.mp3] [MOTIF: Jane, Goldberg Aria phrase]

Narrator (Kore): The Commencement. Narrator (Kore): June twelfth, two thousand five. Stanford's commencement. Steve Jobs did not go to the university. He stands up anyway, and he tells the students about the day he was fired from the company he had founded. SPEECH (Achird: Steve Jobs, at commencement in 2005, candid and quiet): The focus of my entire adult life was gone, and it was devastating. Narrator (Kore): [HARD] The sentence is about Apple in nineteen eighty-five. The use of it, on this lawn, is the cult energy the last hour named in a memo. Mike Markkula had written that a company's image is everything the customer sees, hears, or feels. A founder at a university he did not attend, telling a story of exile and return, is that memo performed as a sermon. He had the honesty, at least, to describe the firing as a devastation, not as a brand.

08. The Seeing Stone

[CUE: singing-bowl chapter chime] [MUSIC: borodin from 0s, ducked] [AMBIENCE: hum.mp3]

Narrator (Kore): The Seeing Stone. Narrator (Kore): Palantir. Say the Elvish first. Palantír. A seeing stone. In Tolkien's story the stones are powerful, and they are not unambiguously good. Sauron uses them to spy, to talk to conspirators, and to bend people who do not know the stones are dangerous to hold. Peter Thiel named the company after that object. Max Chafkin, his biographer, marks the choice. Narrator (Kore): The founding myth is cleaner than the recruiting. Thiel drove to the University Club in Palo Alto one morning to see Alex Karp, a man he knew from Stanford Law School. Karp thought of himself as a liberal in a circle of conservatives. The early team was the student paper come due. Stephen Cohen had edited the Stanford Review. Joe Lonsdale had edited it before him, and had interned at PayPal. Nathan Gettings had worked on fraud at PayPal. Sequoia's Michael Moritz passed. Kleiner Perkins passed. The check that mattered, in two thousand five, was two million dollars from In-Q-Tel, the Central Intelligence Agency's own venture fund, against a rough prototype. About three and a half million in twenty twenty-six grocery money. The venture firms on the road declined a product. The state's venture fund bought a way of seeing. Narrator (Kore): Call it software as state capacity. Not a social network. A tool a government uses when it wants to know who connects to whom, and it would rather rent the tool than build it. The company later moved into Facebook's old headquarters on University Avenue in Palo Alto, and it added a log, a system it called Gotham, that recorded each search, so an analyst could be audited. Karp's pitch was that the privacy trail would make the mining acceptable. Chafkin records a former engineer who sat in meetings

where, right after the privacy talk, a government client would suggest looking up an ex-girlfriend. The employees would remind the client that the search was logged, and then let the search happen. A log is not a refusal. It is a receipt.

09. The Bus

[CUE: singing-bowl chapter chime] [MUSIC: hebrides from 30s, ducked] [AMBIENCE: wind.mp3, crowd.mp3]

Narrator (Kore): The Bus. Narrator (Kore): October, twenty thirteen. A parking lot on El Camino Real, in Palo Alto. El Camino Real, the royal road. A fourteen-seat Mercedes bus, leather seats, purple lights in the ceiling, the name of a brokerage, DeLeon Realty, on the side. Matt Sheehan is on it, a Palo Alto native home from China with a broken ankle. He will write the book this scene comes from, The Transpacific Experiment. The men in the lot smoke and complain, in Mandarin, about business at home. At ten they stub the cigarettes and climb on. Kim Heng, who runs the China work for the firm, stands in the aisle and speaks mostly Mandarin, with English dropped in for the street names. Narrator (Kore): The brochure is the document. A photograph of the Great Wall comes down the page and becomes the Stanford campus, and Hoover Tower, the stack at the middle of the farm, becomes one more guard post on the wall. The wall continues, on the paper, into the Golden Gate Bridge. A university tower, drawn as a Chinese fortification, sold to families who have come to buy a house near the schools. Narrator (Kore): The second stop is a mansion in Menlo Park. Marble stair, limestone patio, a pool. One of the men jokes. First the Mandarin, which Sheehan renders as an obscenity, wo cao. Then the English. SPOKEN (Zubenelgenubi: a buyer on the bus, laughing, in English after the curse): It's such a beautiful place. Why didn't Bo Xilai come here to buy his villa? Narrator (Kore): Bo Xilai was a senior Chinese official who had fallen, in twenty twelve, in a corruption scandal. The joke lands because the bus knows what the house is for. Sheehan's phrase for the use is the new Swiss bank account. A house in Palo Alto holds value outside Chinese politics, outside a currency, and, for some buyers, outside the question of how the money was made. Fathers, he writes, often skip the green card so they will not owe American tax on worldwide income. Mothers and children stay. Or the house sits empty except for the vacations. Narrator (Kore): The numbers, from the National Association of Realtors as Sheehan uses them. Between twenty eleven and twenty fifteen, Chinese residential purchases in the United States more than quadrupled, to twenty-eight point six billion dollars. That is about eighty-five times what Stanford got for the Google shares. In twenty thirteen, fifty-three percent of the reported Chinese purchases were in California. In Palo Alto, during twenty twelve and twenty thirteen, the median house price rose fifty percent, to nearly two million dollars, and it had done that rise in two years. Some local brokers estimated that fifteen to twenty-five percent of the town's sales were going to Chinese buyers. One broker told Sheehan he had just sold a house to a contest of four people from China, and none of them had seen it. Narrator (Kore): The legal door for a number of the families was a visa called EB-five. At the time, five hundred thousand dollars into a project in a high-unemployment area, if the project created ten jobs, put a person in line for a green card. Five hundred thousand dollars in twenty fourteen is about seven hundred thousand in

twenty twenty-six grocery money. In twenty fourteen, nine thousand one hundred twenty-eight of those visas went to Chinese applicants, more than eighty-five percent of the program. The university's town had become a store of value, and the federal immigration code had become part of the financing. Narrator (Kore): Sheehan's own family is the turn, and it is documented, not decorative. His great-grandfather, Tommy Sheehan, a San Francisco longshoreman, used to drive a priest named Peter C. Yorke around the city. In nineteen oh one Yorke keynoted the California Chinese Exclusion Convention. SPEECH (Schedar: Father Peter Yorke, at the 1901 exclusion convention, formal and hard): Their thoughts are not our thoughts. Their blood is not our blood. Their outlook is not our outlook. Narrator (Kore): One century later the great-grandson is on a bus that sells Palo Alto to Chinese families, and the brochure puts the university's tower on the Great Wall. The priest's sentence and the brochure are the same road, read in opposite directions. Narrator (Kore): Hold the older photograph against a newer one, because Sheehan does. In eighteen sixty-nine, at the golden spike, the picture of the finished transcontinental railroad has engineers and white workers. Not a single Chinese face. The men who had done the most dangerous work on the mountain are not in the frame. In twenty thirteen, when a new span of the Bay Bridge opened between San Francisco and Oakland, the mayors were Ed Lee and Jean Quan, both Chinese American, both the children of working people. Quan later said the posing was a joke between them. SPOKEN (Sulafat: Jean Quan, remembering, dry): This time the Chinese were in the middle. Narrator (Kore): Neither photograph is the bus. The bus is the money, which does not pose. It buys.

10. The Citation

[CUE: singing-bowl chapter chime] [MUSIC: aase from 0s, ducked] [AMBIENCE: hum.mp3, telegraph.mp3]

Narrator (Kore): The Citation. Narrator (Kore): The search engine on the other side of the Pacific is not a Stanford export, and the sentence has to be exact. Robin Li did not go to Stanford. He did not go to Berkeley. He did not work at Google. He studied information management at Peking University, and computer science at the State University of New York at Buffalo, finishing in nineteen ninety-four. At a Dow Jones company in New Jersey he built a link-ranking system called RankDex, and he filed a patent on February fifth, nineteen ninety-seven. It was granted in nineteen ninety-nine and assigned to his employer. Narrator (Kore): Stanford's patent on PageRank, Larry Page's, assigned to the Leland Stanford Junior University, discusses that system as prior art. The word is cited. It is not copied. Narrator (Kore): The person who carried Li into the Valley's mythology was his co-founder, Eric Xu, a biology doctorate from Texas A and M and a former postdoc at Berkeley. Xu made a documentary for Chinese state television, a journey into Silicon Valley, and he took Li to meet Jerry Yang, the founder of Yahoo. Xu's line, later, was that he was inspired by seeing an ethnic Chinese person build a company that powerful, and that he was sure Li was inspired too. At a screening of the film at Stanford, Li's wife told Xu she wanted her husband to be an internet entrepreneur. The seed money, one point two million dollars, came half from a firm called Peninsula Capital and half from Integrity Partners. Bloomberg lists Greg Penner,

a Stanford business-school graduate, among the seed investors. Li had trained in Buffalo and then in New Jersey. He had not trained on this bay. Xu had. They founded the company in January two thousand and built it in Beijing. The name is Baidu. Narrator (Kore): Google arrived in the Chinese language in two thousand, was blocked for two weeks in September two thousand two, and by the end of that year had about a quarter of the market. In two thousand six it opened Google dot c n, under Kai-Fu Lee, who had gone to Columbia and Carnegie Mellon, not to Stanford, and who had run research for Microsoft in Beijing. The company censored results in-house, and it put up a disclaimer. Eric Schmidt, the chief executive, described the decision as an evil scale. They were not wild about the restrictions, he said, and they decided that not serving the users at all was the worse evil. Baidu's answer was an advertisement in which a Chinese poet defeats an American in a top hat, and the crowd shouts the last line. SPOKEN (Umbriel: a crowd in an advertisement, in unison, triumphant): Baidu understands Chinese better! Narrator (Kore): By late two thousand nine Baidu had about fifty-eight percent of the market and Google about thirty-six. What pushed Google out was not the ad. It was a hack, Operation Aurora, that reached into the Gmail of human-rights activists, including a twenty-year-old Tibetan activist at Stanford, during Sheehan's last year on the campus. On January twelfth, twenty ten, Google posted "A New Approach to China," and said it would stop censoring. In March it redirected the Chinese service to Hong Kong. By twenty fourteen the services were blocked. Sheehan's summary is the clean one. The twenty ten exit cleared the way for Baidu to own Chinese search. Narrator (Kore): One qualifier, so a proud sentence does not overrun the record. On May seventeenth, nineteen ninety-four, a link between a Chinese physics institute and Stanford's SLAC laboratory gave that institute a full internet connection. China dates its own full-function connection to April twentieth of that year, through a different line. Stanford was in the first week of the Chinese internet. It was not the whole invention of it. Narrator (Kore): The loop closed in both directions. In May twenty fourteen Baidu hired Andrew Ng, a Berkeley doctorate who had helped build deep learning at Stanford and had led Google Brain, the project that trained a network to recognize cats in YouTube videos. Baidu put his lab less than a mile from Google's own campus in Sunnyvale. The company, Sheehan notes, was worth about sixty billion dollars. About eighty-five billion in twenty twenty-six grocery money. The firm that had won the home market came back and hired the person who had taught the machines to see. Narrator (Kore): And Google tried to go back. In twenty eighteen The Intercept reported a secret project, Dragonfly, a censored Chinese search app that would require a sign-in and would share data with a Chinese partner. More than seven hundred forty employees signed a letter. The letter said the objection was not about China. It was about tools that help the powerful oppress the vulnerable, wherever they are. Within about a month, Sheehan writes, the company effectively ended the work. Between those two facts, an ex-Googler named Li Zhifei, a Johns Hopkins doctorate, had gone home and founded a company called Mobvoi. Sheehan noticed two glass arrows set into the office floor. One read California and pointed at the desks. The other read Silicon Valley and pointed at a pair of chairs. Li told him the construction workers could not read the English, and had installed the arrows backward. Google, blocked in the country, then invested about forty to forty-five million dollars in him, in twenty fifteen. Sheehan calls it the parent company's first direct

investment in a Chinese startup since the exit. The arrows pointed the wrong way. The money did not.

11. The Aura

[CUE: singing-bowl chapter chime] [MUSIC: brahms4_2 from 300s, ducked] [AMBIENCE: hum.mp3]

Narrator (Kore): The Aura. Narrator (Kore): Two thousand eight. Sequoia Capital, the firm Don Valentine started on the road in the last hour, hires two investors away from the Stanford endowment. Sebastian Mallaby records it. The job is a new business called Heritage, first to manage the partners' own wealth, then the wealth of the founders Sequoia has backed. Mallaby does not give their names. The fact is the direction of the hire. The university's money managers were worth recruiting onto the road. Narrator (Kore): An endowment is a permanent fund. You spend a slice of the return and you put the rest back, so the principal can keep its purchasing power. Stanford's, on August thirty-first, twenty twenty-five, was forty point eight billion dollars. In that fiscal year it paid out one point nine billion. The endowment is about fifty-five times the founding gift, counted in today's money. Undergraduate tuition in the same report is sixty-five thousand one hundred twenty-seven dollars. Nearly half of undergraduates receive need-based aid, so the sticker is not every family's bill. The road and the treasurer's office have been swapping staff. Narrator (Kore): The aura failed in public at a blood-testing company. Elizabeth Holmes had started as a Stanford undergraduate. Mallaby writes that she persuaded the dean of the engineering school to vouch for her, and that she recruited elder statesmen from the Hoover Institution onto the board. George Shultz, the secretary of state from the last hour, sat among them. His presence was a license. By the middle of the decade the company, Theranos, was valued at nine billion dollars. In twenty fourteen grocery money, that is about twelve point seven billion. It went to zero. Mallaby's line on the money is the one that complicates the postcard. Almost none of what Holmes raised came from Sand Hill Road practitioners. The Walton family put in one hundred fifty million. Rupert Murdoch put in one hundred twenty-one. The DeVos family and the Cox family put in one hundred million each. A venture investor named Tim Draper made a modest angel bet because of a family connection. His father, William Draper the third, had been at Yale with George H. W. Bush, and in Skull and Bones. The Ivy roots of this valley are another story. Narrator (Kore): What the outsiders bought was the credential. Holmes held shares with one hundred votes each, the founder-friendliness pushed, Mallaby writes, past Google and Facebook. Mallaby's phrase for the habit is the plain one. Fake it until you can make it. She faked a machine that did not work. The board of elders made the fake look like a Stanford result. A Hoover letterhead is cheaper to read than a lab notebook. The letterhead was the product. Narrator (Kore): The clubhouse for that product sits on the road, on the university's own land. Theo Baker, who will matter in a moment, describes the Rosewood Hotel in his book. It markets itself as the modern clubhouse for Silicon Valley. It is on Sand Hill Road, on land Stanford owns. A night starts in four figures. Baker's verdict is that the epicenter is a row of dull buildings whose advantage is being close to the campus down the hill. Under the hotel, the tenant may own the building. The university owns the dirt. The bank A. P.

Giannini founded left for Charlotte in nineteen ninety-eight. The clubhouse stayed on the farm.

12. The Freshman

[CUE: singing-bowl chapter chime] [MUSIC: fs_42139.mp3 bed only, no music] [AMBIENCE: fs_42139.mp3] [ERA: radio]

Narrator (Kore): The Freshman. Narrator (Kore): November twenty-ninth, twenty twenty-two. A freshman named Theo Baker publishes, in the Stanford Daily, the first article on questions about the president's science. It is not his first article. A piece on campus social life had run in October. This one is the one that moves the board. He had been tipped, on the first of the month, toward PubPeer, a website where scientists comment on one another's papers. He took the pictures to forensic readers, among them Elisabeth Bik. Then he called his editor. The editor's name is Sam Catania. Baker's book keeps the phone call. SPOKEN (Puck: Sam Catania, a student editor, on the phone, clipped): Holy shit. We need to get going. How quickly can you write this? Narrator (Kore): The board opened a review the same day. The report arrived on July seventeenth, twenty twenty-three, from an outside scientific panel, transmitted by a lawyer, Mark Filip, and addressed, among other places, to three sixty-five Lasuen Street. The findings have to be spoken the way they are written, because the slogan is cruder than the page. On the twelve papers reviewed, the panel concluded that Marc Tessier-Lavigne did not personally engage in research misconduct, the federal term for fabrication, falsification, or plagiarism. It did not find that he directed the manipulation, or that he knew of it when it happened. It also found that several papers exhibit manipulation of research data, and that in at least four of the five papers on which he was a principal author there was apparent manipulation by others, in his lab. Each of those five, the panel wrote, has serious flaws in the presentation of research data. The board, two days later, added its own sentence. When the problems surfaced, he took insufficient steps to correct the scientific record. Narrator (Kore): On July nineteenth he resigned the presidency, effective August thirty-first. Richard Saller became interim president on September first. Tessier-Lavigne remained a tenured professor of biology. His letter to the campus said the report refuted the allegations of fraud and misconduct made against him, and that he was stepping down for the good of the university. Both the letter and the panel are public. A freshman and a student paper started a process that collected more than fifty thousand documents. The president left the office. He kept the chair. That is what governance looked like when it worked, and the limit of what it worked to do. It did not find a forger in the president's own hand. It found a lab, a record left uncorrected, and an office that could not survive the finding. Narrator (Kore): The university also studies the machine, and in June twenty twenty-four that work became the fight. The Stanford Internet Observatory, started in twenty nineteen by Alex Stamos, with Renée DiResta as research director, had been publishing on election rumors. It drew lawsuits and subpoenas from the House Judiciary Committee. Platformer reported that the observatory was being dismantled. Stamos had left. DiResta's contract was not renewed. Staff were told to look for jobs. Stanford disputed the verb. It said the work

would continue under new leadership. Both are on the record. The office had become politically expensive, and the expense is not in dispute.

13. The Advisory Job

[CUE: singing-bowl chapter chime] [MUSIC: brahms1_1 from 420s, ducked] [AMBIENCE: telegraph.mp3, crowd.mp3]

Narrator (Kore): The Advisory Job. Narrator (Kore): October twenty-seventh, twenty twenty-two. Elon Musk completes the purchase of Twitter for forty-four billion dollars. He walks into the headquarters carrying a sink, so that he can say, let that sink in. Walter Isaacson records the entrance. He fires top executives the same day. A post goes out. The bird is freed. Forty-four billion dollars is a little more than Stanford's entire endowment three years later, forty point eight billion. He bought a microphone for more than the university whose file on him, when a lawyer later asked, could not be found. The price is the part that is settled. Narrator (Kore): The political money is a filing, not a theory. Peter Thiel gave fifteen million dollars to a committee called Protect Ohio Values, in three contributions, ten million in March twenty twenty-one and five million more in April twenty twenty-two. The committee existed to support J. D. Vance, who had worked for Thiel, for the United States Senate from Ohio. Vance won the seat that November. In twenty twenty-four he was elected vice president. No coordination has to be alleged. The disclosures are the scene. Fifteen million dollars, from one donor, aimed at one Senate race. A vote is free. Narrator (Kore): Musk's spending in the twenty twenty-four cycle, as Faiz Siddiqui tallied it while writing in December of that year, was more than two hundred seventy-five million dollars through a super PAC called America PAC, toward Trump and other Republicans. That is about eighteen times Thiel's fifteen million. On January twentieth, twenty twenty-five, the same day as the inauguration, an executive order renamed the United States Digital Service as the United States DOGE Service. DOGE is the joke acronym, the Department of Government Efficiency. It is not a department. It is a renamed office in the Executive Office of the President, plus a temporary organization scheduled to end on July fourth, twenty twenty-six. Agency heads were told to give it access to unclassified records and software. Musk's role was a special government employee, a category the law limits to about one hundred thirty days in a year. On May twenty-eighth, twenty twenty-five, the White House confirmed to Reuters that his off-boarding would begin that night. He had led an advisory wrecking crew, and the clock ran out. The companies remained his. The job did not. Narrator (Kore): Untouchability is the distance between the person and the instrument. When the temporary chair ends, the corporations are where he left them. The statute makes the advisor temporary. The cap table does not.

14. The Arithmetic

[CUE: singing-bowl chapter chime] [MUSIC: eroica2 from 240s, ducked] [AMBIENCE: hum.mp3]

Narrator (Kore): The Arithmetic. Narrator (Kore): The chokepoint now is the arithmetic. You can have the data, the graduate students, and the model on a whiteboard, and still be stopped by whoever owns the machines that do the multiplying. Narrator (Kore): Nvidia's fiscal year, ended January twenty-sixth, twenty twenty-five, is the audited number. Revenue was one hundred thirty point five billion dollars, up one hundred fourteen percent from sixty point nine billion the year before. The data-center line, the machines sold for this work, was one hundred fifteen point two billion. One hundred thirty point five billion is a little more than three times Stanford's endowment. Larry Page, at a party in the last hour, said he was not really interested in search. He was making an AI. Artificial intelligence. A machine that does tasks people had assumed required a mind. The training ground he described now has a toll booth, and the booth is a chip. Narrator (Kore): A headline number in the hundreds of billions is a plan, not money spent. The gate moved from the road to the chip. Then the government tried to put a lock on it. Narrator (Kore): October seventh, twenty twenty-two. The United States blocks the sale to China of the most powerful class of these chips, Nvidia's A one hundred and H one hundred, and of the tools that make them. An export control is a freight rate for a machine. Whoever writes it decides who gets to multiply. Nvidia's answer is a new pair of chips, the A eight hundred and the H eight hundred, drawn to sit just outside the line. In October twenty twenty-three the rules close that gap. Nvidia answers with the H twenty. In December twenty twenty-four the controls reach the memory, the packaging, and the equipment. In April twenty twenty-five, selling the H twenty into China requires a license, which Nvidia's own filing treats as a ban in practice. In August the licenses resume. In December twenty twenty-five the president allows a newer chip, the H two hundred, if the United States government takes a twenty-five percent cut of the sale. By May twenty twenty-six about ten Chinese firms are cleared to buy it. The paperwork is the part that is settled. The rule moves. The chip is redrawn. The rule moves again. Narrator (Kore): January twenty twenty-five. OpenAI tells the Financial Times it has seen evidence that a Chinese lab, DeepSeek, used distillation on OpenAI's models, against OpenAI's terms. Distillation, in this use, means training a new model on the answers of an older one, so the student copies the teacher's results without the teacher's data. OpenAI did not publish the evidence. A House committee later called unauthorized distillation highly likely, relying on what OpenAI had provided. On February twelfth, twenty twenty-six, OpenAI sent that committee a memo alleging the practice had continued, through disguised routers and resellers. No court has found this. OpenAI alleges. No finding says the lab took the models. Narrator (Kore): January twenty-ninth, twenty twenty-six. A federal courtroom in San Francisco. Linwei Ding, a former Google engineer, is convicted. He had come to the United States in twenty ten for a master's degree. The government said he uploaded more than five hundred files, more than two thousand pages, of Google's AI infrastructure. The exhibit built for the jury was his own pitch, for a company he called Zhisuan. Quote (Charon, flat documentary): We have experience with Google's ten-thousand-card computational power platform. We just need to replicate and upgrade it. Narrator (Kore): Ten thousand cards. A card, here, is one of those chips. The platform is a room of them. While he was in China, an intern back in Mountain View swiped Ding's badge, so the building would record him as at his desk. The Department of Justice called the verdict the first conviction on AI-related economic espionage charges. Seven counts of espionage, and

seven of stealing trade secrets. The same years produced other sentences that were about secrets and not about spying. In November twenty twenty-five a San Jose resident, Chenguang Gong, received forty-six months for taking missile-detection sensor technology, after a long run of applications to Chinese talent programs. Narrator (Kore): August twentieth, twenty twenty-six. Judge Chhabria throws out all seven espionage counts. The evidence did not show that Ding knew, or intended, that what he did would benefit the Chinese government. The trade-secret counts stand. On September first he is sentenced to twelve months minus a day, a fine of twenty-five thousand dollars, and restitution of more than one hundred eighty-nine thousand. About a year in custody. He took the files. After August the espionage conviction is gone. Matt Sheehan, in the housing book, records the caution that belongs next to the verdict. A string of Chinese American researchers had been charged with espionage only to have the cases prove unfounded. And in twenty eighteen the president said, in private, that almost every student who comes over to this country is a spy. The Ding case is what those two sentences collapse into when a court does the sorting. The theft was proved. The spy charge was not. A badge swipe is not a nationality. Narrator (Kore): Which leaves the line people now repeat as if a court had found it. China can manufacture, and it cannot make California. It is an argument, and the sources line up on both sides. The case for a gap comes from a scholar named Jeffrey Ding, no relation to the engineer. He argues that China's weakness is diffusion, the spread of a technology through the whole economy, more than the moment of invention. A RAND study finds the links from Chinese research to Chinese industry are the weaker ones. The case against the slogan is a measurement. In September twenty twenty-four the Information Technology and Innovation Foundation put China ahead, or level, in nuclear power, electric vehicles, and batteries, near the frontier in robotics, artificial intelligence, and quantum, and behind in semiconductors and in biopharma. Its twenty twenty-six index has China leading seven of ten advanced industries. Dan Wang, in a book called Breakneck, as a Foreign Affairs review renders him, reverses the premise. China is an engineering state, building big at breakneck speed, in contrast to the United States' lawyerly society, blocking everything it can, good and bad. On that account, California is the place that cannot build. Sheehan's rhyme is the nineteen eighties, when Americans said Japan could copy and could not invent, and Japanese investment went on generating jobs, income, and even innovation. The narrow sentence, the one the documents will support, is this. China builds at a scale California no longer can. Whether it can grow the loop this series has been walking, risk capital, an open university, and immigrant talent on the same road, is the question that is still open.

15. The Pattern

[CUE: singing-bowl chapter chime] [AMBIENCE: wind.mp3, crowd.mp3]

Narrator (Kore): The Pattern. Narrator (Kore): January twentieth, twenty twenty-five. Washington. The inauguration was supposed to be outdoors. The cold moved it inside, into the Capitol Rotunda, and the move scrambled the seating. What the photographs show, and what the Associated Press and the Guardian reported, is a row of technology executives with the president's family, ahead of several cabinet nominees. Elon Musk.

Mark Zuckerberg. Jeff Bezos. Sundar Pichai, of Google. Tim Cook was in that company of guests. The head of TikTok was in the room, the Associated Press noted, beside the nominee to lead the intelligence community. The weather is why the chart changed. The pictures are what remain. Proximity is the evidence. It does not require a theory of anyone's heart. Narrator (Kore): The coda to the purchase falls after this hour. On February twenty-sixth, twenty twenty-six, the Warner Bros. Discovery board called Paramount's bid superior. Netflix did not raise. On October sixth, twenty twenty-six, the deal closed. Justice Kagan denied a last-minute petition. The company was renamed Skydance. David Ellison and Ynon Kreiz became co-chief executives. The Ellisons and RedBird hold every voting share. Forty-seven billion dollars of the new equity came from the Ellisons, RedBird, and the sovereign funds of Saudi Arabia, Abu Dhabi, and Qatar. One family. A database bought as an idea. Then CBS, CNN, HBO, Paramount, and Warner Bros. Narrator (Kore): [SILENCE] [HARD] Rail controlled movement. Stanford controls succession. Narrator (Kore): It is. And it connects to me. I did not inherit a railroad, or a fund, or a studio. I inherited the habit of treating the names as scenery. Serra. Jordan. Terman. Page Mill. Sand Hill. Once you hear them as a succession, you cannot go back to hearing them as scenery. That is what these hours were for. The walk after this is only the sidewalk, in the order the names were laid down.

Coda. The Street Names

[CUE: singing-bowl chapter chime] [MUSIC: scottish3 from 0s, ducked] [AMBIENCE: wind.mp3, horses.mp3] [MOTIF: railroad stinger]

Narrator (Kore): The Street Names. A post-roll. Dusk on the farm. The oaks go black first. The signs are already lit, which is the fact this walk is heading toward, and it will not be hurried. Read them in the order the layers were laid down. You can do it on foot in an afternoon. Four centuries of proprietors, one sidewalk. Narrator (Kore): Gálvez. José de Gálvez, the inspector general who sent the expedition north. The administrator, not the priest who walked. The people already living on this ground do not have a sign at this corner. Narrator (Kore): Serra. Junípero Serra's name came off the mall across the front of the campus. Serra Mall became Jane Stanford Way. The street that kept his name is the one that runs out. Narrator (Kore): Lasuen. Fermín de Lasuén, Serra's successor, who carried the missions farther up the coast, including La Purísima. The scientific panel's letter, in twenty twenty-three, was addressed to three sixty-five Lasuen Street. Narrator (Kore): El Camino Real. The royal road, still the main street of the Peninsula. In October twenty thirteen the buyers' bus left from a lot on this road, purple lights in the ceiling, a brochure in the aisle. Narrator (Kore): Alvarado. Juan Bautista Alvarado, the governor who in eighteen thirty-six led a revolt against Mexico City's grip on California. David Starr Jordan, the first president, put the name on the row. Narrator (Kore): Stanford Avenue. The farm, the railroad fortune, the grant that could not be sold. The family could lease it. They could not cash it out. Narrator (Kore): Jane Stanford Way. The rename of Serra Mall took effect on October seventh, twenty nineteen, and the university dedicated it on November fourteenth. The official address became four fifty Jane Stanford Way. Serra Street, from Campus Drive East out to El Camino, kept the old

name. The co-founder is on the front door. The priest is on the exit. Narrator (Kore): Jordan Hall. The first president's name came off the building in October twenty twenty. It is Building four twenty now. Narrator (Kore): Hoover Tower. The archive, and the institution where a secretary of state came back to sit, and then sat on a blood-testing board. On the twenty thirteen brochure, this tower is a guard post on the Great Wall. Narrator (Kore): Terman. Frederick Terman's engineering buildings, at the edge of the farm he helped lease to industry. The research park is the lease, made architectural. Narrator (Kore): Page Mill. William Page's lumber road, before it was the spine of the research park. It still runs from the timber to the chip. Narrator (Kore): Sand Hill. Four minutes from the Quad. The gate. The Rosewood, on the university's land, calling itself the modern clubhouse. A company that cannot get a meeting in one of the low buildings does not merely lack money. It lacks the next introduction. Narrator (Kore): Stand where the royal road meets the farm and the order is the order of the hours. A mission administrator. A priest. A governor. A railroad family. A co-founder, put on the front door in twenty nineteen, while the priest was left on the road out. A tower that a brochure redrew as a wall. A lumber road that became a chip road. A venture road that became a clubhouse on leased ground. The signs do not argue. They accumulate. Narrator (Kore): The golden-spike photograph is the same kind of document, made a century and a half before two mayors stood in the middle of a new bridge and said, this time the Chinese were in the middle. Who is in the frame. Who is not. A street sign is that choice, cast in metal and lifted out of reach. Narrator (Kore): And the names that are not here. The Muwekma Ohlone, whose land the campus occupies. Little on the map, and what recognition exists is recent and small. The Chinese workers of the Central Pacific, absent from the golden-spike photograph, and absent from any building at the scale of the work they did. The people sterilized under the nineteen oh nine law. Nothing. The farm workers. Cesar Chavez has streets in California, real ones, and they are mostly in the Central Valley and in city neighborhoods, not on this walk. The list is the record. What a place commemorates is what it decided to remember, and this peninsula decided to remember the people who held the title. Narrator (Kore): Read a sign as a document and it answers four questions. Who. When. For what. And who decided. British streets carry empire. Parisian streets carry writers. Most American streets carry presidents and trees. This peninsula carries the people who held the title, in the order they held it. Narrator (Kore): [SILENCE] [HARD] Subliminal does not mean hidden. It means beneath the threshold at which something is consciously registered. The street signs are twelve feet up and lit at night. It was never a secret. It was only subliminal.